

VCU118 EVALUATION BOARD HW-U1-VCU118

(XCVU9P-2FLGA2104)

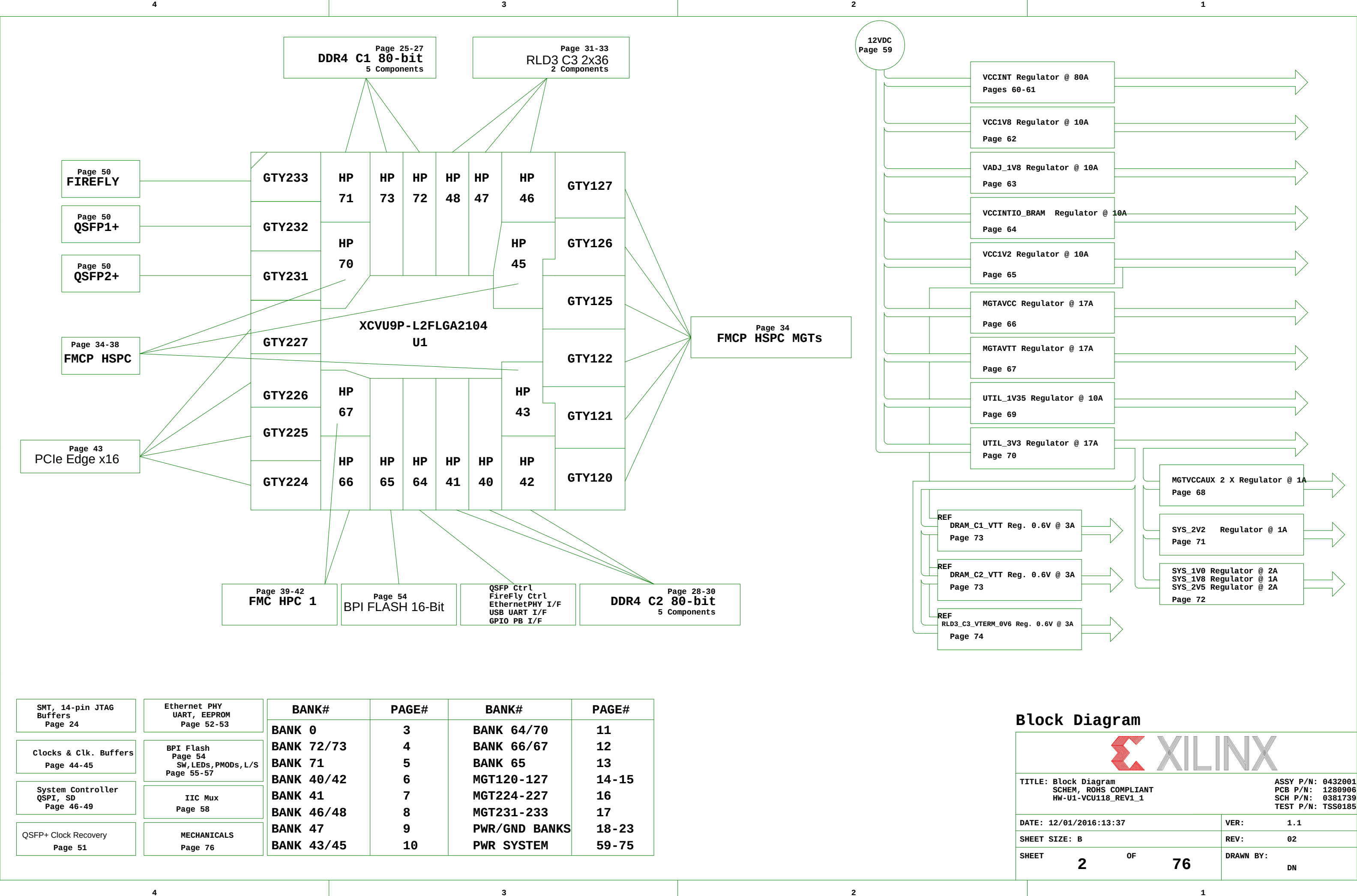
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Title Page			
			
TITLE: Title Page SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37		VER:	1.1
SHEET SIZE: B		REV:	02
SHEET	1	OF	76
		DRAWN BY:	DN



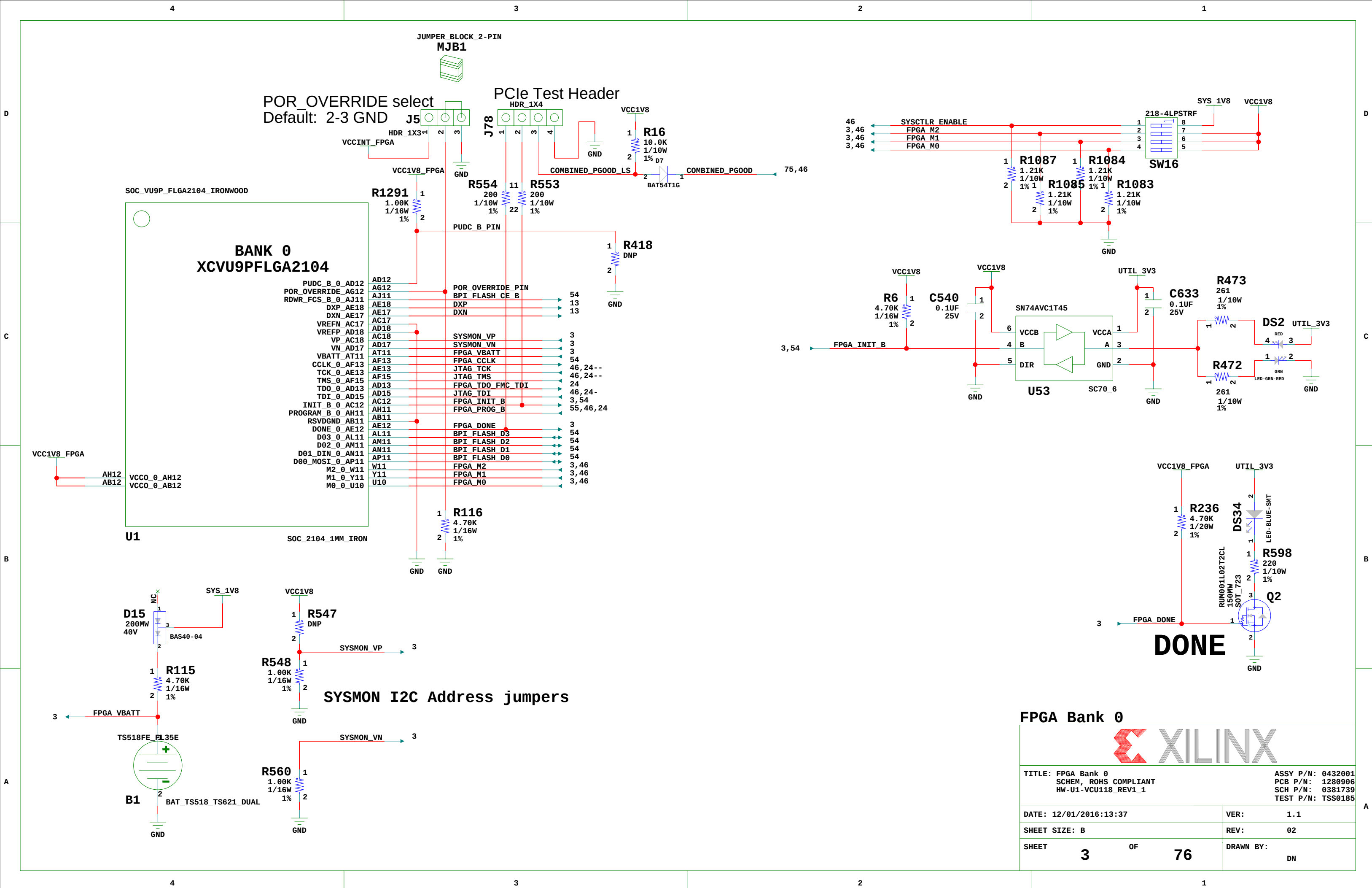
		BANK#	PAGE#	BANK#	PAGE#
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Block Diagram

TITLE: Block Diagram
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 2 OF 76	DRAWN BY: DN



Bank 73 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 73
XCVU9PFLGA2104

IO_T3U_N12_73_A20
IO_L24N_T3U_N11_73_B20
IO_L24P_T3U_N10_73_C20
IO_L23N_T3U_N9_73_D19
IO_L23P_T3U_N8_73_D20
IO_L22N_T3U_N7_DBC_AD0N_73_A18
IO_L22P_T3U_N6_DBC_AD0P_73_A19
IO_L21N_T3L_N5_AD8N_73_C18
IO_L21P_T3L_N4_AD8P_73_C19
IO_L20N_T3L_N3_AD1N_73_C17
IO_L20P_T3L_N2_AD1P_73_D17
IO_L19N_T3L_N1_DBC_AD9N_73_B17
IO_L19P_T3L_N0_DBC_AD9P_73_B18
IO_T2U_N12_73_G16
IO_L18N_T2U_N11_AD2N_73_D16
IO_L18P_T2U_N10_AD2P_73_E17
IO_L17N_T2U_N9_AD10N_73_F20
IO_L17P_T2U_N8_AD10P_73_G20
IO_L16N_T2U_N7_QBC_AD3N_73_E16
IO_L16P_T2U_N6_QBC_AD3P_73_F16
IO_L15N_T2L_N5_AD11N_73_E18
IO_L15P_T2L_N4_AD11P_73_E19
IO_L14N_T2L_N3_GC_73_F18
IO_L14P_T2L_N2_GC_73_F19
IO_L13N_T2L_N1_GC_QBC_73_G17
IO_L13P_T2L_N0_GC_QBC_73_G18
IO_T1U_N12_73_L19
IO_L12N_T1U_N11_GC_73_H18
IO_L12P_T1U_N10_GC_73_H19
IO_L11N_T1U_N9_GC_73_H17
IO_L11P_T1U_N8_GC_73_J17
IO_L10N_T1U_N7_QBC_AD4N_73_J19
IO_L10P_T1U_N6_QBC_AD4P_73_K19
IO_L9N_T1L_N5_AD12N_73_K18
IO_L9P_T1L_N4_AD12P_73_L18
IO_L8N_T1L_N3_AD5N_73_K16
IO_L8P_T1L_N2_AD5P_73_L16
IO_L7N_T1L_N1_QBC_AD13N_73_J16
IO_L7P_T1L_N0_QBC_AD13P_73_K17
IO_T0U_N12_VRP_73_T18
IO_L6N_T0U_N11_AD6N_73_M16
IO_L6P_T0U_N10_AD6P_73_N17
IO_L5N_T0U_N9_AD14N_73_N18
IO_L5P_T0U_N8_AD14P_73_N19
IO_L4N_T0U_N7_DBC_AD7N_73_P16
IO_L4P_T0U_N6_DBC_AD7P_73_P17
IO_L3N_T0L_N5_AD15N_73_M17
IO_L3P_T0L_N4_AD15P_73_M18
IO_L2N_T0L_N3_73_P19
IO_L2P_T0L_N2_73_R19
IO_L1N_T0L_N1_DBC_73_R17
IO_L1P_T0L_N0_DBC_73_R18
VREF_73_T19

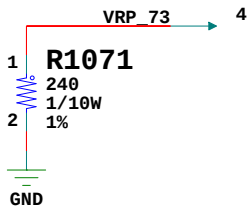
A20 DDR4_C1_TEN → 25-, 26-, 27
B20 DDR4_C1_DQ39 → 26
C20 DDR4_C1_DQ38 → 26
D19 DDR4_C1_DQ37 → 26
D20 DDR4_C1_DQ36 → 26
A18 DDR4_C1_DQS4_C → 26
A19 DDR4_C1_DQS4_T → 26
C18 DDR4_C1_DQ35 → 26
C19 DDR4_C1_DQ34 → 26
C17 DDR4_C1_DQ33 → 26
D17 DDR4_C1_DQ32 → 26
B17 GPIO_DIP_SW1 → 55
B18 DDR4_C1_DM4 → 26
G16 GPIO_DIP_SW2 → 55
D16 DDR4_C1_DQ31 → 25
E17 DDR4_C1_DQ30 → 25
F20 DDR4_C1_DQ29 → 25
G20 DDR4_C1_DQ28 → 25
E16 DDR4_C1_DQS3_C → 25
F16 DDR4_C1_DQS3_T → 25
E18 DDR4_C1_DQ27 → 25
E19 DDR4_C1_DQ26 → 25
F18 DDR4_C1_DQ25 → 25
F19 DDR4_C1_DQ24 → 25
G17 NC
G18 DDR4_C1_DM3 → 25
L19 CPU_RESET → 55
H18 DDR4_C1_DQ23 → 25
H19 DDR4_C1_DQ22 → 25
H17 DDR4_C1_DQ21 → 25
J17 DDR4_C1_DQ20 → 25
J19 DDR4_C1_DQS2_C → 25
K19 DDR4_C1_DQS2_T → 25
K18 DDR4_C1_DQ19 → 25
L18 DDR4_C1_DQ18 → 25
K16 DDR4_C1_DQ17 → 25
L16 DDR4_C1_DQ16 → 25
J16 GPIO_DIP_SW3 → 55
K17 DDR4_C1_DM2 → 25
T18 VRP_73 → 4
M16 DDR4_C1_DQ15 → 25
N17 DDR4_C1_DQ14 → 25
N18 DDR4_C1_DQ13 → 25
N19 DDR4_C1_DQ12 → 25
P16 DDR4_C1_DQS1_C → 25
P17 DDR4_C1_DQS1_T → 25
M17 DDR4_C1_DQ11 → 25
M18 DDR4_C1_DQ10 → 25
P19 DDR4_C1_DQ9 → 25
R19 DDR4_C1_DQ8 → 25
R17 DDR4_C1_ALERT_B → 25-, 26-, 27
R18 DDR4_C1_DM1 → 25
T19 NC

VCC1V2_FPGA

A17 VCC0_73_A17
D18 VCC0_73_D18
G19 VCC0_73_G19
H16 VCC0_73_H16
L17 VCC0_73_L17
P18 VCC0_73_P18

U1

SOC_2104_1MM_IRON



Bank 72 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 72
XCVU9PFLGA2104

IO_T3U_N12_72_D21
IO_L24N_T3U_N11_72_A21
IO_L24P_T3U_N10_72_B21
IO_L23N_T3U_N9_72_B22
IO_L23P_T3U_N8_72_B23
IO_L22N_T3U_N7_DBC_AD0N_72_C22
IO_L22P_T3U_N6_DBC_AD0P_72_D22
IO_L21N_T3L_N5_AD8N_72_C23
IO_L21P_T3L_N4_AD8P_72_C24
IO_L20N_T3L_N3_AD1N_72_A23
IO_L20P_T3L_N2_AD1P_72_A24
IO_L19N_T3L_N1_DBC_AD9N_72_D24
IO_L19P_T3L_N0_DBC_AD9P_72_E24
IO_T2U_N12_72_H20
IO_L18N_T2U_N11_AD2N_72_F23
IO_L18P_T2U_N10_AD2P_72_F24
IO_L17N_T2U_N9_AD10N_72_E21
IO_L17P_T2U_N8_AD10P_72_F21
IO_L16N_T2U_N7_QBC_AD3N_72_G23
IO_L16P_T2U_N6_QBC_AD3P_72_H24
IO_L15N_T2L_N5_AD11N_72_E22
IO_L15P_T2L_N4_AD11P_72_E23
IO_L14N_T2L_N3_GC_72_H22
IO_L14P_T2L_N2_GC_72_H23
IO_L13N_T2L_N1_GC_QBC_72_G21
IO_L13P_T2L_N0_GC_QBC_72_G22
IO_T1U_N12_72_J20
IO_L12N_T1U_N11_GC_72_J22
IO_L12P_T1U_N10_GC_72_K22
IO_L11N_T1U_N9_GC_72_J21
IO_L11P_T1U_N8_GC_72_K21
IO_L10N_T1U_N7_QBC_AD4N_72_L20
IO_L10P_T1U_N6_QBC_AD4P_72_M20
IO_L9N_T1L_N5_AD12N_72_L21
IO_L9P_T1L_N4_AD12P_72_M21
IO_L8N_T1L_N3_AD5N_72_J24
IO_L8P_T1L_N2_AD5P_72_K24
IO_L7N_T1L_N1_QBC_AD13N_72_K23
IO_L7P_T1L_N0_QBC_AD13P_72_L23
IO_T0U_N12_VRP_72_T21
IO_L6N_T0U_N11_AD6N_72_R23
IO_L6P_T0U_N10_AD6P_72_T23
IO_L5N_T0U_N9_AD14N_72_P22
IO_L5P_T0U_N8_AD14P_72_R22
IO_L4N_T0U_N7_DBC_AD7N_72_M22
IO_L4P_T0U_N6_DBC_AD7P_72_N22
IO_L3N_T0L_N5_AD15N_72_P21
IO_L3P_T0L_N4_AD15P_72_R21
IO_L2N_T0L_N3_72_M23
IO_L2P_T0L_N2_72_N23
IO_L1N_T0L_N1_DBC_72_N20
IO_L1P_T0L_N0_DBC_72_P20
VREF_72_T20

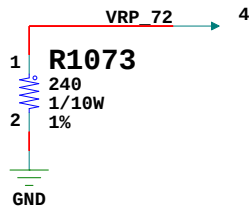
D21 GPIO_DIP_SW4 → 55
A21 DDR4_C1_DQ71 → 27
B21 DDR4_C1_DQ70 → 27
B22 DDR4_C1_DQ69 → 27
B23 DDR4_C1_DQ68 → 27
C22 DDR4_C1_DQS8_C → 27
D22 DDR4_C1_DQS8_T → 27
C23 DDR4_C1_DQ67 → 27
C24 DDR4_C1_DQ66 → 27
A23 DDR4_C1_DQ65 → 27
A24 DDR4_C1_DQ64 → 27
D24 NC
E24 DDR4_C1_DM8 → 27
H20 SI5328_INT_ALM_LS → 56
F23 DDR4_C1_DQ63 → 26
F24 DDR4_C1_DQ62 → 26
E21 DDR4_C1_DQ61 → 26
F21 DDR4_C1_DQ60 → 26
G23 DDR4_C1_DQS7_C → 26
H24 DDR4_C1_DQS7_T → 26
E22 DDR4_C1_DQ59 → 26
E23 DDR4_C1_DQ58 → 26
H22 DDR4_C1_DQ57 → 26
H23 DDR4_C1_DQ56 → 26
G21 NC
G22 DDR4_C1_DM7 → 26
J20 NC
J22 DDR4_C1_DQ55 → 26
K22 DDR4_C1_DQ54 → 26
J21 DDR4_C1_DQ53 → 26
K21 DDR4_C1_DQ52 → 26
L20 DDR4_C1_DQ56_C → 26
M20 DDR4_C1_DQ56_T → 26
L21 DDR4_C1_DQ51 → 26
M21 DDR4_C1_DQ50 → 26
J24 DDR4_C1_DQ49 → 26
K24 DDR4_C1_DQ48 → 26
K23 NC
L23 DDR4_C1_DM6 → 26
T21 VRP_72 → 4
R23 DDR4_C1_DQ47 → 26
T23 DDR4_C1_DQ46 → 26
P22 DDR4_C1_DQ45 → 26
R22 DDR4_C1_DQ44 → 26
M22 DDR4_C1_DQ55_C → 26
N22 DDR4_C1_DQ55_T → 26
P21 DDR4_C1_DQ43 → 26
R21 DDR4_C1_DQ42 → 26
M23 DDR4_C1_DQ41 → 26
N23 DDR4_C1_DQ40 → 26
N20 DDR4_C1_RESET_B → 25-, 26-, 27
P20 DDR4_C1_DM5 → 26
T20 NC

VCC1V2_FPGA

B24 VCC0_72_B24
C21 VCC0_72_C21
F22 VCC0_72_F22
J23 VCC0_72_J23
K20 VCC0_72_K20
N21 VCC0_72_N21
T22 VCC0_72_T22

U1

SOC_2104_1MM_IRON



FPGA Banks 73 72 C1 DDR4 Data



TITLE: FPGA Banks 73 72 C1 DDR4 Data
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 4 OF 76	DRAWN BY: DN

Bank 71 HP

SOC_VU9P_FLGA2104_IRONWOOD

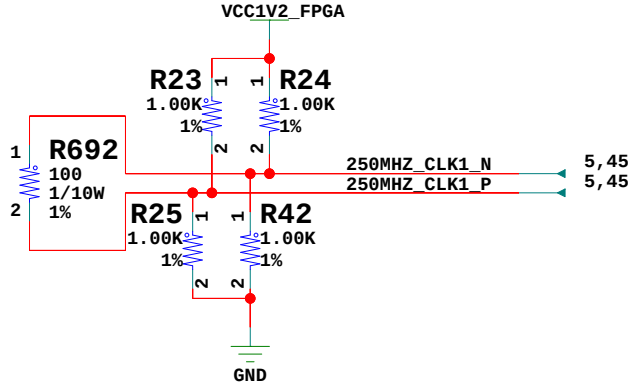
BANK 71
XCVU9PFLGA2104

IO_T3U_N12_71_D14
IO_L24N_T3U_N11_71_B15
IO_L24P_T3U_N10_71_B16
IO_L23N_T3U_N9_71_C14
IO_L23P_T3U_N8_71_C15
IO_L22N_T3U_N7_DBC_AD0N_71_A13
IO_L22P_T3U_N6_DBC_AD0P_71_A14
IO_L21N_T3L_N5_AD8N_71_A15
IO_L21P_T3L_N4_AD8P_71_A16
IO_L20N_T3L_N3_AD1N_71_B12
IO_L20P_T3L_N2_AD1P_71_C12
IO_L19N_T3L_N1_DBC_AD9N_71_B13
IO_L19P_T3L_N0_DBC_AD9P_71_C13
IO_T2U_N12_71_D15
IO_L18N_T2U_N11_AD2N_71_H14
IO_L18P_T2U_N10_AD2P_71_H15
IO_L17N_T2U_N9_AD10N_71_F15
IO_L17P_T2U_N8_AD10P_71_G15
IO_L16N_T2U_N7_QBC_AD3N_71_E14
IO_L16P_T2U_N6_QBC_AD3P_71_F14
IO_L15N_T2L_N5_AD11N_71_G13
IO_L15P_T2L_N4_AD11P_71_H13
IO_L14N_T2L_N3_GC_71_E13
IO_L14P_T2L_N2_GC_71_F13
IO_L13N_T2L_N1_GC_QBC_71_D12
IO_L13P_T2L_N0_GC_QBC_71_E12
IO_T1U_N12_71_A10
IO_L12N_T1U_N11_GC_71_A11
IO_L12P_T1U_N10_GC_71_B11
IO_L11N_T1U_N9_GC_71_B10
IO_L11P_T1U_N8_GC_71_C10
IO_L10N_T1U_N7_QBC_AD4N_71_A8
IO_L10P_T1U_N6_QBC_AD4P_71_A9
IO_L9N_T1L_N5_AD12N_71_B7
IO_L9P_T1L_N4_AD12P_71_B8
IO_L8N_T1L_N3_AD5N_71_C7
IO_L8P_T1L_N2_AD5P_71_D7
IO_L7N_T1L_N1_QBC_AD13N_71_C8
IO_L7P_T1L_N0_QBC_AD13P_71_C9
IO_T0U_N12_VRP_71_D8
IO_L6N_T0U_N11_AD6N_71_D9
IO_L6P_T0U_N10_AD6P_71_E9
IO_L5N_T0U_N9_AD14N_71_G12
IO_L5P_T0U_N8_AD14P_71_H12
IO_L4N_T0U_N7_DBC_AD7N_71_D10
IO_L4P_T0U_N6_DBC_AD7P_71_D11
IO_L3N_T0L_N5_AD15N_71_F9
IO_L3P_T0L_N4_AD15P_71_F10
IO_L2N_T0L_N3_71_E11
IO_L2P_T0L_N2_71_F11
IO_L1N_T0L_N1_DBC_71_G10
IO_L1P_T0L_N0_DBC_71_G11
VREF_71_J15

VCC1V2_FPGA

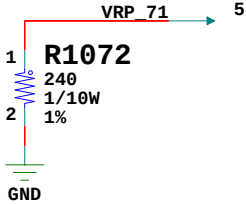
A7 VCCO_71_A7
B14 VCCO_71_B14
C11 VCCO_71_C11
E15 VCCO_71_E15
F12 VCCO_71_F12
G9 VCCO_71_G9

D14 DDR4_C1_A0
B15 DDR4_C1_A1
B16 DDR4_C1_A2
C14 DDR4_C1_A3
C15 DDR4_C1_A4
A13 DDR4_C1_A5
A14 DDR4_C1_A6
A15 DDR4_C1_A7
A16 DDR4_C1_A8
B12 DDR4_C1_A9
C12 DDR4_C1_A10
B13 DDR4_C1_A11
C13 DDR4_C1_A12
D15 DDR4_C1_A13
H14 DDR4_C1_A14_WE_B
H15 DDR4_C1_A15_CAS_B
F15 DDR4_C1_A16_RAS_B
G15 DDR4_C1_BA0
E14 DDR4_C1_CK_C
F14 DDR4_C1_CK_T
G13 DDR4_C1_BA1
H13 DDR4_C1_BG0
E13 DDR4_C1_ACT_B
F13 DDR4_C1_CS_B
D12 250MHZ_CLK1_N
E12 250MHZ_CLK1_P
A10 DDR4_C1_CKE
A11 DDR4_C1_DQ79
B11 DDR4_C1_DQ78
B10 DDR4_C1_DQ77
C10 DDR4_C1_DQ76
A8 DDR4_C1_DQS9_C
A9 DDR4_C1_DQS9_T
B7 DDR4_C1_DQ75
B8 DDR4_C1_DQ74
C7 DDR4_C1_DQ73
D7 DDR4_C1_DQ72
C8 DDR4_C1_ODT
C9 DDR4_C1_DM9
D8 VRP_71
D9 DDR4_C1_DQ7
E9 DDR4_C1_DQ6
G12 DDR4_C1_DQ5
H12 DDR4_C1_DQ4
D10 DDR4_C1_DQS0_C
D11 DDR4_C1_DQS0_T
F9 DDR4_C1_DQ3
F10 DDR4_C1_DQ2
E11 DDR4_C1_DQ1
F11 DDR4_C1_DQ0
G10 DDR4_C1_PAR
G11 DDR4_C1_DM0
J15 NC



U1

SOC_2104_1MM_IRON



FPGA Bank 71 C1 DDR4 Addr Ctrl

TITLE: FPGA Bank 71 C1 DDR4 Addr Ctrl SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 5 OF 76	DRAWN BY: DN

Bank 40 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 40
XCVU9PFLGA2104

IO_T3U_N12_40_AR29
IO_L24N_T3U_N11_40_AN31
IO_L24P_T3U_N10_40_AN30
IO_L23N_T3U_N9_40_AR30
IO_L23P_T3U_N8_40_AP30
IO_L22N_T3U_N7_DBC_AD0N_40_AP32
IO_L22P_T3U_N6_DBC_AD0P_40_AP31
IO_L21N_T3L_N5_AD8N_40_AT30
IO_L21P_T3L_N4_AD8P_40_AT29
IO_L20N_T3L_N3_AD1N_40_AT34
IO_L20P_T3L_N2_AD1P_40_AR33
IO_L19N_T3L_N1_DBC_AD9N_40_AT32
IO_L19P_T3L_N0_DBC_AD9P_40_AR32
IO_T2U_N12_40_AV34
IO_L18N_T2U_N11_AD2N_40_AV31
IO_L18P_T2U_N10_AD2P_40_AU31
IO_L17N_T2U_N9_AD10N_40_AU32
IO_L17P_T2U_N8_AD10P_40_AT31
IO_L16N_T2U_N7_QBC_AD3N_40_AV29
IO_L16P_T2U_N6_QBC_AD3P_40_AU29
IO_L15N_T2L_N5_AD11N_40_AU34
IO_L15P_T2L_N4_AD11P_40_AU33
IO_L14N_T2L_N3_GC_40_AW30
IO_L14P_T2L_N2_GC_40_AV30
IO_L13N_T2L_N1_GC_QBC_40_AW33
IO_L13P_T2L_N0_GC_QBC_40_AV33
IO_T1U_N12_40_AY30
IO_L12N_T1U_N11_GC_40_AY33
IO_L12P_T1U_N10_GC_40_AY32
IO_L11N_T1U_N9_GC_40_AW32
IO_L11P_T1U_N8_GC_40_AW31
IO_L10N_T1U_N7_QBC_AD4N_40_BA34
IO_L10P_T1U_N6_QBC_AD4P_40_AY34
IO_L9N_T1L_N5_AD12N_40_BA31
IO_L9P_T1L_N4_AD12P_40_BA30
IO_L8N_T1L_N3_AD5N_40_BB33
IO_L8P_T1L_N2_AD5P_40_BA32
IO_L7N_T1L_N1_QBC_AD13N_40_BB32
IO_L7P_T1L_N0_QBC_AD13P_40_BB31
IO_T0U_N12_VRP_40_BC30
IO_L6N_T0U_N11_AD6N_40_BD31
IO_L6P_T0U_N10_AD6P_40_BC31
IO_L5N_T0U_N9_AD14N_40_BD33
IO_L5P_T0U_N8_AD14P_40_BC33
IO_L4N_T0U_N7_DBC_AD7N_40_BF31
IO_L4P_T0U_N6_DBC_AD7P_40_BF30
IO_L3N_T0L_N5_AD15N_40_BE33
IO_L3P_T0L_N4_AD15P_40_BD32
IO_L2N_T0L_N3_40_BE30
IO_L2P_T0L_N2_40_BD30
IO_L1N_T0L_N1_DBC_40_BF32
IO_L1P_T0L_N0_DBC_40_BE32
VREF_40_AN29

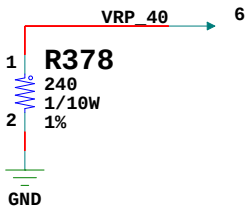
AR29 DDR4_C2_ALERT_B 28-, 29-, 30
AN31 DDR4_C2_DQ31 28
AN30 DDR4_C2_DQ30 28
AR30 DDR4_C2_DQ29 28
AP30 DDR4_C2_DQ28 28
AP32 DDR4_C2_DQS3_C 28
AP31 DDR4_C2_DQS3_T 28
AT30 DDR4_C2_DQ27 28
AT29 DDR4_C2_DQ26 28
AT34 DDR4_C2_DQ25 28
AR33 DDR4_C2_DQ24 28
AT32 GPIO_LED0 55
AR32 DDR4_C2_DM3 28
AV34 GPIO_LED1 55
AV31 DDR4_C2_DQ23 28
AU31 DDR4_C2_DQ22 28
AU32 DDR4_C2_DQ21 28
AT31 DDR4_C2_DQ20 28
AV29 DDR4_C2_DQS2_C 28
AU29 DDR4_C2_DQS2_T 28
AU34 DDR4_C2_DQ19 28
AU33 DDR4_C2_DQ18 28
AW30 DDR4_C2_DQ17 28
AV30 DDR4_C2_DQ16 28
AW33 FAN_FAIL_LS_B 13
AV33 DDR4_C2_DM2 28
AY30 GPIO_LED2 55
AY33 DDR4_C2_DQ15 28
AY32 DDR4_C2_DQ14 28
AW32 DDR4_C2_DQ13 28
AW31 DDR4_C2_DQ12 28
BA34 DDR4_C2_DQS1_C 28
AY34 DDR4_C2_DQS1_T 28
BA31 DDR4_C2_DQ11 28
BA30 DDR4_C2_DQ10 28
BB33 DDR4_C2_DQ9 28
BA32 DDR4_C2_DQ8 28
BB32 GPIO_LED3 55
BB31 DDR4_C2_DM1 28
BC30 VRP_40 6
BD31 DDR4_C2_DQ7 28
BC31 DDR4_C2_DQ6 28
BD33 DDR4_C2_DQ5 28
BC33 DDR4_C2_DQ4 28
BF31 DDR4_C2_DQS0_C 28
BF30 DDR4_C2_DQS0_T 28
BE33 DDR4_C2_DQ3 28
BD32 DDR4_C2_DQ2 28
BE30 DDR4_C2_DQ1 28
BD30 DDR4_C2_DQ0 28
BF32 GPIO_LED4 55
BE32 DDR4_C2_DM0 28
AN29 NC

VCC1V2_FPGA

AR31
AV32
AW29
BA33
BB30
BE31
VCCO_40_AR31
VCCO_40_AV32
VCCO_40_AW29
VCCO_40_BA33
VCCO_40_BB30
VCCO_40_BE31

U1

SOC_2104_1MM_IRON



Bank 42 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 42
XCVU9PFLGA2104

IO_T3U_N12_42_AU37
IO_L24N_T3U_N11_42_AV39
IO_L24P_T3U_N10_42_AV38
IO_L23N_T3U_N9_42_AU39
IO_L23P_T3U_N8_42_AU38
IO_L22N_T3U_N7_DBC_AD0N_42_AW38
IO_L22P_T3U_N6_DBC_AD0P_42_AW37
IO_L21N_T3L_N5_AD8N_42_AV40
IO_L21P_T3L_N4_AD8P_42_AU40
IO_L20N_T3L_N3_AD1N_42_AW36
IO_L20P_T3L_N2_AD1P_42_AW35
IO_L19N_T3L_N1_DBC_AD9N_42_AV36
IO_L19P_T3L_N0_DBC_AD9P_42_AV35
IO_T2U_N12_42_AY35
IO_L18N_T2U_N11_AD2N_42_AY39
IO_L18P_T2U_N10_AD2P_42_AY38
IO_L17N_T2U_N9_AD10N_42_AY40
IO_L17P_T2U_N8_AD10P_42_AW40
IO_L16N_T2U_N7_QBC_AD3N_42_BA36
IO_L16P_T2U_N6_QBC_AD3P_42_BA35
IO_L15N_T2L_N5_AD11N_42_BA40
IO_L15P_T2L_N4_AD11P_42_BA39
IO_L14N_T2L_N3_GC_42_BB37
IO_L14P_T2L_N2_GC_42_BB36
IO_L13N_T2L_N1_GC_QBC_42_BA37
IO_L13P_T2L_N0_GC_QBC_42_AY37
IO_T1U_N12_42_BC40
IO_L12N_T1U_N11_GC_42_BD38
IO_L12P_T1U_N10_GC_42_BC38
IO_L11N_T1U_N9_GC_42_BB39
IO_L11P_T1U_N8_GC_42_BB38
IO_L10N_T1U_N7_QBC_AD4N_42_BF39
IO_L10P_T1U_N6_QBC_AD4P_42_BE39
IO_L9N_T1L_N5_AD12N_42_BD40
IO_L9P_T1L_N4_AD12P_42_BC39
IO_L8N_T1L_N3_AD5N_42_BE38
IO_L8P_T1L_N2_AD5P_42_BD37
IO_L7N_T1L_N1_QBC_AD13N_42_BF40
IO_L7P_T1L_N0_QBC_AD13P_42_BE40
IO_T0U_N12_VRP_42_BB34
IO_L6N_T0U_N11_AD6N_42_BF37
IO_L6P_T0U_N10_AD6P_42_BF36
IO_L5N_T0U_N9_AD14N_42_BE37
IO_L5P_T0U_N8_AD14P_42_BD36
IO_L4N_T0U_N7_DBC_AD7N_42_BF35
IO_L4P_T0U_N6_DBC_AD7P_42_BE35
IO_L3N_T0L_N5_AD15N_42_BC36
IO_L3P_T0L_N4_AD15P_42_BC35
IO_L2N_T0L_N3_42_BF34
IO_L2P_T0L_N2_42_BE34
IO_L1N_T0L_N1_DBC_42_BD35
IO_L1P_T0L_N0_DBC_42_BC34
VREF_42_AU36

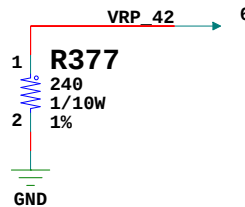
AU37 GPIO_LED5 55
AV39 DDR4_C2_DQ63 29
AV38 DDR4_C2_DQ62 29
AU39 DDR4_C2_DQ61 29
AU38 DDR4_C2_DQ60 29
AW38 DDR4_C2_DQS7_C 29
AW37 DDR4_C2_DQS7_T 29
AV40 DDR4_C2_DQ59 29
AU40 DDR4_C2_DQ58 29
AW36 DDR4_C2_DQ57 29
AW35 DDR4_C2_DQ56 29
AV36 GPIO_LED6 55
AV35 DDR4_C2_DM7 29
AY35 DDR4_C2_TEN 28-, 29-, 30
AY39 DDR4_C2_DQ55 29
AY38 DDR4_C2_DQ54 29
AY40 DDR4_C2_DQ53 29
AW40 DDR4_C2_DQ52 29
BA36 DDR4_C2_DQ56_C 29
BA35 DDR4_C2_DQ56_T 29
BA40 DDR4_C2_DQ51 29
BA39 DDR4_C2_DQ50 29
BB37 DDR4_C2_DQ49 29
BB36 DDR4_C2_DQ48 29
BA37 GPIO_LED7 55
AY37 DDR4_C2_DM6 29
BC40 FAN_OT_LS_B 13
BD38 DDR4_C2_DQ47 29
BC38 DDR4_C2_DQ46 29
BB39 DDR4_C2_DQ45 29
BB38 DDR4_C2_DQ44 29
BF39 DDR4_C2_DQS5_C 29
BE39 DDR4_C2_DQS5_T 29
BD40 DDR4_C2_DQ43 29
BC39 DDR4_C2_DQ42 29
BE38 DDR4_C2_DQ41 29
BD37 DDR4_C2_DQ40 29
BF40 MAXIM_CABLE_LS_B 56
BE40 DDR4_C2_DM5 29
BB34 VRP_42 6
BF37 DDR4_C2_DQ39 29
BF36 DDR4_C2_DQ38 29
BE37 DDR4_C2_DQ37 29
BD36 DDR4_C2_DQ36 29
BF35 DDR4_C2_DQ54_C 29
BE35 DDR4_C2_DQ54_T 29
BC36 DDR4_C2_DQ35 29
BC35 DDR4_C2_DQ34 29
BF34 DDR4_C2_DQ33 29
BE34 DDR4_C2_DQ32 29
BD35 DDR4_C2_RESET_B 28-, 29-, 30
BC34 DDR4_C2_DM4 29
AU36 NC

VCC1V2_FPGA

AU35
AW39
AY36
BB40
BC37
BD34
BF38
VCCO_42_AU35
VCCO_42_AW39
VCCO_42_AY36
VCCO_42_BB40
VCCO_42_BC37
VCCO_42_BD34
VCCO_42_BF38

U1

SOC_2104_1MM_IRON



FPGA Banks 40 42 C2 DDR4 Data



TITLE: FPGA Banks 40 42 C2 DDR4 Data SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37		VER:	1.1
SHEET SIZE: B		REV:	02
SHEET	6 OF 76	DRAWN BY: DN	

Bank 41 HP

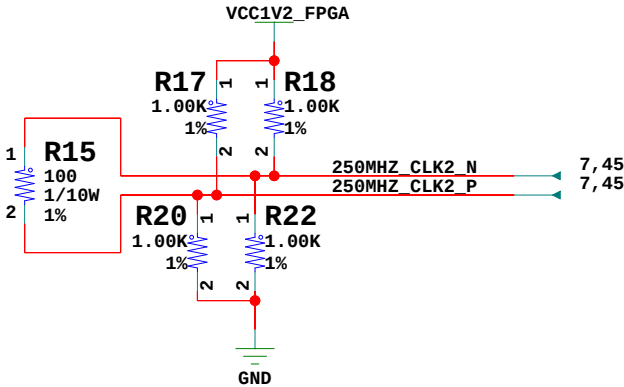
SOC_VU9P_FLGA2104_IRONWOOD

BANK 41
XCVU9PFLGA2104

IO_T3U_N12_41	AN25
IO_L24N_T3U_N11_41	AM27
IO_L24P_T3U_N10_41	AL27
IO_L23N_T3U_N9_41	AP26
IO_L23P_T3U_N8_41	AP25
IO_L22N_T3U_N7_DBC_AD0N_41	AN28
IO_L22P_T3U_N6_DBC_AD0P_41	AM28
IO_L21N_T3L_N5_AD8N_41	AP28
IO_L21P_T3L_N4_AD8P_41	AP27
IO_L20N_T3L_N3_AD1N_41	AN26
IO_L20P_T3L_N2_AD1P_41	AM26
IO_L19N_T3L_N1_DBC_AD9N_41	AR28
IO_L19P_T3L_N0_DBC_AD9P_41	AR27
IO_T2U_N12_41	AV25
IO_L18N_T2U_N11_AD2N_41	AT25
IO_L18P_T2U_N10_AD2P_41	AR25
IO_L17N_T2U_N9_AD10N_41	AU28
IO_L17P_T2U_N8_AD10P_41	AU27
IO_L16N_T2U_N7_QBC_AD3N_41	AT27
IO_L16P_T2U_N6_QBC_AD3P_41	AT26
IO_L15N_T2L_N5_AD11N_41	AW28
IO_L15P_T2L_N4_AD11P_41	AV28
IO_L14N_T2L_N3_GC_41	AV26
IO_L14P_T2L_N2_GC_41	AU26
IO_L13N_T2L_N1_GC_QBC_41	AW27
IO_L13P_T2L_N0_GC_QBC_41	AW26
IO_T1U_N12_41	AY29
IO_L12N_T1U_N11_GC_41	BB27
IO_L12P_T1U_N10_GC_41	BA27
IO_L11N_T1U_N9_GC_41	AY28
IO_L11P_T1U_N8_GC_41	AY27
IO_L10N_T1U_N7_QBC_AD4N_41	BB26
IO_L10P_T1U_N6_QBC_AD4P_41	BA26
IO_L9N_T1L_N5_AD12N_41	BC28
IO_L9P_T1L_N4_AD12P_41	BB28
IO_L8N_T1L_N3_AD5N_41	BC26
IO_L8P_T1L_N2_AD5P_41	BC25
IO_L7N_T1L_N1_QBC_AD13N_41	BB29
IO_L7P_T1L_N0_QBC_AD13P_41	BA29
IO_T0U_N12_VRP_41	BC29
IO_L6N_T0U_N11_AD6N_41	BD26
IO_L6P_T0U_N10_AD6P_41	BD25
IO_L5N_T0U_N9_AD14N_41	BE27
IO_L5P_T0U_N8_AD14P_41	BD27
IO_L4N_T0U_N7_DBC_AD7N_41	BF25
IO_L4P_T0U_N6_DBC_AD7P_41	BE25
IO_L3N_T0L_N5_AD15N_41	BE28
IO_L3P_T0L_N4_AD15P_41	BD28
IO_L2N_T0L_N3_41	BF27
IO_L2P_T0L_N2_41	BF26
IO_L1N_T0L_N1_DBC_41	BF29
IO_L1P_T0L_N0_DBC_41	BE29
VREF_41	AL26

AN25	DDR4_C2_ACT_B	28-,29-,30
AM27	DDR4_C2_A0	28-,29-,30
AL27	DDR4_C2_A1	28-,29-,30
AP26	DDR4_C2_A2	28-,29-,30
AP25	DDR4_C2_A3	28-,29-,30
AN28	DDR4_C2_A4	28-,29-,30
AM28	DDR4_C2_A5	28-,29-,30
AP28	DDR4_C2_A6	28-,29-,30
AP27	DDR4_C2_A7	28-,29-,30
AN26	DDR4_C2_A8	28-,29-,30
AM26	DDR4_C2_A9	28-,29-,30
AR28	DDR4_C2_A10	28-,29-,30
AR27	DDR4_C2_A11	28-,29-,30
AV25	DDR4_C2_A12	28-,29-,30
AT25	DDR4_C2_A13	28-,29-,30
AR25	DDR4_C2_BA0	28-,29-,30
AU28	DDR4_C2_BA1	28-,29-,30
AU27	DDR4_C2_B60	28-,29-,30
AT27	DDR4_C2_CK_C	28-,29-,30
AT26	DDR4_C2_CK_T	28-,29-,30
AW28	DDR4_C2_CKE	28-,29-,30
AV28	DDR4_C2_A14_WE_B	28-,29-,30
AV26	DDR4_C2_A16_RAS_B	28-,29-,30
AU26	DDR4_C2_A15_CAS_B	28-,29-,30
AW27	250MHZ_CLK2_N	7,45
AW26	250MHZ_CLK2_P	7,45
AY29	DDR4_C2_CS_B	28-,29-,30
BB27	DDR4_C2_DQ79	30
BA27	DDR4_C2_DQ78	30
AY28	DDR4_C2_DQ77	30
AY27	DDR4_C2_DQ76	30
BB26	DDR4_C2_DQS9_C	30
BA26	DDR4_C2_DQS9_T	30
BC28	DDR4_C2_DQ75	30
BB28	DDR4_C2_DQ74	30
BC26	DDR4_C2_DQ73	30
BC25	DDR4_C2_DQ72	30
BB29	DDR4_C2_ODT	28-,29-,30
BA29	DDR4_C2_DM9	30
BC29	VRP_41	7
BD26	DDR4_C2_DQ71	30
BD25	DDR4_C2_DQ70	30
BE27	DDR4_C2_DQ69	30
BD27	DDR4_C2_DQ68	30
BF25	DDR4_C2_DQS8_C	30
BE25	DDR4_C2_DQS8_T	30
BE28	DDR4_C2_DQ67	30
BD28	DDR4_C2_DQ66	30
BF27	DDR4_C2_DQ65	30
BF26	DDR4_C2_DQ64	30
BF29	DDR4_C2_PAR	28-,29-,30
BE29	DDR4_C2_DM8	30
AL26	NC	

PCB LAYOUT DIRECTIVE:
TERMINATION RESISTOR
AND BIAS RESISTOR
NETWORK TO BE PLACED
IN "FLY-BY" CONFIGURATION

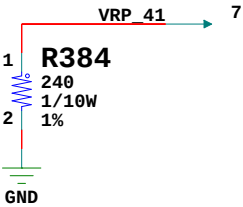


VCC1V2_FPGA

AN27	VCC0_41_AN27
AT28	VCC0_41_AT28
AU25	VCC0_41_AU25
AY26	VCC0_41_AY26
BC27	VCC0_41_BC27
BF28	VCC0_41_BF28

U1

SOC_2104_1MM_IRON



FPGA Bank 41 C2 DDR4 Addr Ctrl

TITLE: FPGA Bank 41 C2 DDR4 Addr Ctrl SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 7 OF 76	DRAWN BY: DN

Bank 46 HP

SOC_VU9P_FLGA2104_IRONWOOD

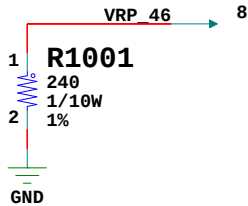
BANK 46
XCVU9PFLGA2104

IO_T3U_N12_46_D36
IO_L24N_T3U_N11_46_A35
IO_L24P_T3U_N10_46_A34
IO_L23N_T3U_N9_46_A36
IO_L23P_T3U_N8_46_B35
IO_L22N_T3U_N7_DBC_AD0N_46_B37
IO_L22P_T3U_N6_DBC_AD0P_46_B36
IO_L21N_T3L_N5_AD8N_46_C34
IO_L21P_T3L_N4_AD8P_46_D34
IO_L20N_T3L_N3_AD1N_46_C35
IO_L20P_T3L_N2_AD1P_46_D35
IO_L19N_T3L_N1_DBC_AD9N_46_C37
IO_L19P_T3L_N0_DBC_AD9P_46_D37
IO_T2U_N12_46_C38
IO_L18N_T2U_N11_AD2N_46_A38
IO_L18P_T2U_N10_AD2P_46_B38
IO_L17N_T2U_N9_AD10N_46_C40
IO_L17P_T2U_N8_AD10P_46_D40
IO_L16N_T2U_N7_QBC_AD3N_46_A40
IO_L16P_T2U_N6_QBC_AD3P_46_A39
IO_L15N_T2L_N5_AD11N_46_B40
IO_L15P_T2L_N4_AD11P_46_C39
IO_L14N_T2L_N3_GC_46_E38
IO_L14P_T2L_N2_GC_46_E37
IO_L13N_T2L_N1_GC_QBC_46_D39
IO_L13P_T2L_N0_GC_QBC_46_E39
IO_T1U_N12_46_E36
IO_L12N_T1U_N11_GC_46_G37
IO_L12P_T1U_N10_GC_46_G36
IO_L11N_T1U_N9_GC_46_F36
IO_L11P_T1U_N8_GC_46_F35
IO_L10N_T1U_N7_QBC_AD4N_46_G35
IO_L10P_T1U_N6_QBC_AD4P_46_H34
IO_L9N_T1L_N5_AD12N_46_H37
IO_L9P_T1L_N4_AD12P_46_J36
IO_L8N_T1L_N3_AD5N_46_H35
IO_L8P_T1L_N2_AD5P_46_J35
IO_L7N_T1L_N1_QBC_AD13N_46_E34
IO_L7P_T1L_N0_QBC_AD13P_46_F34
IO_T0U_N12_VRP_46_K38
IO_L6N_T0U_N11_AD6N_46_F39
IO_L6P_T0U_N10_AD6P_46_F38
IO_L5N_T0U_N9_AD14N_46_J37
IO_L5P_T0U_N8_AD14P_46_K37
IO_L4N_T0U_N7_DBC_AD7N_46_G38
IO_L4P_T0U_N6_DBC_AD7P_46_H38
IO_L3N_T0L_N5_AD15N_46_F40
IO_L3P_T0L_N4_AD15P_46_G40
IO_L2N_T0L_N3_46_H40
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IO_L1P_T0L_N0_DBC_46_J39
VREF_46_J34

D36 NC
A35 RLD3 C3 72B DM1
A34 RLD3 C3 72B DQ35
A36 RLD3 C3 72B DQ34
B35 RLD3 C3 72B DQ33
B37 RLD3 C3 72B DQ32
B36 RLD3 C3 72B DQ31
C34 RLD3 C3 72B DQ30
D34 RLD3 C3 72B DQ29
C35 RLD3 C3 72B DQ28
D35 RLD3 C3 72B DQ27
C37 RLD3 C3 72B QK3 N
D37 RLD3 C3 72B QK3 P
C38 NC
A38 RLD3 C3 72B QVLD1
B38 RLD3 C3 72B DQ26
C40 RLD3 C3 72B DQ25
D40 RLD3 C3 72B DQ24
A40 RLD3 C3 72B DQ23
A39 RLD3 C3 72B DQ22
B40 RLD3 C3 72B DQ21
C39 RLD3 C3 72B DQ20
E38 RLD3 C3 72B DQ19
E37 RLD3 C3 72B DQ18
D39 RLD3 C3 72B QK2 N
E39 RLD3 C3 72B QK2 P
E36 NC
G37 RLD3 C3 72B QVLD0
G36 RLD3 C3 72B DQ17
F36 RLD3 C3 72B DQ16
F35 RLD3 C3 72B DQ15
G35 RLD3 C3 72B DQ14
H34 RLD3 C3 72B DQ13
H37 RLD3 C3 72B DQ12
J36 RLD3 C3 72B DQ11
H35 RLD3 C3 72B DQ10
J35 RLD3 C3 72B DQ9
E34 RLD3 C3 72B QK1 N
F34 RLD3 C3 72B QK1 P
K38 VRP 46
F39 RLD3 C3 72B DM0
F38 RLD3 C3 72B DQ8
J37 RLD3 C3 72B DQ7
K37 RLD3 C3 72B DQ6
G38 RLD3 C3 72B DQ5
H38 RLD3 C3 72B DQ4
F40 RLD3 C3 72B DQ3
G40 RLD3 C3 72B DQ2
H40 RLD3 C3 72B DQ1
H39 RLD3 C3 72B DQ0
J40 RLD3 C3 72B QK0 N
J39 RLD3 C3 72B QK0 P
J34 NC

U1

SOC_2104_1MM_IRON



Bank 48 HP

SOC_VU9P_FLGA2104_IRONWOOD

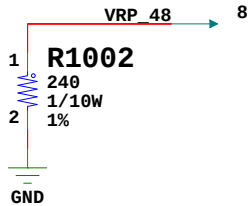
BANK 48
XCVU9PFLGA2104

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IO_L21P_T3L_N4_AD8P_48_C27
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IO_L19P_T3L_N0_DBC_AD9P_48_D27
IO_T2U_N12_48_H25
IO_L18N_T2U_N11_AD2N_48_F25
IO_L18P_T2U_N10_AD2P_48_G25
IO_L17N_T2U_N9_AD10N_48_E27
IO_L17P_T2U_N8_AD10P_48_E26
IO_L16N_T2U_N7_QBC_AD3N_48_G28
IO_L16P_T2U_N6_QBC_AD3P_48_H28
IO_L15N_T2L_N5_AD11N_48_E28
IO_L15P_T2L_N4_AD11P_48_F28
IO_L14N_T2L_N3_GC_48_G27
IO_L14P_T2L_N2_GC_48_H27
IO_L13N_T2L_N1_GC_QBC_48_F26
IO_L13P_T2L_N0_GC_QBC_48_G26
IO_T1U_N12_48_J25
IO_L12N_T1U_N11_GC_48_J27
IO_L12P_T1U_N10_GC_48_K27
IO_L11N_T1U_N9_GC_48_J26
IO_L11P_T1U_N8_GC_48_K26
IO_L10N_T1U_N7_QBC_AD4N_48_L25
IO_L10P_T1U_N6_QBC_AD4P_48_L24
IO_L9N_T1L_N5_AD12N_48_K28
IO_L9P_T1L_N4_AD12P_48_L28
IO_L8N_T1L_N3_AD5N_48_L26
IO_L8P_T1L_N2_AD5P_48_M25
IO_L7N_T1L_N1_QBC_AD13N_48_M28
IO_L7P_T1L_N0_QBC_AD13P_48_M27
IO_T0U_N12_VRP_48_M26
IO_L6N_T0U_N11_AD6N_48_N24
IO_L6P_T0U_N10_AD6P_48_P24
IO_L5N_T0U_N9_AD14N_48_N27
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IO_L3P_T0L_N4_AD15P_48_R27
IO_L2N_T0L_N3_48_R24
IO_L2P_T0L_N2_48_T24
IO_L1N_T0L_N1_DBC_48_R26
IO_L1P_T0L_N0_DBC_48_T26
VREF_48_T25

A25 NC
B25 RLD3 C3 72B DM3
C25 RLD3 C3 72B DQ71
D26 RLD3 C3 72B DQ70
D25 RLD3 C3 72B DQ69
A26 RLD3 C3 72B DQ68
B26 RLD3 C3 72B DQ67
B27 RLD3 C3 72B DQ66
C27 RLD3 C3 72B DQ65
A28 RLD3 C3 72B DQ64
B28 RLD3 C3 72B DQ63
C28 RLD3 C3 72B QK7 N
D27 RLD3 C3 72B QK7 P
H25 NC
F25 RLD3 C3 72B QVLD3
G25 RLD3 C3 72B DQ62
E27 RLD3 C3 72B DQ61
E26 RLD3 C3 72B DQ60
G28 RLD3 C3 72B DQ59
H28 RLD3 C3 72B DQ58
E28 RLD3 C3 72B DQ57
F28 RLD3 C3 72B DQ56
G27 RLD3 C3 72B DQ55
H27 RLD3 C3 72B DQ54
F26 RLD3 C3 72B QK6 N
G26 RLD3 C3 72B QK6 P
J25 NC
J27 RLD3 C3 72B QVLD2
K27 RLD3 C3 72B DQ53
J26 RLD3 C3 72B DQ52
K26 RLD3 C3 72B DQ51
L25 RLD3 C3 72B DQ50
L24 RLD3 C3 72B DQ49
K28 RLD3 C3 72B DQ48
L28 RLD3 C3 72B DQ47
L26 RLD3 C3 72B DQ46
M25 RLD3 C3 72B DQ45
M28 RLD3 C3 72B QK5 N
M27 RLD3 C3 72B QK5 P
M26 VRP 48
N24 RLD3 C3 72B DM2
P24 RLD3 C3 72B DQ44
N27 RLD3 C3 72B DQ43
P26 RLD3 C3 72B DQ42
N25 RLD3 C3 72B DQ41
P25 RLD3 C3 72B DQ40
P27 RLD3 C3 72B DQ39
R27 RLD3 C3 72B DQ38
R24 RLD3 C3 72B DQ37
T24 RLD3 C3 72B DQ36
R26 RLD3 C3 72B QK4 N
T26 RLD3 C3 72B QK4 P
T25 NC

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FPGA Banks 46 48 C3 RLD3 Data

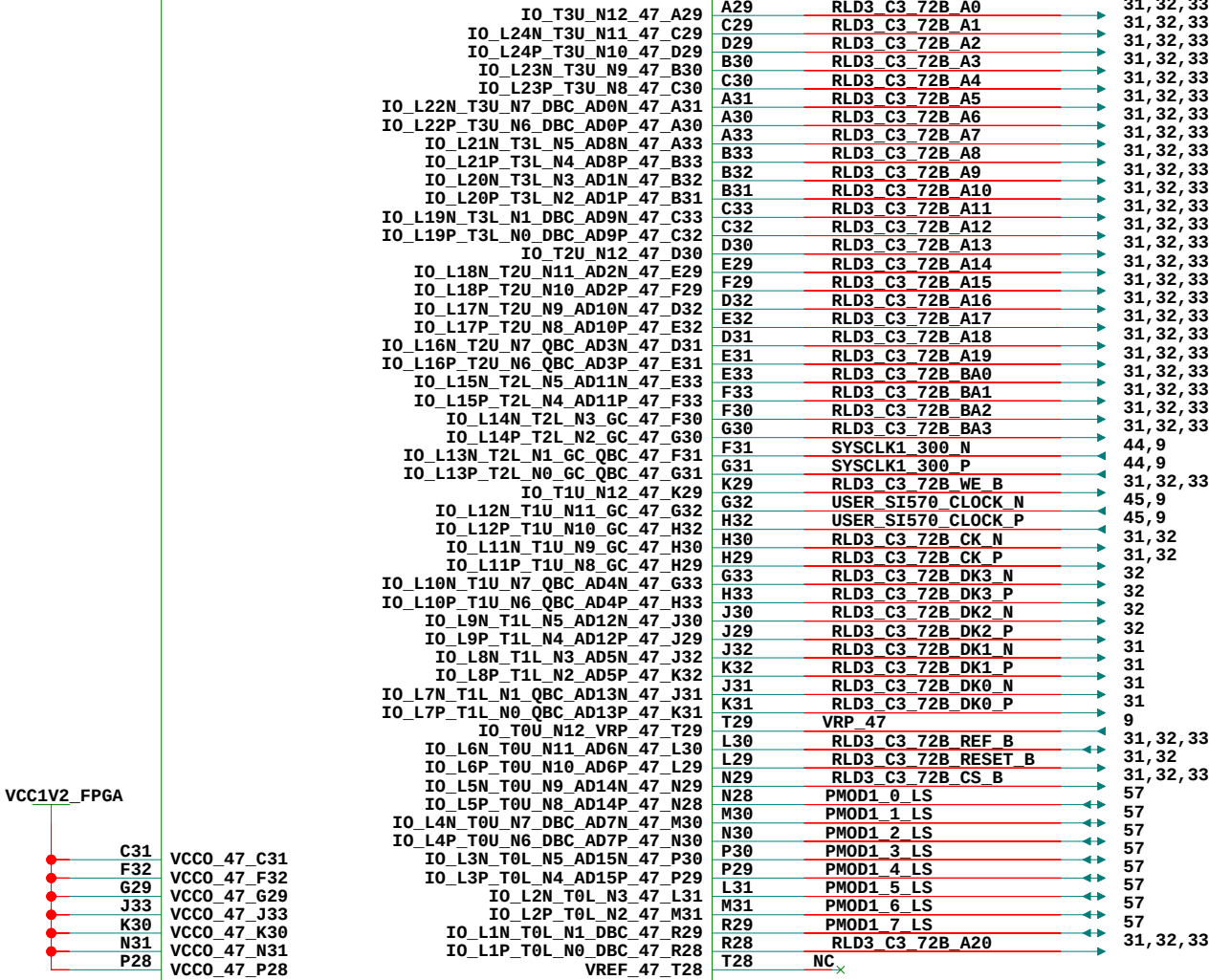


TITLE: FPGA Banks 46 48 C3 RLD3 Data
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 8 OF 76	DRAWN BY: DN

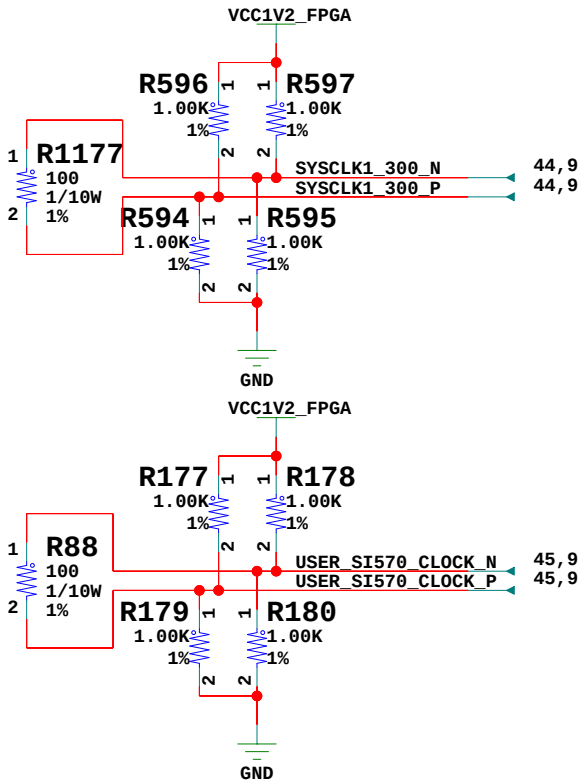
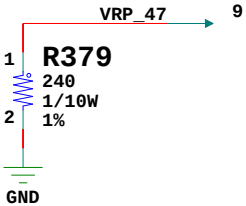
SOC_VU9P_FLGA2104_IRONWOOD

BANK 47
XCVU9PFLGA2104



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SOC_2104_1MM_IRON



FPGA Bank 47 C3 RLD3 Addr Ctrl



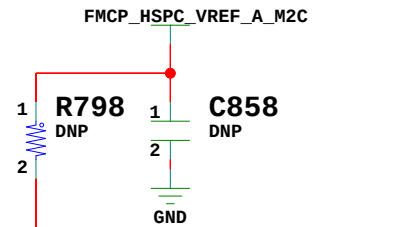
TITLE: FPGA Bank 47 C3 RLD3 Addr Ctrl
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 9 OF 76	DRAWN BY: DN

Layout: Place resistor and capacitor for VREF
Underneath the FPGA via array
right next to the via

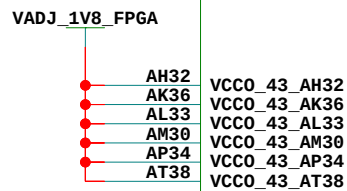
Bank 43 HP

SOC_VU9P_FLGA2104_IRONWOOD



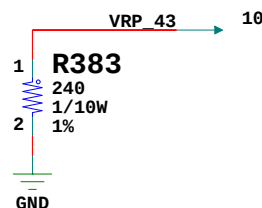
BANK 43
XCVU9PFLGA2104

	IO_T30_N12_43_AH30	AH30	NC																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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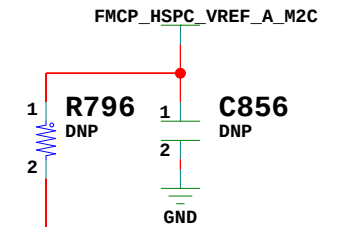
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SQC_2104_1MM_IRON



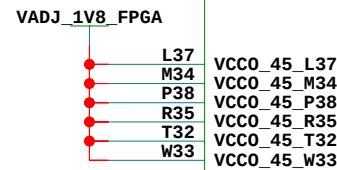
Bank 45 HP

S0C_VU9P_FLGA2104_IRONWOOD



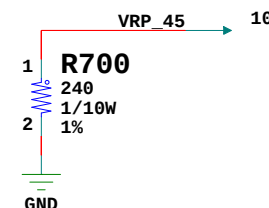
BANK 45
XCVU9PFLGA2104

	I0_T3U_N12_45_K36	K36 NC x					
	I0_L24N_T3U_N11_45_L35	L35		FMCP_HSPC_LA21_N	<=>	36	
	I0_L24P_T3U_N10_45_M35	M35		FMCP_HSPC_LA21_P	<=>	36	
	I0_L23N_T3U_N9_45_M32	M32		FMCP_HSPC_LA20_N	<=>	35	
	I0_L23P_T3U_N8_45_N32	N32		FMCP_HSPC_LA20_P	<=>	35	
I0_L22N_T3U_N7_DBC_AD0N_45_M33	M33			FMCP_HSPC_LA19_N	<=>	36	
I0_L22P_T3U_N6_DBC_AD0P_45_N33	N33			FMCP_HSPC_LA19_P	<=>	36	
I0_L21N_T3L_N5_AD8N_45_K33	K33			FMCP_HSPC_LA32_N	<=>	36	
I0_L21P_T3L_N4_AD8P_45_L33	L33			FMCP_HSPC_LA32_P	<=>	36	
I0_L20N_T3L_N3_AD1N_45_N35	N35			FMCP_HSPC_LA22_N	<=>	35	
I0_L20P_T3L_N2_AD1P_45_N34	N34			FMCP_HSPC_LA22_P	<=>	35	
I0_L19N_T3L_N1_DBC_AD9N_45_K34	K34			FMCP_HSPC_LA33_N	<=>	35	
I0_L19P_T3L_N0_DBC_AD9P_45_L34	L34			FMCP_HSPC_LA33_P	<=>	35	
	I0_T2U_N12_45_R36	R36 NC x					
I0_L18N_T2U_N11_AD2N_45_M38	M38			FMCP_HSPC_LA30_N	<=>	36	
I0_L18P_T2U_N10_AD2P_45_N38	N38			FMCP_HSPC_LA30_P	<=>	36	
I0_L17N_T2U_N9_AD10N_45_L36	L36			FMCP_HSPC_LA28_N	<=>	36	
I0_L17P_T2U_N8_AD10P_45_M36	M36			FMCP_HSPC_LA28_P	<=>	36	
I0_L16N_T2U_N7_QBC_AD3N_45_N37	N37			FMCP_HSPC_LA31_N	<=>	35	
I0_L16P_T2U_N6_QBC_AD3P_45_P37	P37			FMCP_HSPC_LA31_P	<=>	35	
I0_L15N_T2L_N5_AD11N_45_L38	L38 NC x						
I0_L15P_T2L_N4_AD11P_45_M37	M37 NC x						
I0_L14N_T2L_N3_GC_45_P36	P36			FMCP_HSPC_CLK1_M2C_N	<=>	35	
I0_L14P_T2L_N2_GC_45_P35	P35			FMCP_HSPC_CLK1_M2C_P	<=>	35	
I0_L13N_T2L_N1_GC_QBC_45_P34	P34			FMCP_HSPC_LA17_CC_N	<=>	34	
I0_L13P_T2L_N0_GC_QBC_45_R34	R34			FMCP_HSPC_LA17_CC_P	<=>	34	
	I0_T1U_N12_45_V30	V30 NC x					
I0_L12N_T1U_N11_GC_45_R33	R33 NC x						
I0_L12P_T1U_N10_GC_45_T33	T33 NC x						
I0_L11N_T1U_N9_GC_45_P32	P32			USER_SMA_CLOCK_N	<=>	10,	
I0_L11P_T1U_N8_GC_45_R32	R32			USER_SMA_CLOCK_P	<=>	10,	
I0_L10N_T1U_N7_QBC_AD4N_45_P31	P31			FMCP_HSPC_LA18_CC_N	<=>	34	
I0_L10P_T1U_N6_QBC_AD4P_45_R31	R31			FMCP_HSPC_LA18_CC_P	<=>	34	
I0_L9N_T1L_N5_AD12N_45_W31	W31 NC x						
I0_L9P_T1L_N4_AD12P_45_Y31	Y31 NC x						
I0_L8N_T1L_N3_AD5N_45_U32	U32 NC x						
I0_L8P_T1L_N2_AD5P_45_U31	U31 NC x						
I0_L7N_T1L_N1_QBC_AD13N_45_T31	T31 NC x						
I0_L7P_T1U_N0_QBC_AD13P_45_T30	T30 NC x						
	I0_T0U_N12_VRP_45_Y33	Y33 VRP_45					10
I0_L6N_T0U_N11_AD6N_45_T35	T35			FMCP_HSPC_LA24_N	<=>	36	
I0_L6P_T0U_N10_AD6P_45_T34	T34			FMCP_HSPC_LA24_P	<=>	36	
I0_L5N_T0U_N9_AD14N_45_V34	V34			FMCP_HSPC_LA27_N	<=>	34	
I0_L5P_T0U_N8_AD14P_45_V33	V33			FMCP_HSPC_LA27_P	<=>	34	
I0_L4N_T0U_N7_DBC_AD7N_45_T36	T36			FMCP_HSPC_LA29_N	<=>	35	
I0_L4P_T0U_N6_DBC_AD7P_45_U35	U35			FMCP_HSPC_LA29_P	<=>	35	
I0_L3N_T0L_N5_AD15N_45_W34	W34			FMCP_HSPC_LA25_N	<=>	35	
I0_L3P_T0L_N4_AD15P_45_Y34	Y34			FMCP_HSPC_LA25_P	<=>	35	
I0_L2N_T0L_N3_45_U33	U33			FMCP_HSPC_LA26_N	<=>	34	
I0_L2P_T0L_N2_45_V32	V32			FMCP_HSPC_LA26_P	<=>	34	
I0_L1N_T0L_N1_DBC_45_W32	W32			FMCP_HSPC_LA23_N	<=>	34	
I0_L1P_T0L_N0_DBC_45_Y32	Y32			FMCP_HSPC_LA23_P	<=>	34	
	VREF_45_U30	U30 VREF_45					



U1

SQC_2104_1MM_IRON



FPGA Banks 43 45 FMCP HSPC



TITLE: FPGA Banks 43 45 FMCP HSPC SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185
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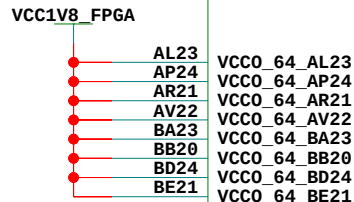
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 10 OF 76	DRAWN BY: DN

Bank 64 HP

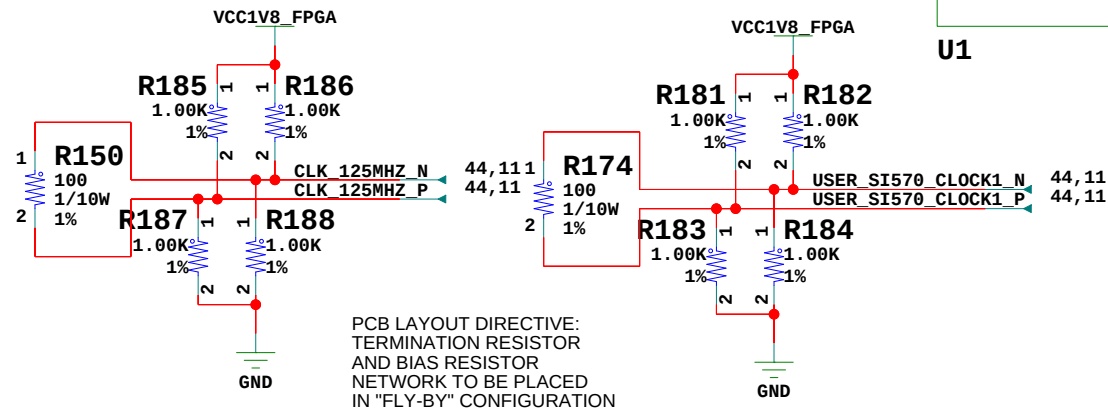
SOC_VU9P_FLGA2104_IRONWOOD

BANK 64
XCVU9PFLGA2104

IO_T3U_N12_64_AL25	AL25	IIC_MUX_RESET_B	58
IO_L24N_T3U_N11_64_AM24	AM24	IIC_MAIN_SCL	47, 58
IO_L24P_T3U_N10_64_AL24	AL24	IIC_MAIN_SDA	47, 58
IO_L23N_T3U_N9_64_AM21	AM21	QSFP1_MODSELL_LS	56
IO_L23P_T3U_N8_64_AL21	AL21	QSFP1_MODPRSL_LS	56
IO_L22N_T3U_N7_DBC_AD0N_64_AM22	AM22	QSFP1_RECCLK_N	51
IO_L22P_T3U_N6_DBC_AD0P_64_AM23	AM23	QSFP1_RECCLK_P	51
IO_L21N_T3L_N5_AD8N_64_AP21	AP21	QSFP1_INTL_LS	56
IO_L21P_T3L_N4_AD8P_64_AN21	AN21	QSFP1_LPMODE_LS	56
IO_L20N_T3L_N3_AD1N_64_AN23	AN23	QSFP2_MODSELL_LS	56
IO_L20P_T3L_N2_AD1P_64_AN24	AN24	QSFP2_MODPRSL_LS	56
IO_L19N_T3L_N1_DBC_AD9N_64_AP22	AP22	QSFP2_RECCLK_N	51
IO_L19P_T3L_N0_DBC_AD9P_64_AP23	AP23	QSFP2_RECCLK_P	51
IO_T2U_N12_64_AT21	AT21	QSFP2_INTL_LS	56
IO_L18N_T2U_N11_AD2N_64_AT24	AT24	QSFP2_LPMODE_LS	56
IO_L18P_T2U_N10_AD2P_64_AR24	AR24	PHY1_PDWN_B_I_INT_B_Q	52
IO_L17N_T2U_N9_AD10N_64_AR22	AR22	PHY1_GPIO_0	52
IO_L17P_T2U_N8_AD10P_64_AR23	AR23	PHY1_MDIO	52
IO_L16N_T2U_N7_QBC_AD3N_64_AV24	AV24	PHY1_SGMII_OUT_N	52
IO_L16P_T2U_N6_QBC_AD3P_64_AU24	AU24	PHY1_SGMII_OUT_P	52
IO_L15N_T2L_N5_AD11N_64_AV21	AV21	PHY1_SGMII_IN_N	52
IO_L15P_T2L_N4_AD11P_64_AU21	AU21	PHY1_SGMII_IN_P	52
IO_L14N_T2L_N3_GC_64_AV23	AV23	PHY1_MDC	52
IO_L14P_T2L_N2_GC_64_AU23	AU23	PHY1_CLKOUT	52
IO_L13N_T2L_N1_GC_QBC_64_AU22	AU22	PHY1_SGMII_CLK_N	52
IO_L13P_T2L_N0_GC_QBC_64_AT22	AT22	PHY1_SGMII_CLK_P	52
IO_T1U_N12_64_BA21	BA21	PHY1_RESET_B	52
IO_L12N_T1U_N11_GC_64_AW22	AW22	USER_SI570_CLOCK1_N	44, 11
IO_L12P_T1U_N10_GC_64_AW23	AW23	USER_SI570_CLOCK1_P	44, 11
IO_L11N_T1U_N9_GC_64_AY23	AY23	CLK_125MHZ_N	44, 11
IO_L11P_T1U_N8_GC_64_AY24	AY24	CLK_125MHZ_P	44, 11
IO_L10N_T1U_N7_QBC_AD4N_64_BA22	BA22	QSFP1_RESETL_LS	56
IO_L10P_T1U_N6_QBC_AD4P_64_AY22	AY22	QSFP2_RESETL_LS	56
IO_L9N_T1L_N5_AD12N_64_AY25	AY25	USB_UART_RTS	53
IO_L9P_T1L_N4_AD12P_64_AW25	AW25	USB_UART_TX	53
IO_L8N_T1L_N3_AD5N_64_BB21	BB21	USB_UART_RX	53
IO_L8P_T1L_N2_AD5P_64_BB22	BB22	USB_UART_CTS	53
IO_L7N_T1L_N1_QBC_AD13N_64_BA24	BA24	SYSCTLR_GPIO_7	47
IO_L7P_T1L_N0_QBC_AD13P_64_BA25	BA25	SYSCTLR_GPIO_6	47
IO_T0U_N12_VRP_64_BC24	BC24	PCIE_WAKE_B_LS	56
IO_L6N_T0U_N11_AD6N_64_BD21	BD21	SYSCTLR_GPIO_5	47
IO_L6P_T0U_N10_AD6P_64_BC21	BC21	SI5328_RST_LS_B	56
IO_L5N_T0U_N9_AD14N_64_BB23	BB23	PMBUS_ALERT_FPGA	59
IO_L5P_T0U_N8_AD14P_64_BB24	BB24	GPIO_SW_N	47, 55
IO_L4N_T0U_N7_DBC_AD7N_64_BF22	BF22	GPIO_SW_W	47, 55
IO_L4P_T0U_N6_DBC_AD7P_64_BE22	BE22	GPIO_SW_S	47, 55
IO_L3N_T0L_N5_AD15N_64_BE23	BE23	GPIO_SW_E	47, 55
IO_L3P_T0L_N4_AD15P_64_BD23	BD23	GPIO_SW_C	47, 55
IO_L2N_T0L_N3_64_BD22	BD22	FIREFLY_MODPRS_LS_B	56
IO_L2P_T0L_N2_64_BC23	BC23	FIREFLY_MODSEL_LS_B	56
IO_L1N_T0L_N1_DBC_64_BF24	BF24	FIREFLY_INT_LS_B	56
IO_L1P_T0L_N0_DBC_64_BE24	BE24	FIREFLY_RESET_LS_B	56
VREF_64_AL22	AL22	NC	



U1 SOC_2104_1MM_IRON



PCB LAYOUT DIRECTIVE:
TERMINATION RESISTOR
AND BIAS RESISTOR
NETWORK TO BE PLACED
IN "FLY-BY" CONFIGURATION

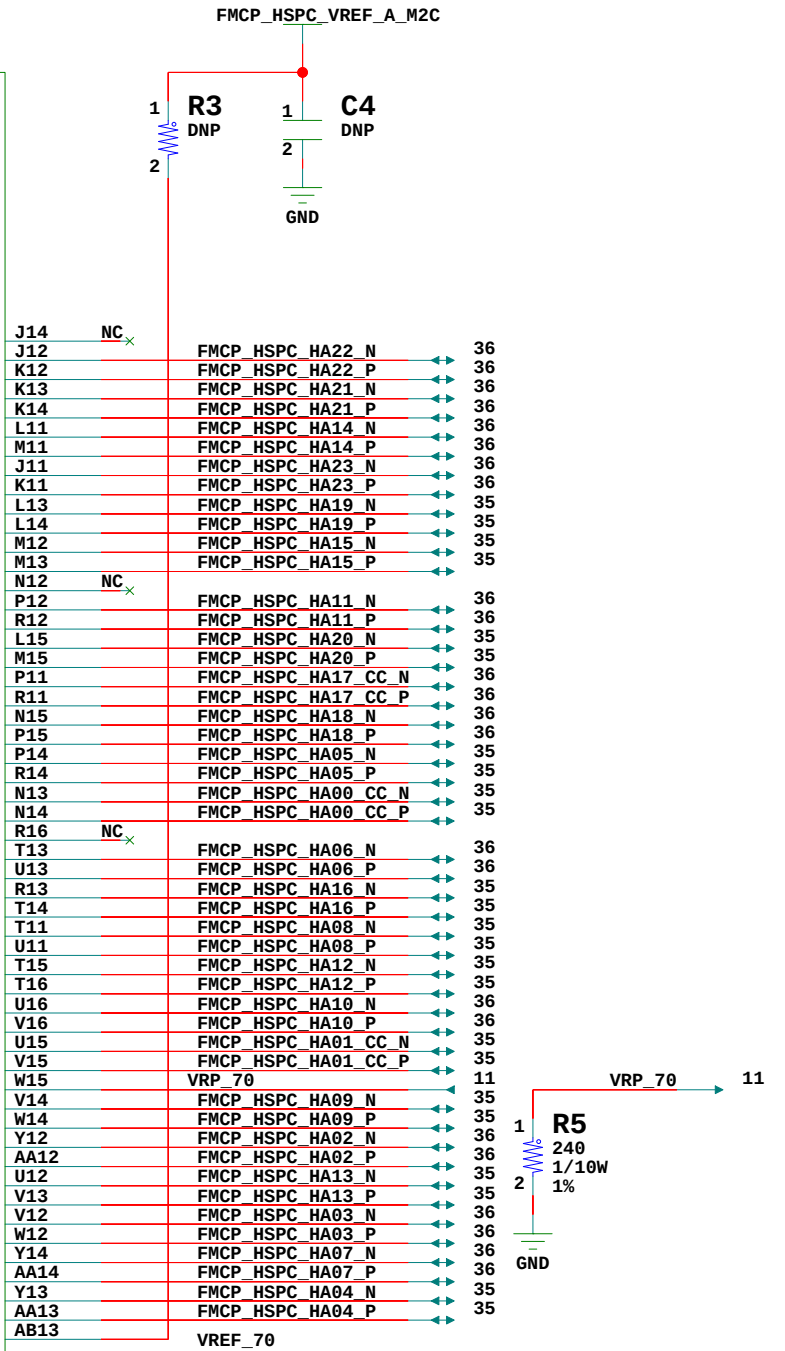
Layout: Place resistor and capacitor for VREF
Underneath the FPGA via array
right next to the via

Bank 70 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 70
XCVU9PFLGA2104

IO_T3U_N12_70_J14	J14	NC	
IO_L24N_T3U_N11_70_J12	J12	FMCP_HSPC_HA22_N	36
IO_L24P_T3U_N10_70_K12	K12	FMCP_HSPC_HA22_P	36
IO_L23N_T3U_N9_70_K13	K13	FMCP_HSPC_HA21_N	36
IO_L23P_T3U_N8_70_K14	K14	FMCP_HSPC_HA21_P	36
IO_L22N_T3U_N7_DBC_AD0N_70_L11	L11	FMCP_HSPC_HA14_N	36
IO_L22P_T3U_N6_DBC_AD0P_70_M11	M11	FMCP_HSPC_HA14_P	36
IO_L21N_T3L_N5_AD8N_70_J11	J11	FMCP_HSPC_HA23_N	36
IO_L21P_T3L_N4_AD8P_70_K11	K11	FMCP_HSPC_HA23_P	36
IO_L20N_T3L_N3_AD1N_70_L13	L13	FMCP_HSPC_HA19_N	35
IO_L20P_T3L_N2_AD1P_70_L14	L14	FMCP_HSPC_HA19_P	35
IO_L19N_T3L_N1_DBC_AD9N_70_M12	M12	FMCP_HSPC_HA15_N	35
IO_L19P_T3L_N0_DBC_AD9P_70_M13	M13	FMCP_HSPC_HA15_P	35
IO_T2U_N12_70_N12	N12	NC	
IO_L18N_T2U_N11_AD2N_70_P12	P12	FMCP_HSPC_HA11_N	36
IO_L18P_T2U_N10_AD2P_70_R12	R12	FMCP_HSPC_HA11_P	36
IO_L17N_T2U_N9_AD10N_70_L15	L15	FMCP_HSPC_HA20_N	35
IO_L17P_T2U_N8_AD10P_70_M15	M15	FMCP_HSPC_HA20_P	36
IO_L16N_T2U_N7_QBC_AD3N_70_P11	P11	FMCP_HSPC_HA17_CC_N	36
IO_L16P_T2U_N6_QBC_AD3P_70_R11	R11	FMCP_HSPC_HA17_CC_P	36
IO_L15N_T2L_N5_AD11N_70_N15	N15	FMCP_HSPC_HA18_N	36
IO_L15P_T2L_N4_AD11P_70_P15	P15	FMCP_HSPC_HA18_P	36
IO_L14N_T2L_N3_GC_70_P14	P14	FMCP_HSPC_HA05_N	35
IO_L14P_T2L_N2_GC_70_R14	R14	FMCP_HSPC_HA05_P	35
IO_L13N_T2L_N1_GC_QBC_70_N13	N13	FMCP_HSPC_HA00_CC_N	35
IO_L13P_T2L_N0_GC_QBC_70_N14	N14	FMCP_HSPC_HA00_CC_P	35
IO_T1U_N12_70_R16	R16	NC	
IO_L12N_T1U_N11_GC_70_T13	T13	FMCP_HSPC_HA06_N	36
IO_L12P_T1U_N10_GC_70_U13	U13	FMCP_HSPC_HA06_P	36
IO_L11N_T1U_N9_GC_70_R13	R13	FMCP_HSPC_HA16_N	35
IO_L11P_T1U_N8_GC_70_T14	T14	FMCP_HSPC_HA16_P	35
IO_L10N_T1U_N7_QBC_AD4N_70_T11	T11	FMCP_HSPC_HA08_N	35
IO_L10P_T1U_N6_QBC_AD4P_70_U11	U11	FMCP_HSPC_HA08_P	35
IO_L9N_T1L_N5_AD12N_70_T15	T15	FMCP_HSPC_HA12_N	35
IO_L9P_T1L_N4_AD12P_70_T16	T16	FMCP_HSPC_HA12_P	35
IO_L8N_T1L_N3_AD5N_70_U16	U16	FMCP_HSPC_HA10_N	36
IO_L8P_T1L_N2_AD5P_70_V16	V16	FMCP_HSPC_HA10_P	36
IO_L7N_T1L_N1_QBC_AD13N_70_U15	U15	FMCP_HSPC_HA01_CC_N	35
IO_L7P_T1L_N0_QBC_AD13P_70_V15	V15	FMCP_HSPC_HA01_CC_P	35
IO_T0U_N12_VRP_70_W15	W15	VRP_70	11
IO_L6N_T0U_N11_AD6N_70_V14	V14	FMCP_HSPC_HA09_N	35
IO_L6P_T0U_N10_AD6P_70_W14	W14	FMCP_HSPC_HA09_P	35
IO_L5N_T0U_N9_AD14N_70_Y12	Y12	FMCP_HSPC_HA02_N	36
IO_L5P_T0U_N8_AD14P_70_AA12	AA12	FMCP_HSPC_HA02_P	36
IO_L4N_T0U_N7_DBC_AD7N_70_U12	U12	FMCP_HSPC_HA13_N	35
IO_L4P_T0U_N6_DBC_AD7P_70_V13	V13	FMCP_HSPC_HA13_P	35
IO_L3N_T0L_N5_AD15N_70_V12	V12	FMCP_HSPC_HA03_N	36
IO_L3P_T0L_N4_AD15P_70_W12	W12	FMCP_HSPC_HA03_P	36
IO_L2N_T0L_N3_70_Y14	Y14	FMCP_HSPC_HA07_N	36
IO_L2P_T0L_N2_70_AA14	AA14	FMCP_HSPC_HA07_P	36
IO_L1N_T0L_N1_DBC_70_Y13	Y13	FMCP_HSPC_HA04_N	35
IO_L1P_T0L_N0_DBC_70_AA13	AA13	FMCP_HSPC_HA04_P	35
VREF_70_AB13	AB13	VREF_70	



U1 SOC_2104_1MM_IRON
FPGA Banks 64 GPIO 70 FMCP HSPC



TITLE: FPGA Banks 64 GPIO 70 FMCP HSPC SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER:	1.1	
SHEET SIZE: B	REV:	02	
SHEET 11 OF 76	DRAWN BY:	DN	

Layout: Place resistor and capacitor for VREF
Underneath the FPGA via array
right next to the via

Bank 47 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 66
XCVU9PFLGA2104

IO_T3U_N12_66_BB11
IO_L24N_T3U_N11_66_BB12
IO_L24P_T3U_N10_66_BB13
IO_L23N_T3U_N9_66_BB14
IO_L23P_T3U_N8_66_BA14
IO_L22N_T3U_N7_DBC_AD0N_66_BA15
IO_L22P_T3U_N6_DBC_AD0P_66_BA16
IO_L21N_T3L_N5_AD8N_66_BD15
IO_L21P_T3L_N4_AD8P_66_BC15
IO_L20N_T3L_N3_AD1N_66_BC16
IO_L20P_T3L_N2_AD1P_66_BB16
IO_L19N_T3L_N1_DBC_AD9N_66_BC13
IO_L19P_T3L_N0_DBC_AD9P_66_BC14
IO_T2U_N12_66_BA10
IO_L18N_T2U_N11_AD2N_66_AW7
IO_L18P_T2U_N10_AD2P_66_AW8
IO_L17N_T2U_N9_AD10N_66_AY7
IO_L17P_T2U_N8_AD10P_66_AY8
IO_L16N_T2U_N7_QBC_AD3N_66_AV8
IO_L16P_T2U_N6_QBC_AD3P_66_AV9
IO_L15N_T2L_N5_AD11N_66_BB7
IO_L15P_T2L_N4_AD11P_66_BA7
IO_L14N_T2L_N3_GC_66_BB8
IO_L14P_T2L_N2_GC_66_BB9
IO_L13N_T2L_N1_GC_QBC_66_BA9
IO_L13P_T2L_N0_GC_QBC_66_AY9
IO_T1U_N12_66_BF7
IO_L12N_T1U_N11_GC_66_BC8
IO_L12P_T1U_N10_GC_66_BC9
IO_L11N_T1U_N9_GC_66_BD10
IO_L11P_T1U_N8_GC_66_BC10
IO_L10N_T1U_N7_QBC_AD4N_66_BF9
IO_L10P_T1U_N6_QBC_AD4P_66_BF10
IO_L9N_T1L_N5_AD12N_66_BE9
IO_L9P_T1L_N4_AD12P_66_BE10
IO_L8N_T1L_N3_AD5N_66_BE7
IO_L8P_T1L_N2_AD5P_66_BE8
IO_L7N_T1L_N1_QBC_AD13N_66_BD7
IO_L7P_T1L_N0_QBC_AD13P_66_BD8
IO_T0U_N12_VRP_66_BD16
IO_L6N_T0U_N11_AD6N_66_BF15
IO_L6P_T0U_N10_AD6P_66_BE15
IO_L5N_T0U_N9_AD14N_66_BF14
IO_L5P_T0U_N8_AD14P_66_BE14
IO_L4N_T0U_N7_DBC_AD7N_66_BE13
IO_L4P_T0U_N6_DBC_AD7P_66_BD13
IO_L3N_T0L_N5_AD15N_66_BD11
IO_L3P_T0L_N4_AD15P_66_BC11
IO_L2N_T0L_N3_66_BF11
IO_L2P_T0L_N2_66_BF12
IO_L1N_T0L_N1_DBC_66_BE12
IO_L1P_T0L_N0_DBC_66_BD12
VREF_66_BA11

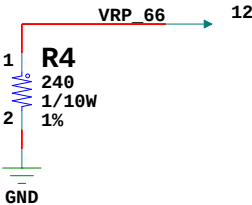
BB11 NC
BB12 FMC HPC1 LA10 N 39
BB13 FMC HPC1 LA10 P 39
BB14 FMC HPC1 LA09 N 39
BA14 FMC HPC1 LA09 P 39
BA15 FMC HPC1 LA11 N 41
BA16 FMC HPC1 LA11 P 41
BD15 FMC HPC1 LA07 N 41
BC15 FMC HPC1 LA07 P 41
BC16 FMC HPC1 LA15 N 41
BB16 FMC HPC1 LA15 P 41
BC13 FMC HPC1 LA12 N 40
BC14 FMC HPC1 LA12 P 40
BA10 NC
AW7 FMC HPC1 LA14 N 39
AW8 FMC HPC1 LA14 P 39
AY7 FMC HPC1 LA13 N 39
AY8 FMC HPC1 LA13 P 39
AV8 FMC HPC1 LA16 N 40
AV9 FMC HPC1 LA16 P 40
BB7 FMC HPC1 PRSNT M2C B LS 56
BA7 FMC HPC1 PG M2C LS 56
BB8 NC
BB9 NC
BA9 FMC HPC1 LA00 CC N 40
AY9 FMC HPC1 LA00 CC P 40
BF7 NC
BC8 FMC HPC1 CLK0 M2C N 41
BC9 FMC HPC1 CLK0 M2C P 41
BD10 NC
BC10 NC
BF9 FMC HPC1 LA01 CC N 39
BF10 FMC HPC1 LA01 CC P 39
BE9 NC
BE10 NC
BE7 NC
BE8 NC
BD7 NC
BD8 NC
BD16 VRP_66 12
BF15 FMC HPC1 LA08 N 40
BE15 FMC HPC1 LA08 P 40
BF14 FMC HPC1 LA05 N 39
BE14 FMC HPC1 LA05 P 39
BE13 FMC HPC1 LA06 N 39
BD13 FMC HPC1 LA06 P 39
BD11 FMC HPC1 LA02 N 41
BC11 FMC HPC1 LA02 P 41
BF11 FMC HPC1 LA04 N 41
BF12 FMC HPC1 LA04 P 41
BE12 FMC HPC1 LA03 N 40
BD12 FMC HPC1 LA03 P 40
BA11 VREF_66

VADJ_1V8_FPGA

AW9 VCCO_66_AW9
BB10 VCCO_66_BB10
BC7 VCCO_66_BC7
BD14 VCCO_66_BD14
BE11 VCCO_66_BE11
BF8 VCCO_66_BF8

U1

SOC_2104_1MM_IRON



Bank 48 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 67
XCVU9PFLGA2104

IO_T3U_N12_67_AM16
IO_L24N_T3U_N11_67_AK13
IO_L24P_T3U_N10_67_AK14
IO_L23N_T3U_N9_67_AM12
IO_L23P_T3U_N8_67_AM13
IO_L22N_T3U_N7_DBC_AD0N_67_AJ12
IO_L22P_T3U_N6_DBC_AD0P_67_AJ13
IO_L21N_T3L_N5_AD8N_67_AL12
IO_L21P_T3L_N4_AD8P_67_AK12
IO_L20N_T3L_N3_AD1N_67_AL15
IO_L20P_T3L_N2_AD1P_67_AK15
IO_L19N_T3L_N1_DBC_AD9N_67_AM14
IO_L19P_T3L_N0_DBC_AD9P_67_AL14
IO_T2U_N12_67_AR15
IO_L18N_T2U_N11_AD2N_67_AP15
IO_L18P_T2U_N10_AD2P_67_AN15
IO_L17N_T2U_N9_AD10N_67_AP16
IO_L17P_T2U_N8_AD10P_67_AN16
IO_L16N_T2U_N7_QBC_AD3N_67_AR12
IO_L16P_T2U_N6_QBC_AD3P_67_AP12
IO_L15N_T2L_N5_AD11N_67_AN13
IO_L15P_T2L_N4_AD11P_67_AN14
IO_L14N_T2L_N3_GC_67_AR13
IO_L14P_T2L_N2_GC_67_AP13
IO_L13N_T2L_N1_GC_QBC_67_AT14
IO_L13P_T2L_N0_GC_QBC_67_AR14
IO_T1U_N12_67_AW16
IO_L12N_T1U_N11_GC_67_AV13
IO_L12P_T1U_N10_GC_67_AV14
IO_L11N_T1U_N9_GC_67_AU13
IO_L11P_T1U_N8_GC_67_AU14
IO_L10N_T1U_N7_QBC_AD4N_67_AY14
IO_L10P_T1U_N6_QBC_AD4P_67_AY15
IO_L9N_T1L_N5_AD12N_67_AW15
IO_L9P_T1L_N4_AD12P_67_AV15
IO_L8N_T1L_N3_AD5N_67_AU16
IO_L8P_T1L_N2_AD5P_67_AU16
IO_L7N_T1L_N1_QBC_AD13N_67_AT15
IO_L7P_T1L_N0_QBC_AD13P_67_AT16
IO_T0U_N12_VRP_67_BA12
IO_L6N_T0U_N11_AD6N_67_AV11
IO_L6P_T0U_N10_AD6P_67_AU11
IO_L5N_T0U_N9_AD14N_67_AY13
IO_L5P_T0U_N8_AD14P_67_AW13
IO_L4N_T0U_N7_DBC_AD7N_67_AW10
IO_L4P_T0U_N6_DBC_AD7P_67_AV10
IO_L3N_T0L_N5_AD15N_67_AY10
IO_L3P_T0L_N4_AD15P_67_AW11
IO_L2N_T0L_N3_67_AY12
IO_L2P_T0L_N2_67_AW12
IO_L1N_T0L_N1_DBC_67_AU12
IO_L1P_T0L_N0_DBC_67_AT12
VREF_67_AL16

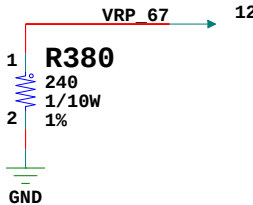
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AK13 FMC HPC1 LA33 N 40
AK14 FMC HPC1 LA33 P 40
AM12 FMC HPC1 LA31 N 40
AM13 FMC HPC1 LA31 P 40
AJ12 FMC HPC1 LA32 N 41
AJ13 FMC HPC1 LA32 P 41
AL12 FMC HPC1 LA30 N 41
AK12 FMC HPC1 LA30 P 41
AL15 FMC HPC1 LA26 N 39
AK15 FMC HPC1 LA26 P 39
AM14 FMC HPC1 LA27 N 39
AL14 FMC HPC1 LA27 P 39
AR15 NC
AP15 FMC HPC1 LA29 N 40
AN15 FMC HPC1 LA29 P 40
AP16 FMC HPC1 LA23 N 39
AN16 FMC HPC1 LA23 P 39
AR12 FMC HPC1 LA18 CC N 39
AP12 FMC HPC1 LA18 CC P 39
AN13 NC
AN14 NC
AR13 FMC HPC1 LA24 N 41
AP13 FMC HPC1 LA24 P 41
AT14 FMC HPC1 LA17 CC N 39
AR14 FMC HPC1 LA17 CC P 39
AW16 NC
AV13 FMC HPC1 CLK1 M2C N 40
AV14 FMC HPC1 CLK1 M2C P 40
AU13 NC
AU14 NC
AY14 PMOD0_0_LS 57
AY15 PMOD0_1_LS 57
AW15 PMOD0_2_LS 57
AV15 PMOD0_3_LS 57
AV16 PMOD0_4_LS 57
AU16 PMOD0_5_LS 57
AT15 PMOD0_6_LS 57
AT16 PMOD0_7_LS 57
BA12 VRP_67 12
AV11 FMC HPC1 LA21 N 41
AU11 FMC HPC1 LA21 P 41
AY13 FMC HPC1 LA22 N 40
AW13 FMC HPC1 LA22 P 40
AW10 FMC HPC1 LA28 N 41
AV10 FMC HPC1 LA28 P 41
AY10 FMC HPC1 LA20 N 40
AW11 FMC HPC1 LA20 P 40
AY12 FMC HPC1 LA19 N 41
AW12 FMC HPC1 LA19 P 41
AU12 FMC HPC1 LA25 N 40
AT12 FMC HPC1 LA25 P 40
AL16 VREF_67

VADJ_1V8_FPGA

AL13 VCCO_67_AL13
AP14 VCCO_67_AP14
AU15 VCCO_67_AU15
AV12 VCCO_67_AV12
AY16 VCCO_67_AY16
BA13 VCCO_67_BA13

U1

SOC_2104_1MM_IRON



FPGA Banks 66 67 FMC HPC1



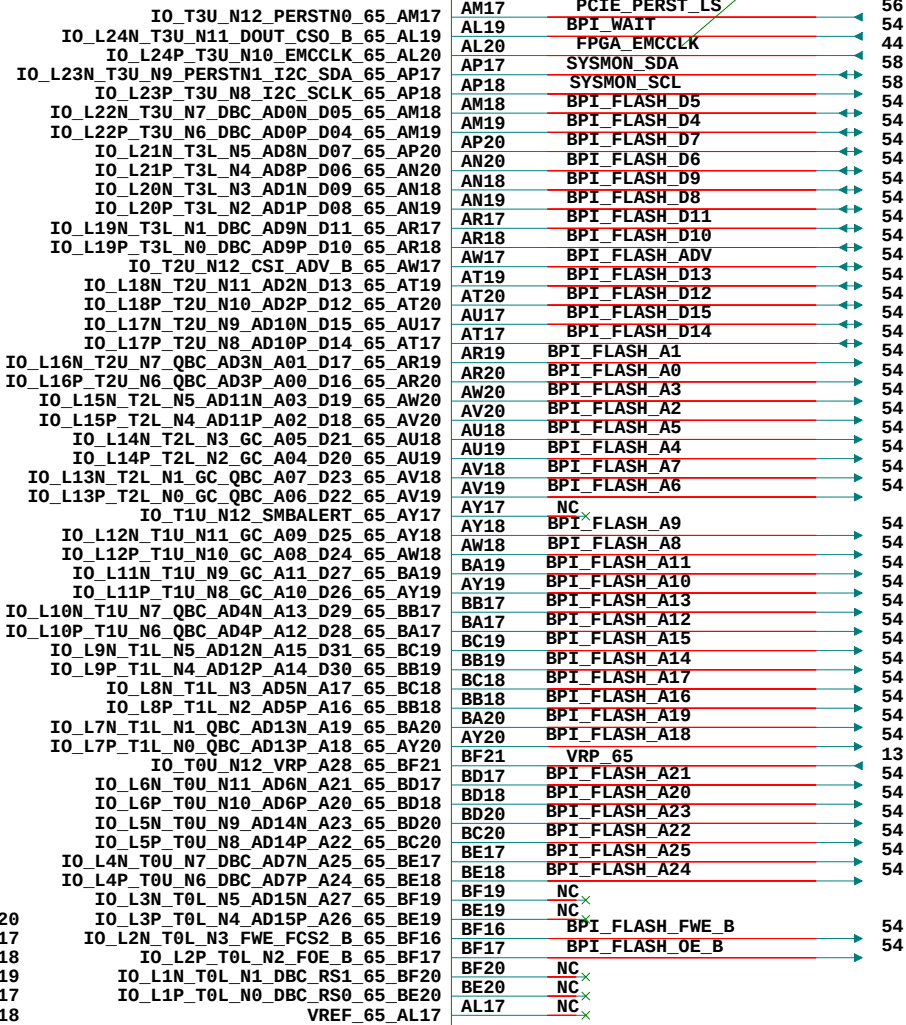
TITLE: FPGA Banks 66 67 FMC HPC1
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 12 OF 76	DRAWN BY: DN

Bank 65 HP

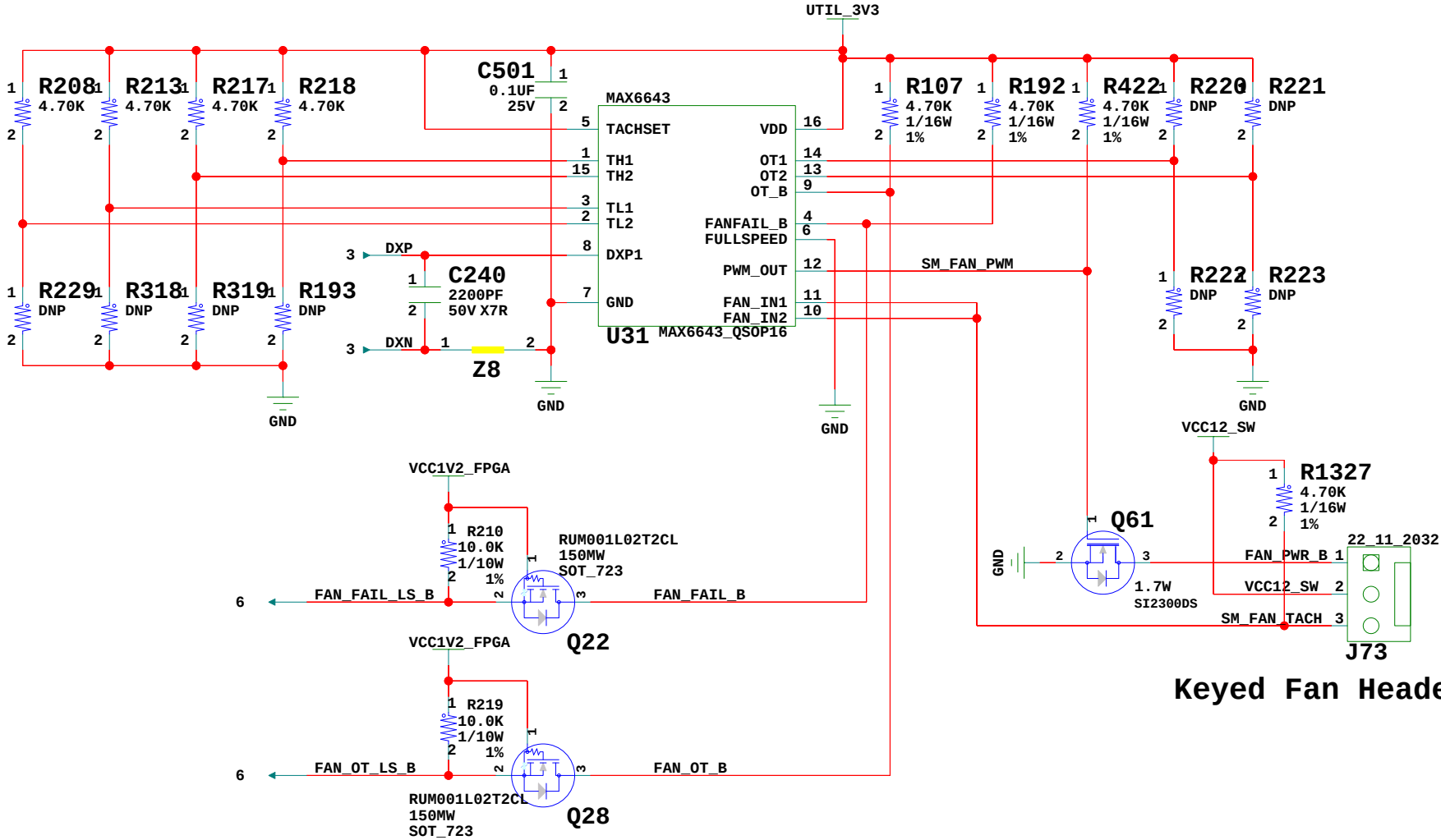
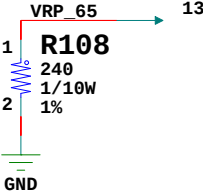
SOC_VU9P_FLGA2104_IRONWOOD

BANK 65
XCVU9PFLGA2104



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SOC_2104_1MM_IRON



FPGA Bank 65 BPI Flash IF

TITLE: FPGA Bank 65 BPI Flash IF SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:39	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 13 OF 76	DRAWN BY: DN

SOC_VU9P_FLGA2104_IRONWOOD

BANK 120
XCVU9PFLGA2104

MGTYTXP0_120	BD42	BD42	FMCP_HSPC_DP20_C2M_P	37
MGTYTXN0_120	BD43	BD43	FMCP_HSPC_DP20_C2M_N	37
MGTYRXP0_120	BC45	BC45	FMCP_HSPC_DP20_M2C_P	37
MGTYRXN0_120	BC46	BC46	FMCP_HSPC_DP20_M2C_N	37
MGTYTXP1_120	BB42	BB42	FMCP_HSPC_DP21_C2M_P	37
MGTYTXN1_120	BB43	BB43	FMCP_HSPC_DP21_C2M_N	37
MGTYRXP1_120	BA45	BA45	FMCP_HSPC_DP21_M2C_P	37
MGTYRXN1_120	BA46	BA46	FMCP_HSPC_DP21_M2C_N	37
MGTYTXP2_120	AY42	AY42	FMCP_HSPC_DP22_C2M_P	37
MGTYTXN2_120	AY43	AY43	FMCP_HSPC_DP22_C2M_N	37
MGTYRXP2_120	AW45	AW45	FMCP_HSPC_DP22_M2C_P	37
MGTYRXN2_120	AW46	AW46	FMCP_HSPC_DP22_M2C_N	37
MGTYTXP3_120	AV42	AV42	FMCP_HSPC_DP23_C2M_P	37
MGTYTXN3_120	AV43	AV43	FMCP_HSPC_DP23_C2M_N	37
MGTYRXP3_120	AU45	AU45	FMCP_HSPC_DP23_M2C_P	37
MGTYRXN3_120	AU46	AU46	FMCP_HSPC_DP23_M2C_N	37
MGTREFCLK0P_120	AN40	AN40	FMCP_HSPC_GBTCLK5_M2C_C_P	14
MGTREFCLK0N_120	AN41	AN41	FMCP_HSPC_GBTCLK5_M2C_C_N	14
MGTREFCLK1P_120	AM38	AM38	FMCP_HSPC_GBT1_5_P	14
MGTREFCLK1N_120	AM39	AM39	FMCP_HSPC_GBT1_5_N	14

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SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK 121
XCVU9PFLGA2104

MGTYTXP0_121	AT42	AT42	FMCP_HSPC_DP0_C2M_P	34
MGTYTXN0_121	AT43	AT43	FMCP_HSPC_DP0_C2M_N	34
MGTYRXP0_121	AR45	AR45	FMCP_HSPC_DP0_M2C_P	34
MGTYRXN0_121	AR46	AR46	FMCP_HSPC_DP0_M2C_N	34
MGTYTXP1_121	AP42	AP42	FMCP_HSPC_DP1_C2M_P	34
MGTYTXN1_121	AP43	AP43	FMCP_HSPC_DP1_C2M_N	34
MGTYRXP1_121	AN45	AN45	FMCP_HSPC_DP1_M2C_P	34
MGTYRXN1_121	AN46	AN46	FMCP_HSPC_DP1_M2C_N	34
MGTYTXP2_121	AM42	AM42	FMCP_HSPC_DP2_C2M_P	34
MGTYTXN2_121	AM43	AM43	FMCP_HSPC_DP2_C2M_N	34
MGTYRXP2_121	AL45	AL45	FMCP_HSPC_DP2_M2C_P	34
MGTYRXN2_121	AL46	AL46	FMCP_HSPC_DP2_M2C_N	34
MGTYTXP3_121	AL40	AL40	FMCP_HSPC_DP3_C2M_P	34
MGTYTXN3_121	AL41	AL41	FMCP_HSPC_DP3_C2M_N	34
MGTYRXP3_121	AJ45	AJ45	FMCP_HSPC_DP3_M2C_P	34
MGTYRXN3_121	AJ46	AJ46	FMCP_HSPC_DP3_M2C_N	34
MGTREFCLK0P_121	AK38	AK38	FMCP_HSPC_GBT0_0_P	14
MGTREFCLK0N_121	AK39	AK39	FMCP_HSPC_GBT0_0_N	14
MGTREFCLK1P_121	AH38	AH38	FMCP_HSPC_GBT1_0_P	14
MGTREFCLK1N_121	AH39	AH39	FMCP_HSPC_GBT1_0_N	14
MGTRREF_LS_BF43	BF43	MGTRREF_121		
MGTAVTTRCAL_LS_BF42	BF42			

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SOC_2104_1MM_IRON

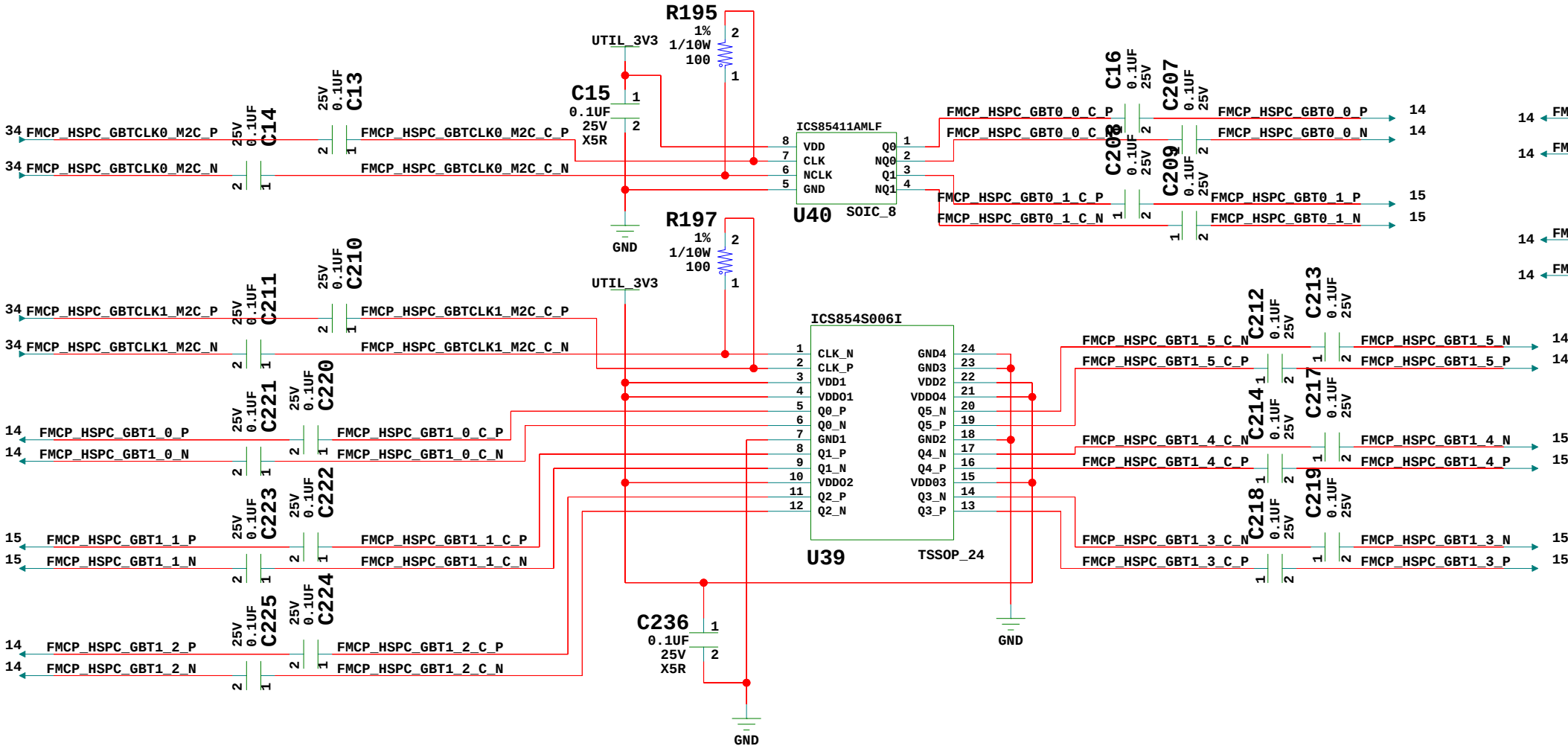
SOC_VU9P_FLGA2104_IRONWOOD

BANK 122
XCVU9PFLGA2104

MGTYTXP0_122	AK42	AK42	FMCP_HSPC_DP8_C2M_P	34
MGTYTXN0_122	AK43	AK43	FMCP_HSPC_DP8_C2M_N	34
MGTYRXP0_122	AG45	AG45	FMCP_HSPC_DP8_M2C_P	34
MGTYRXN0_122	AG46	AG46	FMCP_HSPC_DP8_M2C_N	34
MGTYTXP1_122	AJ40	AJ40	FMCP_HSPC_DP9_C2M_P	34
MGTYTXN1_122	AJ41	AJ41	FMCP_HSPC_DP9_C2M_N	34
MGTYRXP1_122	AF43	AF43	FMCP_HSPC_DP9_M2C_P	34
MGTYRXN1_122	AF44	AF44	FMCP_HSPC_DP9_M2C_N	34
MGTYTXP2_122	AG40	AG40	FMCP_HSPC_DP10_C2M_P	37
MGTYTXN2_122	AG41	AG41	FMCP_HSPC_DP10_C2M_N	37
MGTYRXP2_122	AE45	AE45	FMCP_HSPC_DP10_M2C_P	37
MGTYRXN2_122	AE46	AE46	FMCP_HSPC_DP10_M2C_N	37
MGTYTXP3_122	AE40	AE40	FMCP_HSPC_DP11_C2M_P	37
MGTYTXN3_122	AE41	AE41	FMCP_HSPC_DP11_C2M_N	37
MGTYRXP3_122	AD43	AD43	FMCP_HSPC_DP11_M2C_P	37
MGTYRXN3_122	AD44	AD44	FMCP_HSPC_DP11_M2C_N	37
MGTREFCLK0P_122	AF38	AF38	FMCP_HSPC_GBTCLK2_M2C_C_P	14
MGTREFCLK0N_122	AF39	AF39	FMCP_HSPC_GBTCLK2_M2C_C_N	14
MGTREFCLK1P_122	AD38	AD38	FMCP_HSPC_GBT1_2_P	14
MGTREFCLK1N_122	AD39	AD39	FMCP_HSPC_GBT1_2_N	14

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SOC_2104_1MM_IRON



SOC_VU9P_FLGA2104_IRONWOOD

BANK 125
XCVU9PFLGA2104

MGITYXP0_125	AC40	FMCP	HSPC	DP12	C2M	P	37
MGITYXN0_125	AC41	FMCP	HSPC	DP12	C2M	P	37
MGTYRXP0_125	AC45	FMCP	HSPC	DP12	M2C	P	37
MGTYRXN0_125	AC46	FMCP	HSPC	DP12	M2C	N	37
MGITYXP1_125	AA40	FMCP	HSPC	DP13	C2M	P	37
MGITYXN1_125	AA41	FMCP	HSPC	DP13	C2M	N	37
MGTYRXP1_125	AB43	FMCP	HSPC	DP13	M2C	P	37
MGTYRXN1_125	AB44	FMCP	HSPC	DP13	M2C	N	37
MGITYXP2_125	W40	FMCP	HSPC	DP14	C2M	P	37
MGITYXN2_125	W41	FMCP	HSPC	DP14	C2M	N	37
MGTYRXP2_125	AA45	FMCP	HSPC	DP14	M2C	P	37
MGTYRXN2_125	AA46	FMCP	HSPC	DP14	M2C	N	37
MGITYXP3_125	U40	FMCP	HSPC	DP15	C2M	P	37
MGITYXN3_125	U41	FMCP	HSPC	DP15	C2M	N	37
MGTYRXP3_125	Y43	FMCP	HSPC	DP15	M2C	P	37
MGTYRXN3_125	Y44	FMCP	HSPC	DP15	M2C	N	37
MGTRFCLK0P_125	AB38	FMCP	HSPC	GBTCLK3	M2C	C	15
MGTRFCLK0N_125	AB39	FMCP	HSPC	GBTCLK3	M2C	C	15
MGTRFCLK1P_125	Y38	FMCP	HSPC	GBT1	3	P	14
MGTRFCLK1N_125	Y39	FMCP	HSPC	GBT1	3	N	14

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SOC 2104 1MM IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK 126
XCVU9PFLGA2104

MGTYTXP0_126_T42	T42	FMCP	HSPC	DP4	C2M	P	34
MGTYTXN0_126_T43	T43	FMCP	HSPC	DP4	C2M	N	34
MGTYRXP0_126_W45	W45	FMCP	HSPC	DP4	M2C	P	34
MGTYRXN0_126_W46	W46	FMCP	HSPC	DP4	M2C	N	34
MGTYTXP1_126_P42	P42	FMCP	HSPC	DP5	C2M	P	34
MGTYTXN1_126_P43	P43	FMCP	HSPC	DP5	C2M	N	34
MGTYRXP1_126_U45	U45	FMCP	HSPC	DP5	M2C	P	34
MGTYRXN1_126_U46	U46	FMCP	HSPC	DP5	M2C	N	34
MGTYTXP2_126_M42	M42	FMCP	HSPC	DP6	C2M	P	34
MGTYTXN2_126_M43	M43	FMCP	HSPC	DP6	C2M	N	34
MGTYRXP2_126_R45	R45	FMCP	HSPC	DP6	M2C	P	34
MGTYRXN2_126_R46	R46	FMCP	HSPC	DP6	M2C	N	34
MGTYTXP3_126_K42	K42	FMCP	HSPC	DP7	C2M	P	34
MGTYTXN3_126_K43	K43	FMCP	HSPC	DP7	C2M	N	34
MGTYRXP3_126_N45	N45	FMCP	HSPC	DP7	M2C	P	34
MGTYRXN3_126_N46	N46	FMCP	HSPC	DP7	M2C	N	34
MGTREFCLK0P_126_V38	V38	FMCP	HSPC	GBT0	1	P	14
MGTREFCLK0N_126_V39	V39	FMCP	HSPC	GBT0	1	N	14
MGTREFCLK1P_126_T38	T38	FMCP	HSPC	GBT1	1	P	14
MGTREFCLK1N_126_T39	T39	FMCP	HSPC	GBT1	1	N	14
MGTRREF LN L41	L41	MGTRREF					
MGTAVTTRCAL LN L40	L40						

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SOC 2104 1MM IRON

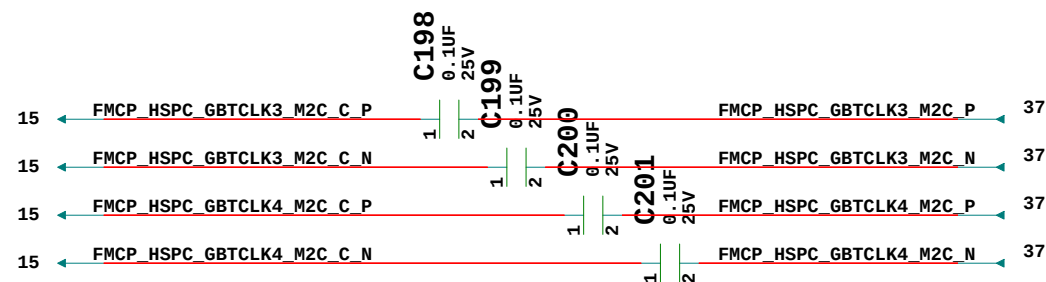
SOC_VU9P_FLGA2104_IRONWOOD

BANK 127
XCVU9PFLGA2104

MGTTYXP0_127_H42	H42	FMCP	HSPC	DP16	C2M	P	37
MGTTYXN0_127_H43	H43	FMCP	HSPC	DP16	C2M	N	37
MGTYRXP0_127_L45	L45	FMCP	HSPC	DP16	M2C	N	37
MGTYRXN0_127_L46	L46	FMCP	HSPC	DP16	M2C	N	37
MGTTYXP1_127_F42	F42	FMCP	HSPC	DP17	C2M	P	37
MGTTYXN1_127_F43	F43	FMCP	HSPC	DP17	C2M	N	37
MGTYRX1_127_J45	J45	FMCP	HSPC	DP17	M2C	P	37
MGTYRXN1_127_J46	J46	FMCP	HSPC	DP17	M2C	N	37
MGTTYXP2_127_D42	D42	FMCP	HSPC	DP18	C2M	P	37
MGTTYXN2_127_D43	D43	FMCP	HSPC	DP18	C2M	N	37
MGTYRX2_127_G45	G45	FMCP	HSPC	DP18	M2C	P	37
MGTYRXN2_127_G46	G46	FMCP	HSPC	DP18	M2C	N	37
MGTTYXP3_127_B42	B42	FMCP	HSPC	DP19	C2M	P	37
MGTTYXN3_127_B43	B43	FMCP	HSPC	DP19	C2M	N	37
MGTYRX3_127_E45	E45	FMCP	HSPC	DP19	M2C	P	37
MGTYRXN3_127_E46	E46	FMCP	HSPC	DP19	M2C	N	37
MGTFREFCLK0P_127_R40	R40	FMCP	HSPC	GBTCLK4	M2C	C P	15
MGTFREFCLK0N_127_R41	R41	FMCP	HSPC	GBTCLK4	M2C	C N	15
MGTFREFCLK1P_127_N40	N40	FMCP	HSPC	GBT1	4	P	14
MGTFREFCLK1N_127_N41	N41	FMCP	HSPC	GBT1	4	N	14

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SOC 2104 1MM IRON



R176 100 1/10W 1% MGTAVTT_FPGA

FPGA Banks 125 126 127



**TITLE: FPGA Banks 125 126 127
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1 1**

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER:	1.1
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SHEET SIZE: B

REV:	02
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SHEET **15** OF **76**

DRAWN BY:

DN

SOC_VU9P_FLGA2104_IRONWOOD

BANK 224
XCVU9PFLGA2104

MGTYTXP0_224_BE5	BE5	PCIE TX15 P	43
MGTYTXN0_224_BE4	BE4	PCIE TX15 N	43
MGTYRXP0_224_BB2	BB2	PCIE RX15 P	43
MGTYRXN0_224_BB1	BB1	PCIE RX15 N	43
MGTYTXP1_224_BC5	BC5	PCIE TX14 P	43
MGTYTXN1_224_BC4	BC4	PCIE TX14 N	43
MGTYRXP1_224_AY2	AY2	PCIE RX14 P	43
MGTYRXN1_224_AY1	AY1	PCIE RX14 N	43
MGTYTXP2_224_BA5	BA5	PCIE TX13 P	43
MGTYTXN2_224_BA4	BA4	PCIE TX13 N	43
MGTYRXP2_224_AV2	AV2	PCIE RX13 P	43
MGTYRXN2_224_AV1	AV1	PCIE RX13 N	43
MGTYTXP3_224_AW5	AW5	PCIE TX12 P	43
MGTYTXN3_224_AW4	AW4	PCIE TX12 N	43
MGTYRXP3_224_AT2	AT2	PCIE RX12 P	43
MGTYRXN3_224_AT1	AT1	PCIE RX12 N	43
MGTREFCLK0P_224_AR9	AR9	NC	
MGTREFCLK0N_224_AR8	AR8	NC	
MGTREFCLK1P_224_AN9	AN9	NC	
MGTREFCLK1N_224_AN8	AN8	NC	

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SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK 225
XCVU9PFLGA2104

MGTYTXP0_225_AU5	AU5	PCIE TX11 P	43
MGTYTXN0_225_AU4	AU4	PCIE TX11 N	43
MGTYRXP0_225_AP2	AP2	PCIE RX11 P	43
MGTYRXN0_225_AP1	AP1	PCIE RX11 N	43
MGTYTXP1_225_AT7	AT7	PCIE TX10 P	43
MGTYTXN1_225_AT6	AT6	PCIE TX10 N	43
MGTYRXP1_225_AM2	AM2	PCIE RX10 P	43
MGTYRXN1_225_AM1	AM1	PCIE RX10 N	43
MGTYTXP2_225_AR5	AR5	PCIE TX9 P	43
MGTYTXN2_225_AR4	AR4	PCIE TX9 N	43
MGTYRXP2_225_AK2	AK2	PCIE RX9 P	43
MGTYRXN2_225_AK1	AK1	PCIE RX9 N	43
MGTYTXP3_225_AP7	AP7	PCIE TX8 P	43
MGTYTXN3_225_AP6	AP6	PCIE TX8 N	43
MGTYRXP3_225_AJ4	AJ4	PCIE RX8 P	43
MGTYRXN3_225_AJ3	AJ3	PCIE RX8 N	43
MGTREFCLK0P_225_AL9	AL9	PCIE CLK1 P	16
MGTREFCLK0N_225_AL8	AL8	PCIE CLK1 N	16
MGTREFCLK1P_225_AJ9	AJ9	MGT SI570 CLOCK1 C P	16
MGTREFCLK1N_225_AJ8	AJ8	MGT SI570 CLOCK1 C N	16

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SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK 226
XCVU9PFLGA2104

MGTYTXP0_226_AN5	AN5	PCIE TX7 P	43
MGTYTXN0_226_AN4	AN4	PCIE TX7 N	43
MGTYRXP0_226_AH2	AH2	PCIE RX7 P	43
MGTYRXN0_226_AH1	AH1	PCIE RX7 N	43
MGTYTXP1_226_AM7	AM7	PCIE TX6 P	43
MGTYTXN1_226_AM6	AM6	PCIE TX6 N	43
MGTYRXP1_226_AG4	AG4	PCIE RX6 P	43
MGTYRXN1_226_AG3	AG3	PCIE RX6 N	43
MGTYTXP2_226_AK7	AK7	PCIE TX5 P	43
MGTYTXN2_226_AK6	AK6	PCIE TX5 N	43
MGTYRXP2_226_AF2	AF2	PCIE RX5 P	43
MGTYRXN2_226_AF1	AF1	PCIE RX5 N	43
MGTYTXP3_226_AH7	AH7	PCIE TX4 P	43
MGTYTXN3_226_AH6	AH6	PCIE TX4 N	43
MGTYRXP3_226_AE4	AE4	PCIE RX4 P	43
MGTYRXN3_226_AE3	AE3	PCIE RX4 N	43
MGTREFCLK0P_226_AG9	AG9		
MGTREFCLK0N_226_AG8	AG8		
MGTREFCLK1P_226_AE9	AE9	NC	
MGTREFCLK1N_226_AE8	AE8	NC	
MGTAVTTTRCAL_RS_BD3	BD3		
MGTAVTTTRCAL_RS_BD2	BD2	MGTTRREF_226	

U1

SOC_2104_1MM_IRON

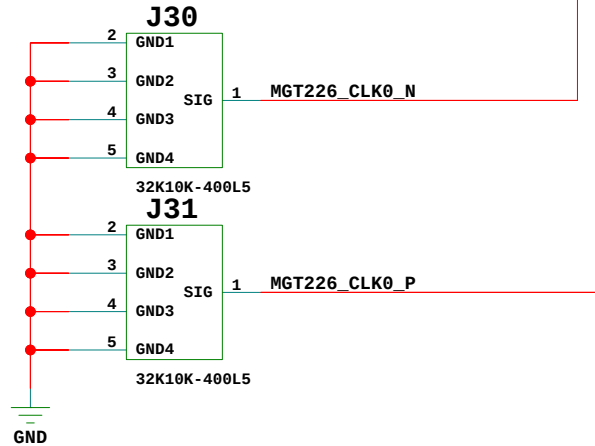
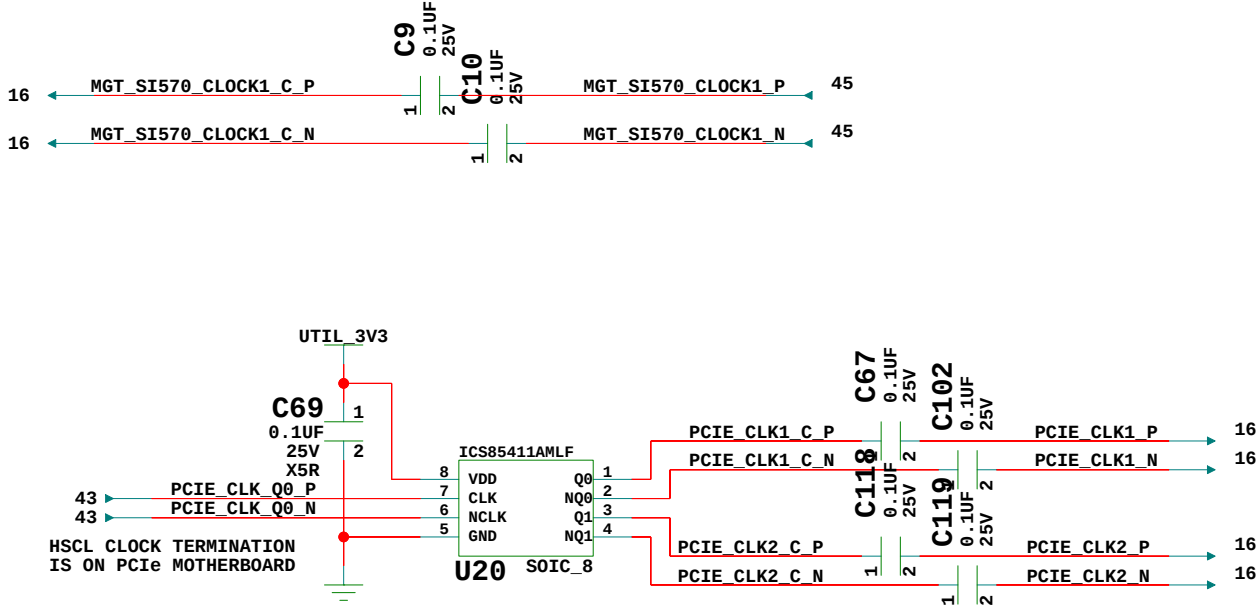
SOC_VU9P_FLGA2104_IRONWOOD

BANK 227
XCVU9PFLGA2104

MGTYTXP0_227_AF7	AF7	PCIE TX3 P	43
MGTYTXN0_227_AF6	AF6	PCIE TX3 N	43
MGTYRXP0_227_AD2	AD2	PCIE RX3 P	43
MGTYRXN0_227_AD1	AD1	PCIE RX3 N	43
MGTYTXP1_227_AD7	AD7	PCIE TX2 P	43
MGTYTXN1_227_AD6	AD6	PCIE TX2 N	43
MGTYRXP1_227_AC4	AC4	PCIE RX2 P	43
MGTYRXN1_227_AC3	AC3	PCIE RX2 N	43
MGTYTXP2_227_AB7	AB7	PCIE TX1 P	43
MGTYTXN2_227_AB6	AB6	PCIE TX1 N	43
MGTYRXP2_227_AB2	AB2	PCIE RX1 P	43
MGTYRXN2_227_AB1	AB1	PCIE RX1 N	43
MGTYTXP3_227_Y7	Y7	PCIE TX0 P	43
MGTYTXN3_227_Y6	Y6	PCIE TX0 N	43
MGTYRXP3_227_AA4	AA4	PCIE RX0 P	43
MGTYRXN3_227_AA3	AA3	PCIE RX0 N	43
MGTREFCLK0P_227_AC9	AC9	PCIE CLK2 P	16
MGTREFCLK0N_227_AC8	AC8	PCIE CLK2 N	16
MGTREFCLK1P_227_AA9	AA9	NC	
MGTREFCLK1N_227_AA8	AA8	NC	

U1

SOC_2104_1MM_IRON



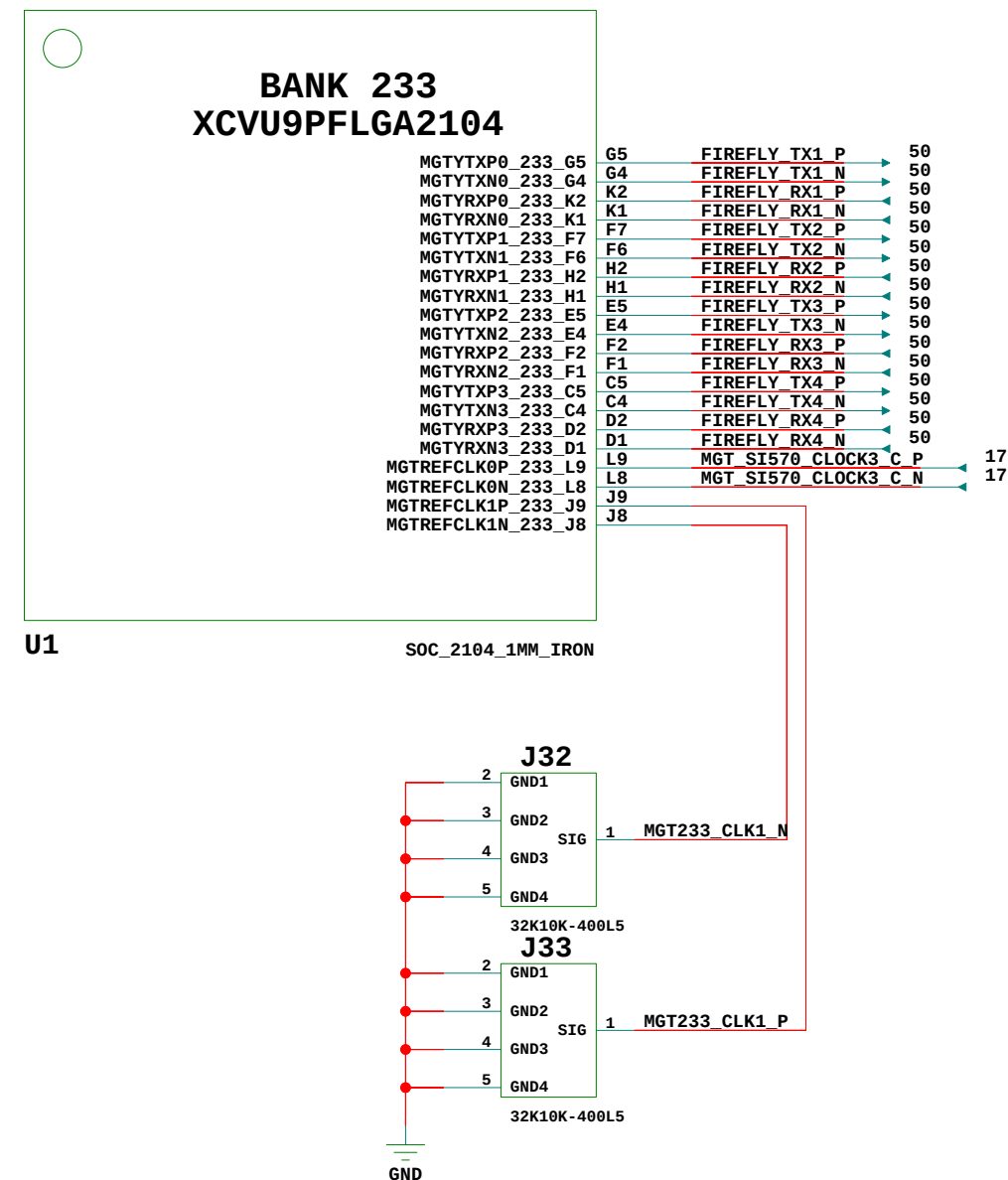
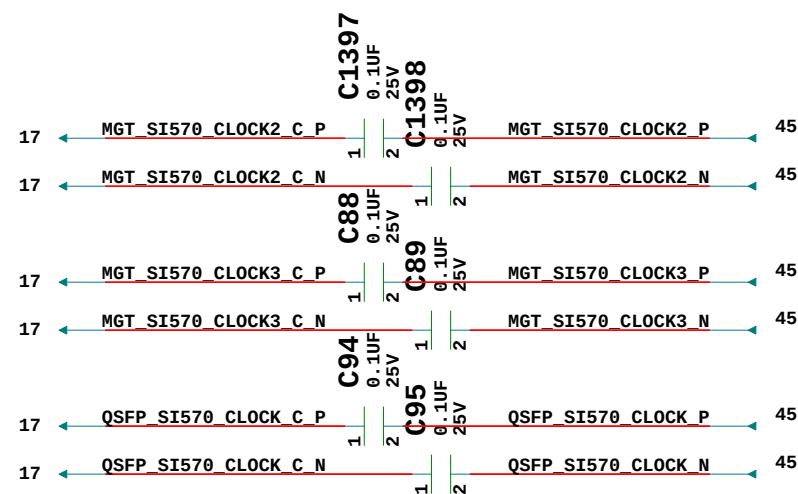
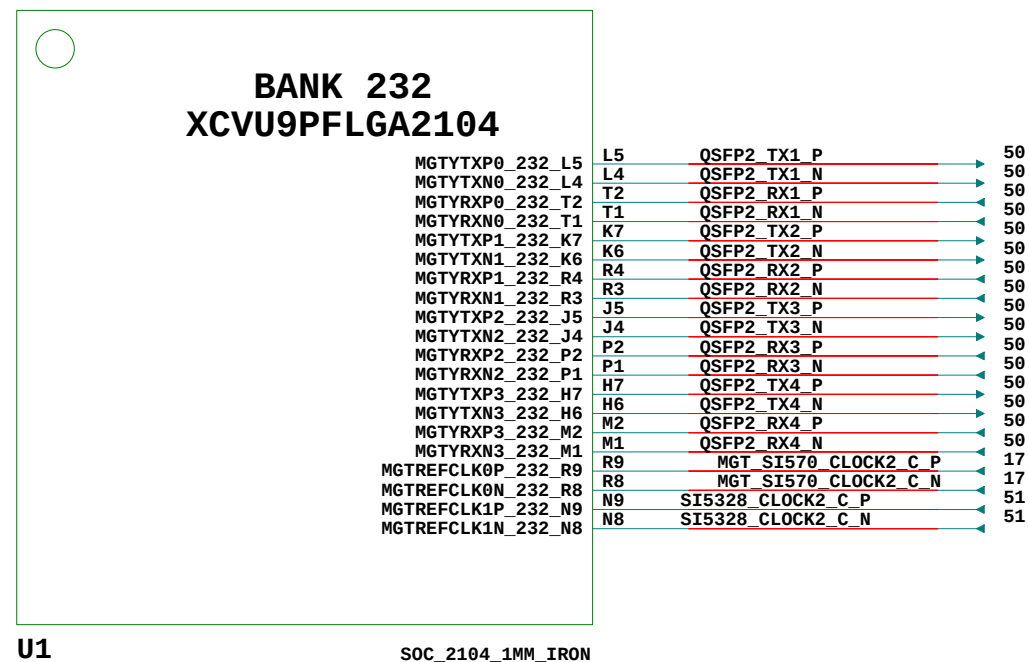
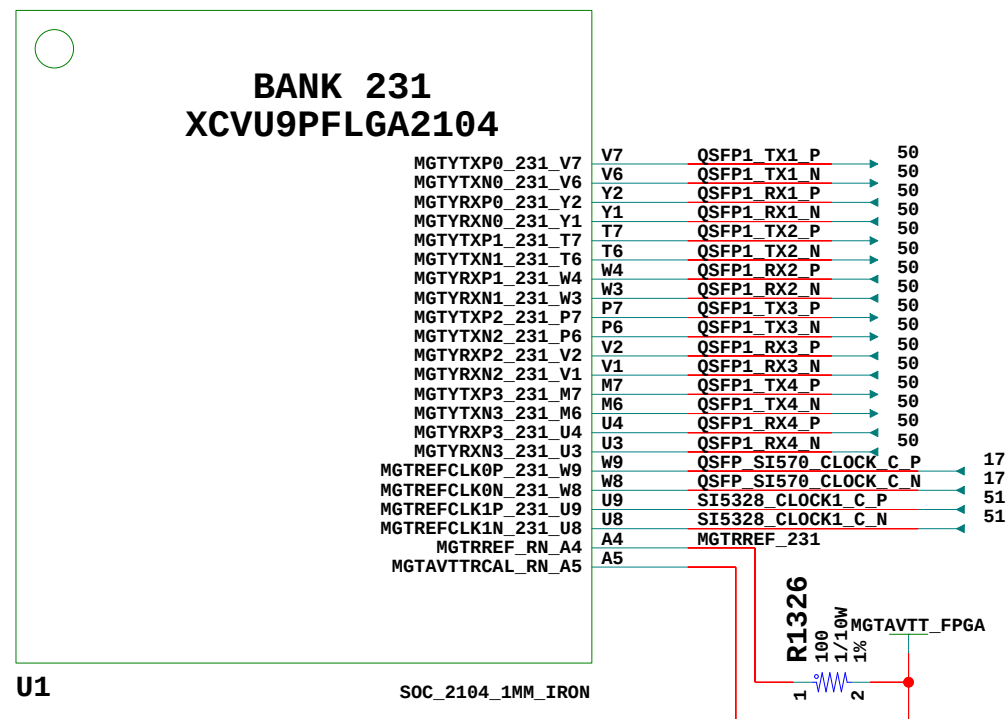
FPGA Banks 224 225 226 227



TITLE: FPGA Banks 224 225 226 227
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 16 OF 76	DRAWN BY: DN



FPGA Banks 231 232 233



TITLE: FPGA Banks 231 232 233
SCHEM, ROHS COMPLIANT
HW-U1-VCU118 REV1 1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER:	1.1
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SHEET SIZE: B

REV:	02
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SHEET **17** OF **76**

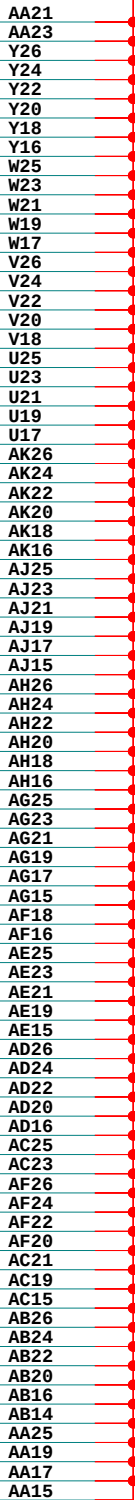
DRAWN BY: **DN**

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCINT
XCVU9PFLGA2104

VCCINT_AA21
VCCINT_AA23
VCCINT_Y26
VCCINT_Y24
VCCINT_Y22
VCCINT_Y20
VCCINT_Y18
VCCINT_Y16
VCCINT_W25
VCCINT_W23
VCCINT_W21
VCCINT_W19
VCCINT_W17
VCCINT_V26
VCCINT_V24
VCCINT_V22
VCCINT_V20
VCCINT_V18
VCCINT_U25
VCCINT_U23
VCCINT_U21
VCCINT_U19
VCCINT_U17
VCCINT_AK26
VCCINT_AK24
VCCINT_AK22
VCCINT_AK20
VCCINT_AK18
VCCINT_AK16
VCCINT_AJ25
VCCINT_AJ23
VCCINT_AJ21
VCCINT_AJ19
VCCINT_AJ17
VCCINT_AJ15
VCCINT_AH26
VCCINT_AH24
VCCINT_AH22
VCCINT_AH20
VCCINT_AH18
VCCINT_AH16
VCCINT_AG25
VCCINT_AG23
VCCINT_AG21
VCCINT_AG19
VCCINT_AG17
VCCINT_AG15
VCCINT_AF18
VCCINT_AF16
VCCINT_AE25
VCCINT_AE23
VCCINT_AE21
VCCINT_AE19
VCCINT_AE15
VCCINT_AD26
VCCINT_AD24
VCCINT_AD22
VCCINT_AD20
VCCINT_AD16
VCCINT_AC25
VCCINT_AC23
VCCINT_AF26
VCCINT_AF24
VCCINT_AF22
VCCINT_AF20
VCCINT_AC21
VCCINT_AC19
VCCINT_AC15
VCCINT_AB26
VCCINT_AB24
VCCINT_AB22
VCCINT_AB20
VCCINT_AB16
VCCINT_AB14
VCCINT_AA25
VCCINT_AA19
VCCINT_AA17
VCCINT_AA15

VCCINT_FPGA



U1

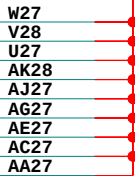
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCINT_IO
XCVU9PFLGA2104

VCCINT_IO_W27
VCCINT_IO_V28
VCCINT_IO_U27
VCCINT_IO_AK28
VCCINT_IO_AJ27
VCCINT_IO_AG27
VCCINT_IO_AE27
VCCINT_IO_AC27
VCCINT_IO_AA27

VCCINTIO_BRAM_FPGA



U1

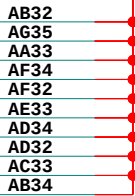
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCAUX
XCVU9PFLGA2104

VCCAUX_AB32
VCCAUX_AG35
VCCAUX_AA33
VCCAUX_AF34
VCCAUX_AF32
VCCAUX_AE33
VCCAUX_AD34
VCCAUX_AD32
VCCAUX_AC33
VCCAUX_AB34

VCC1V8_FPGA



U1

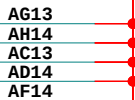
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCBRAM
XCVU9PFLGA2104

VCCBRAM_AG13
VCCBRAM_AH14
VCCBRAM_AC13
VCCBRAM_AD14
VCCBRAM_AF14

VCCINTIO_BRAM_FPGA



U1

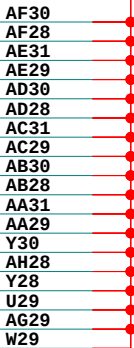
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCAUX_IO
XCVU9PFLGA2104

VCCAUX_IO_AF30
VCCAUX_IO_AF28
VCCAUX_IO_AE31
VCCAUX_IO_AE29
VCCAUX_IO_AD30
VCCAUX_IO_AD28
VCCAUX_IO_AC31
VCCAUX_IO_AC29
VCCAUX_IO_AB30
VCCAUX_IO_AB28
VCCAUX_IO_AA31
VCCAUX_IO_AA29
VCCAUX_IO_Y30
VCCAUX_IO_AH28
VCCAUX_IO_Y28
VCCAUX_IO_U29
VCCAUX_IO_AG29
VCCAUX_IO_W29

VCC1V8_FPGA



U1

SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCADC
XCVU9PFLGA2104

VCCADC_AB18

AB18

VCC1V8_FPGA

FERRITE-600

L13

FPGA_SYSMON_VCC

U1

SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK GNDADC
XCVU9PFLGA2104

GNDADC_AB17

AB17

FERRITE-600

L87

SYSMON_AGND

GND

U1

SOC_2104_1MM_IRON

FPGA Power 1



TITLE: FPGA Power 1
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER: 1.1

SHEET SIZE: B

REV: 02

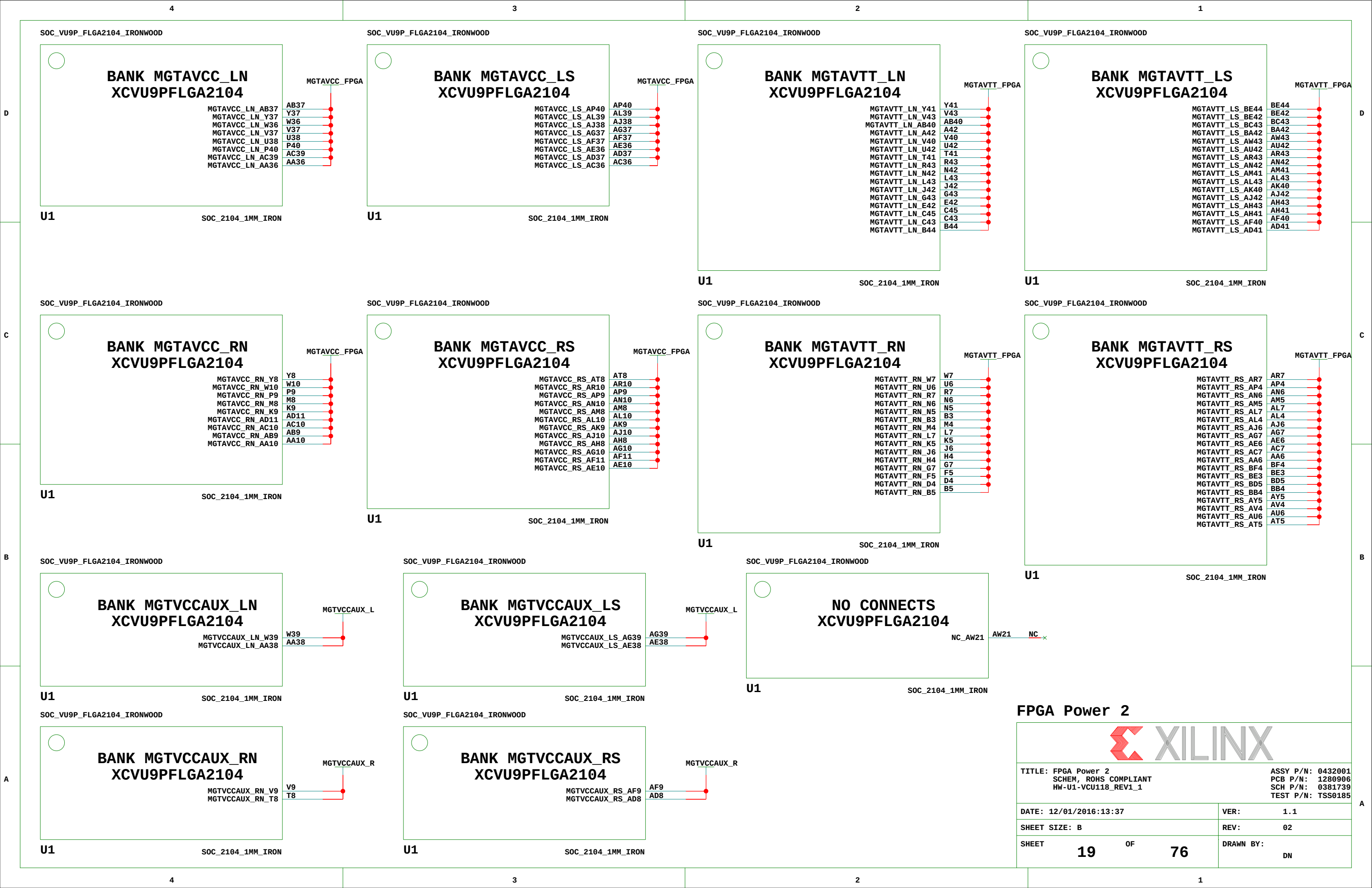
SHEET

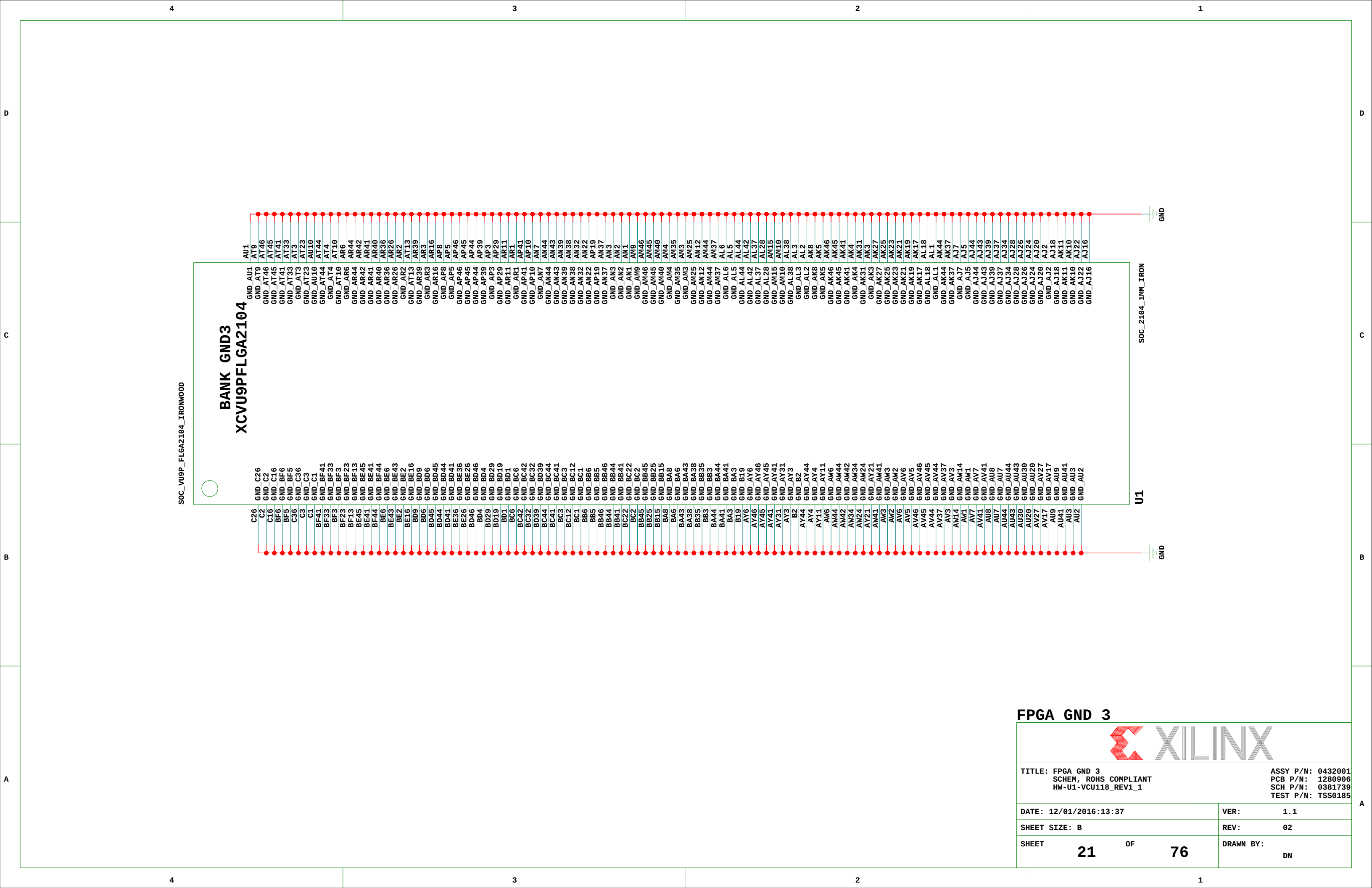
OF

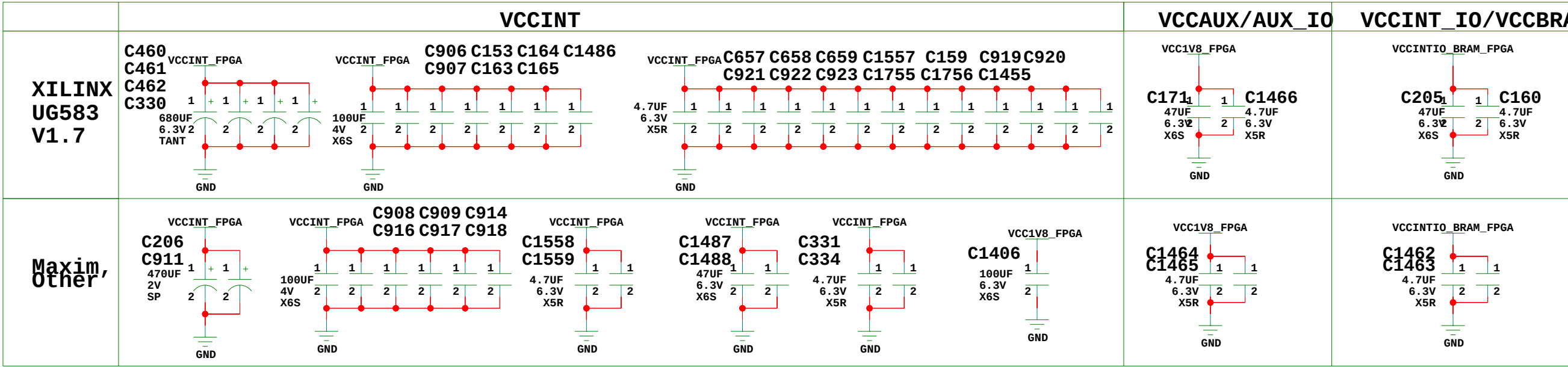
76

DRAWN BY:

DN

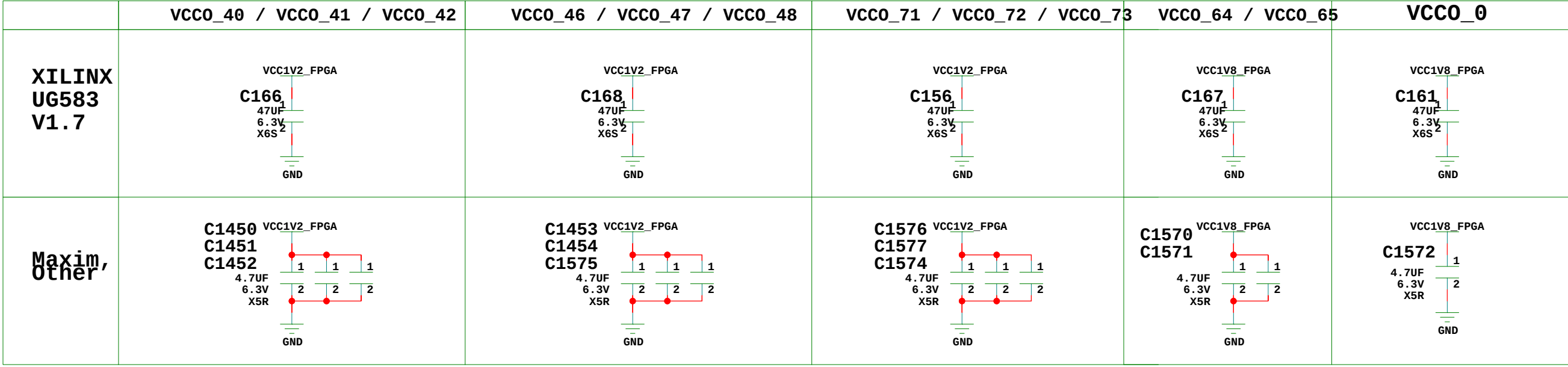




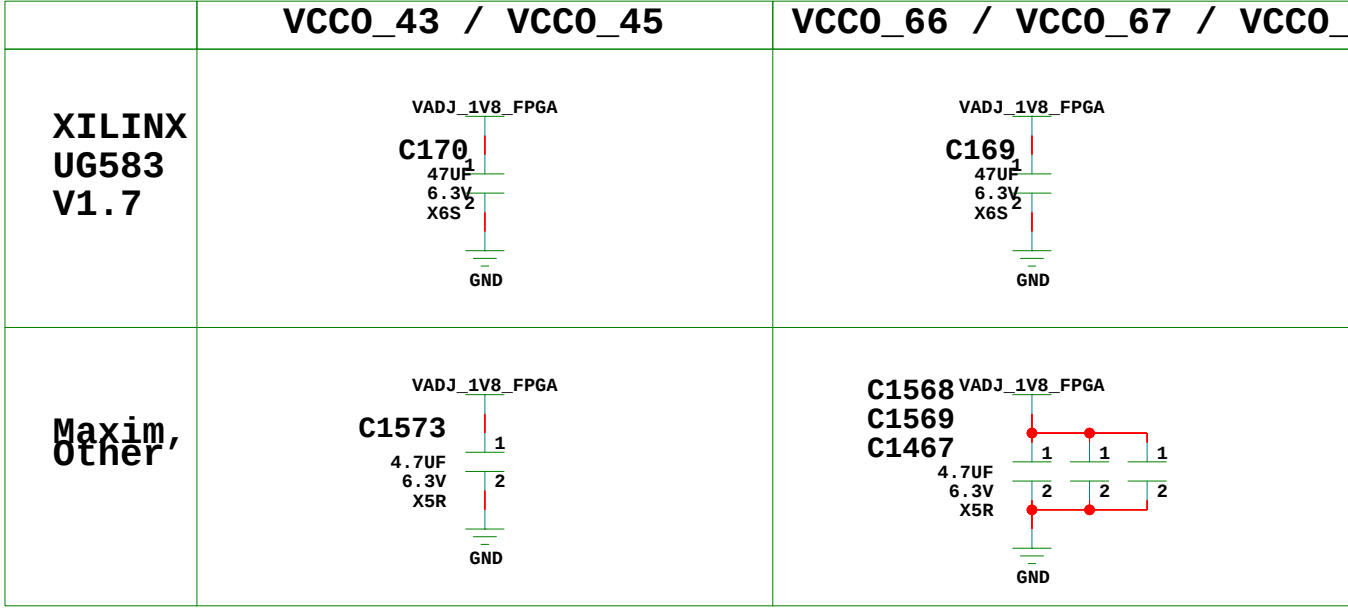


VCCINT_FPGA
VCC1V8_FPGA
VCCINTIO_BRAM_FPGA

VADJ_1V8_FPGA
VCC1V2_FPGA
UTIL_1V35
UTIL_3V3



VCC0_40 1.2V
VCC0_41 1.2V
VCC0_42 1.2V
VCC0_46 1.2V
VCC0_47 1.2V
VCC0_48 1.2V
VCC0_71 1.2V
VCC0_72 1.2V
VCC0_73 1.2V
VCC0_64 1.8V
VCC0_65 1.8V
VCC0_0 1.8V



VCC0_43 1.8V(Vadj)
VCC0_45 1.8V(Vadj)
VCC0_66 1.8V(Vadj)
VCC0_67 1.8V(Vadj)
VCC0_70 1.8V(Vadj)

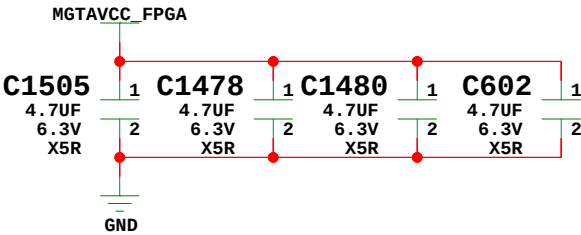
FPGA Decoupling 1

TITLE: FPGA Decoupling 1 SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 22 OF 76	DRAWN BY: DN

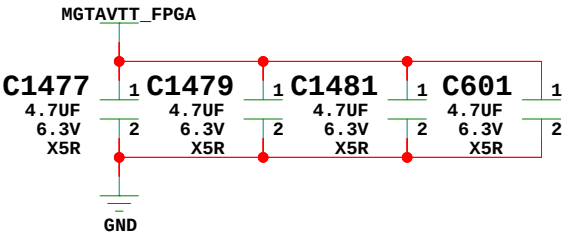
MGTY - 2 PWR. GROUPS LEFT, 2 PWR. GROUPS RIGHT EACH RAIL

XILINX
UG578
V1.0

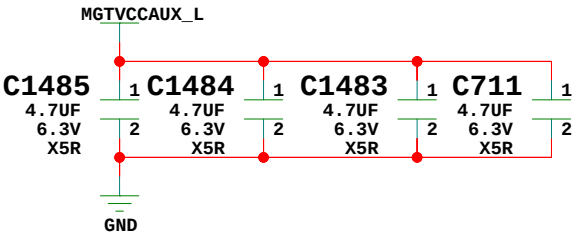
MGTAVCC



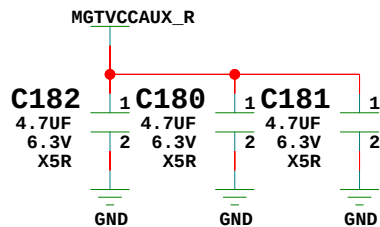
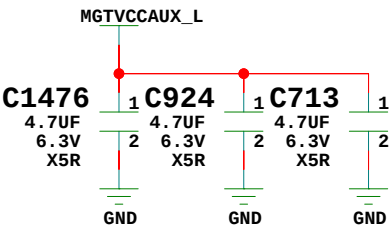
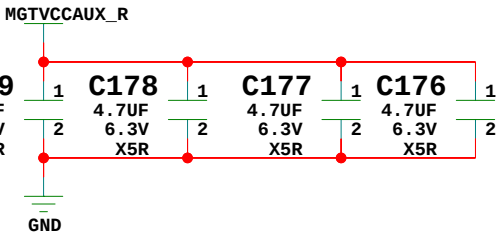
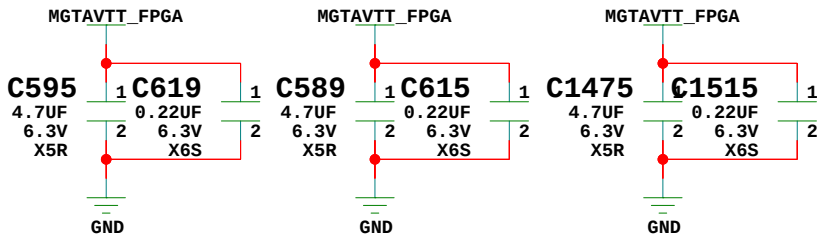
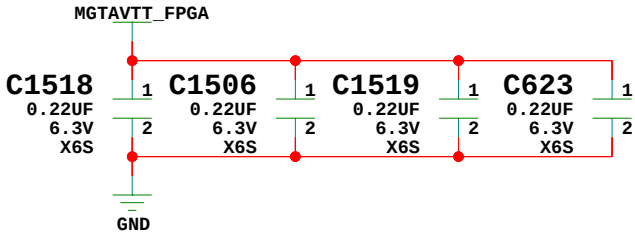
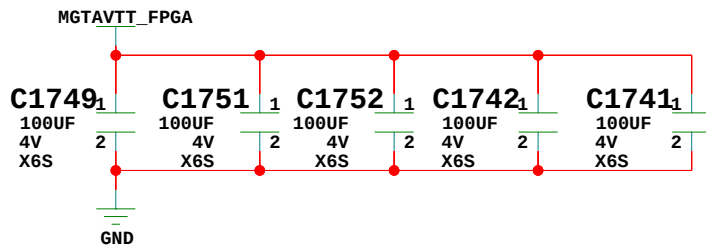
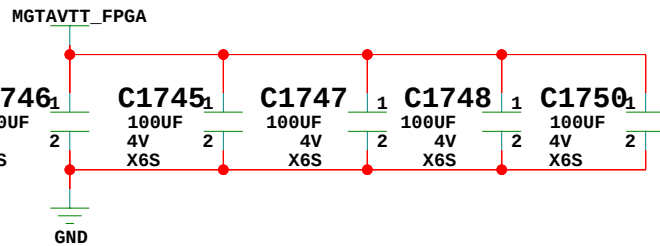
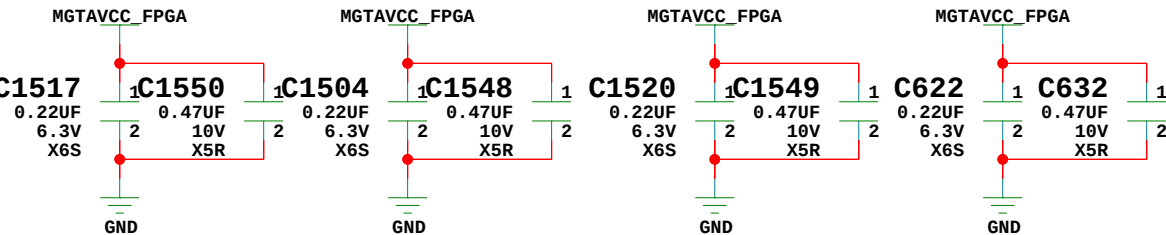
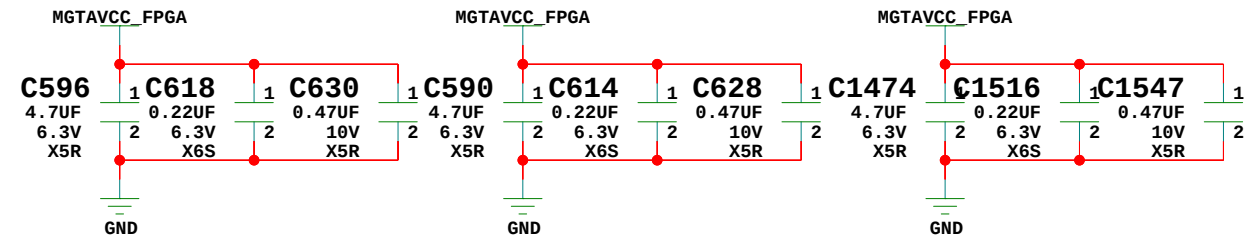
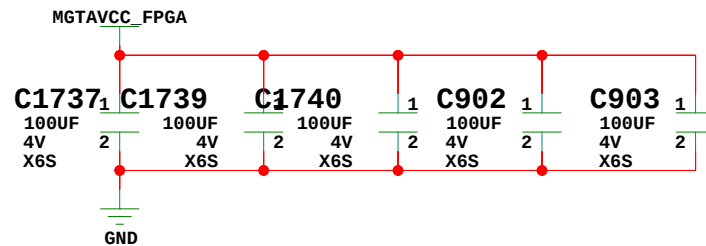
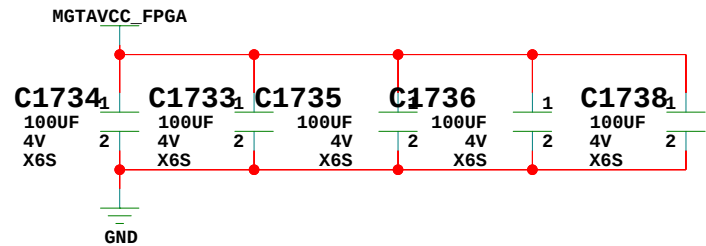
MGTAVTT



MGTVCCAUX



Maxim,
Other



FPGA Decoupling 2



TITLE: FPGA Decoupling 2
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER: 1.1

SHEET SIZE: B

REV: 02

SHEET

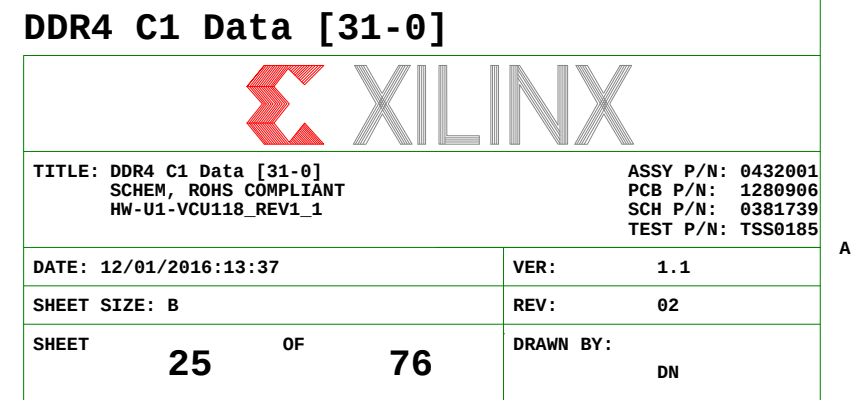
23

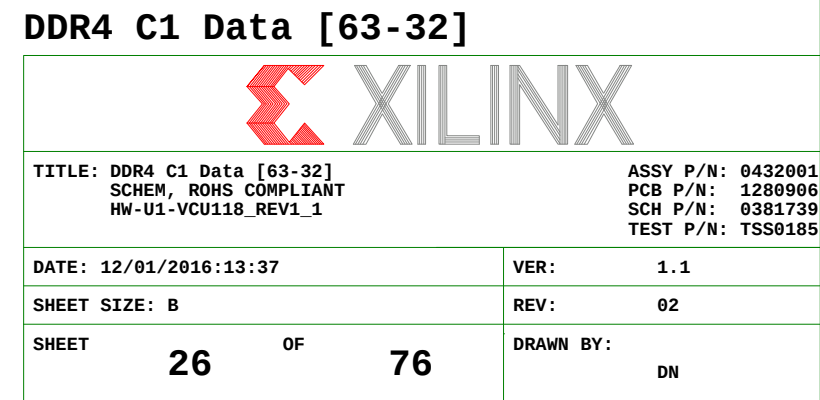
OF

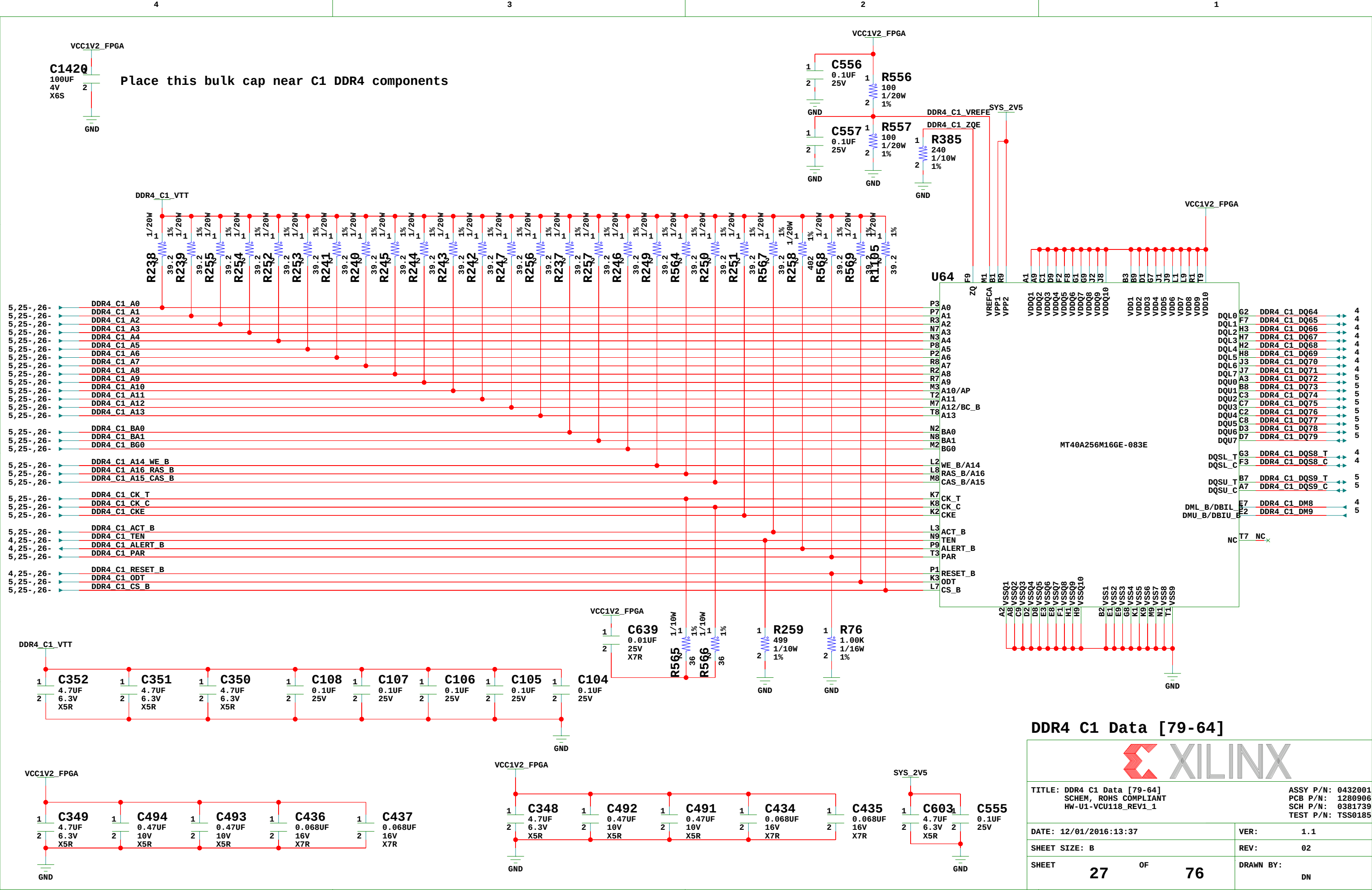
76

DRAWN BY:

DN

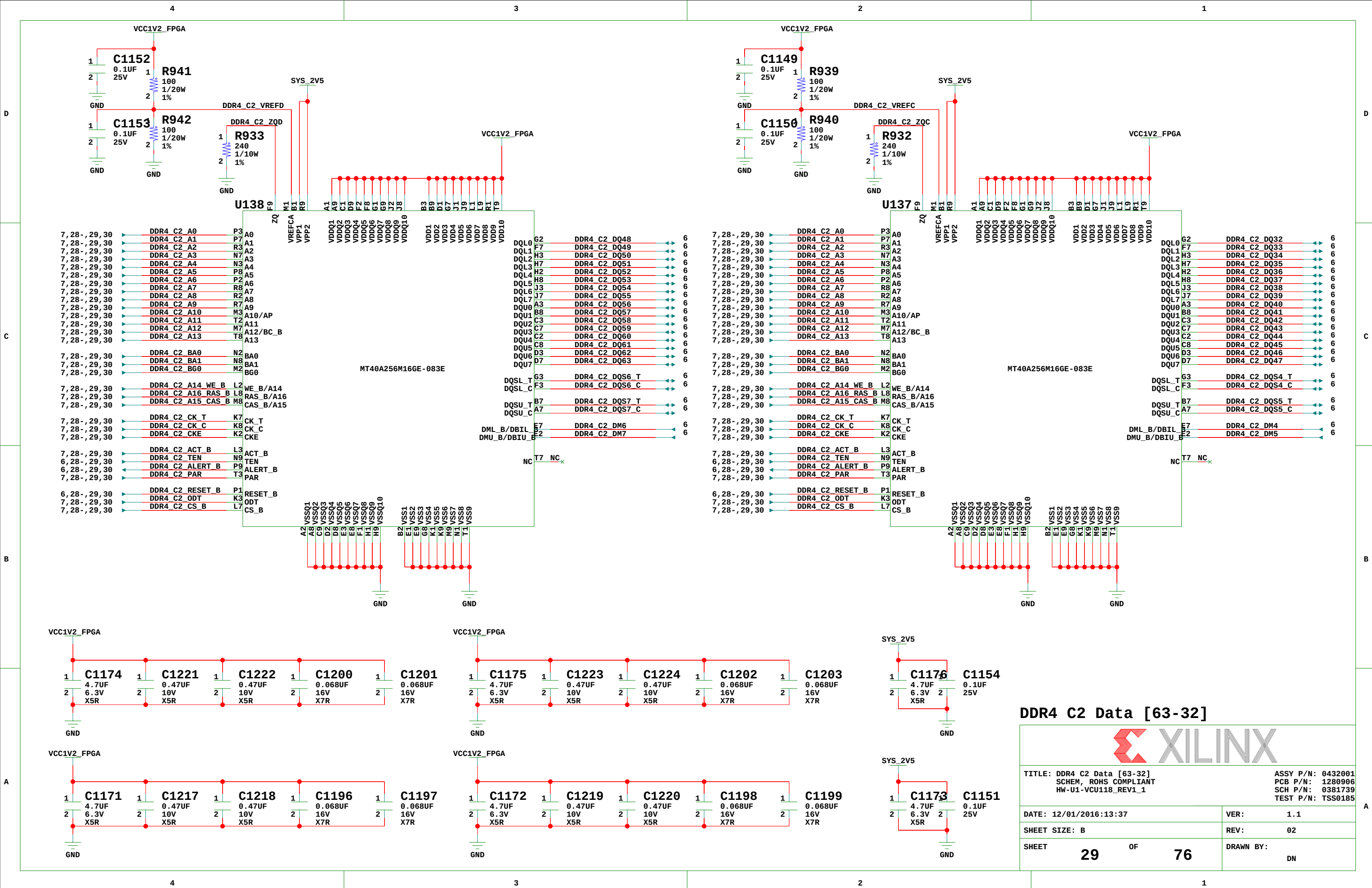


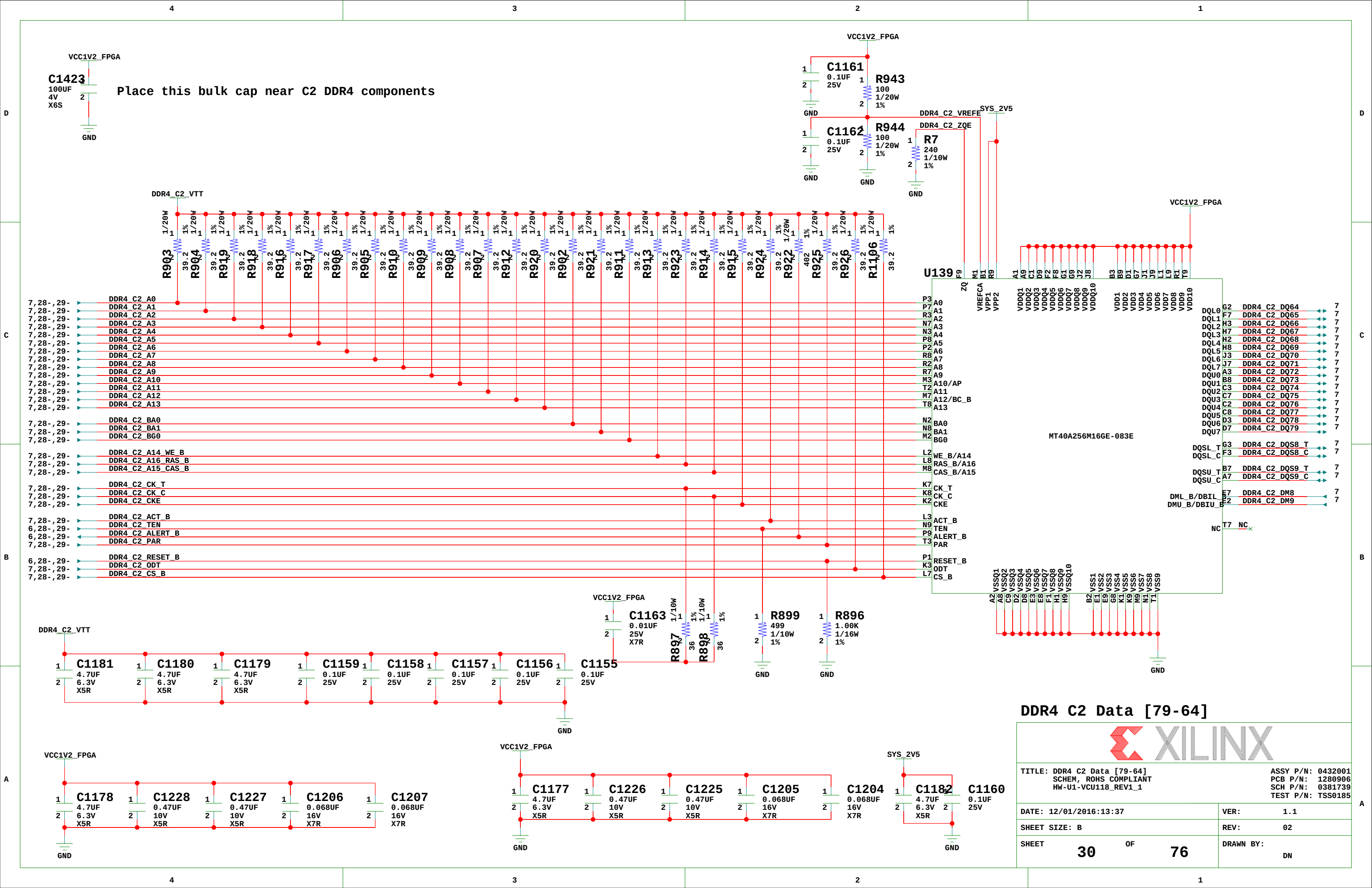


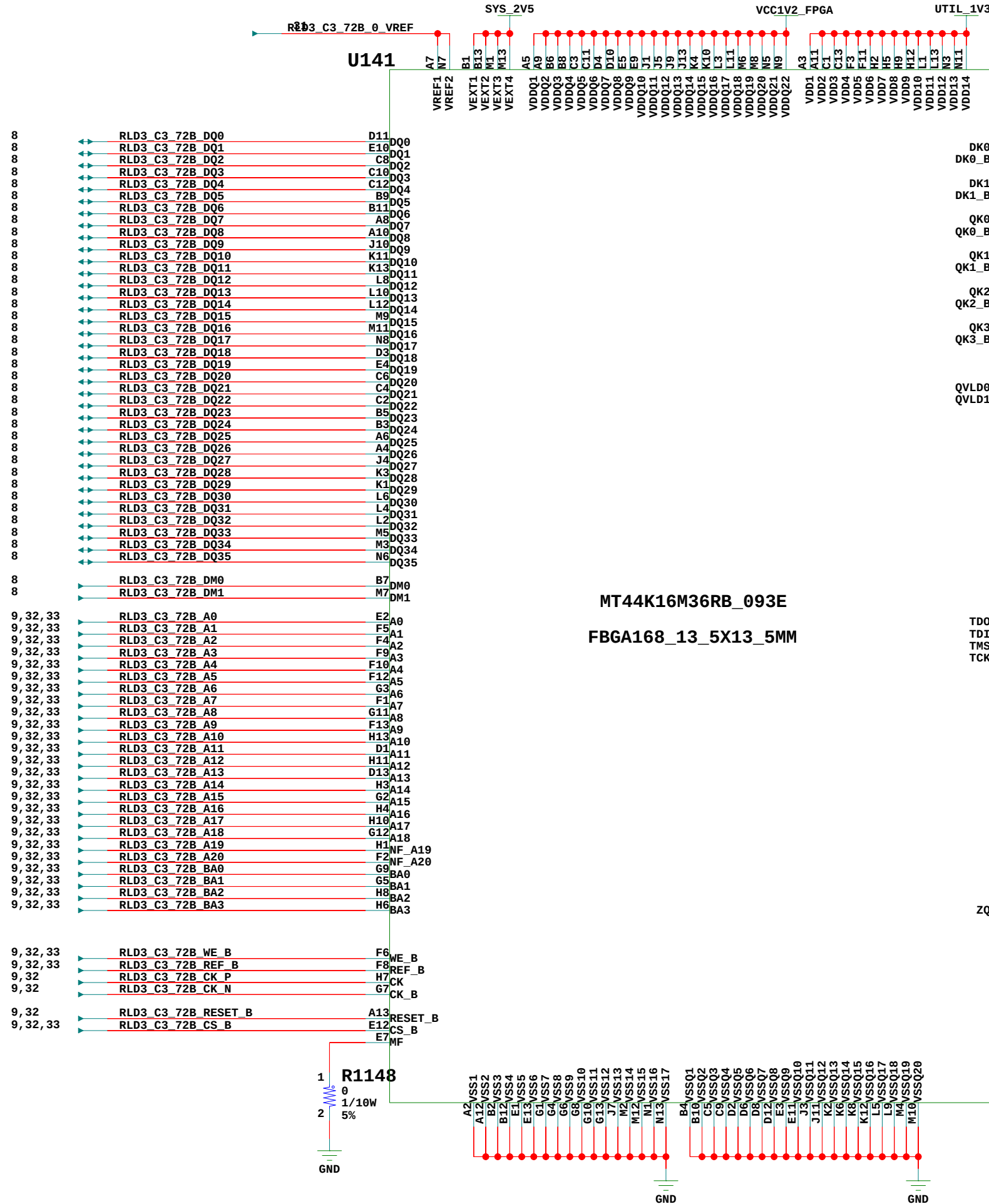


DDR4 C1 Data [79-64]

TITLE: DDR4 C1 Data [79-64] SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 27 OF 76	DRAWN BY: DN

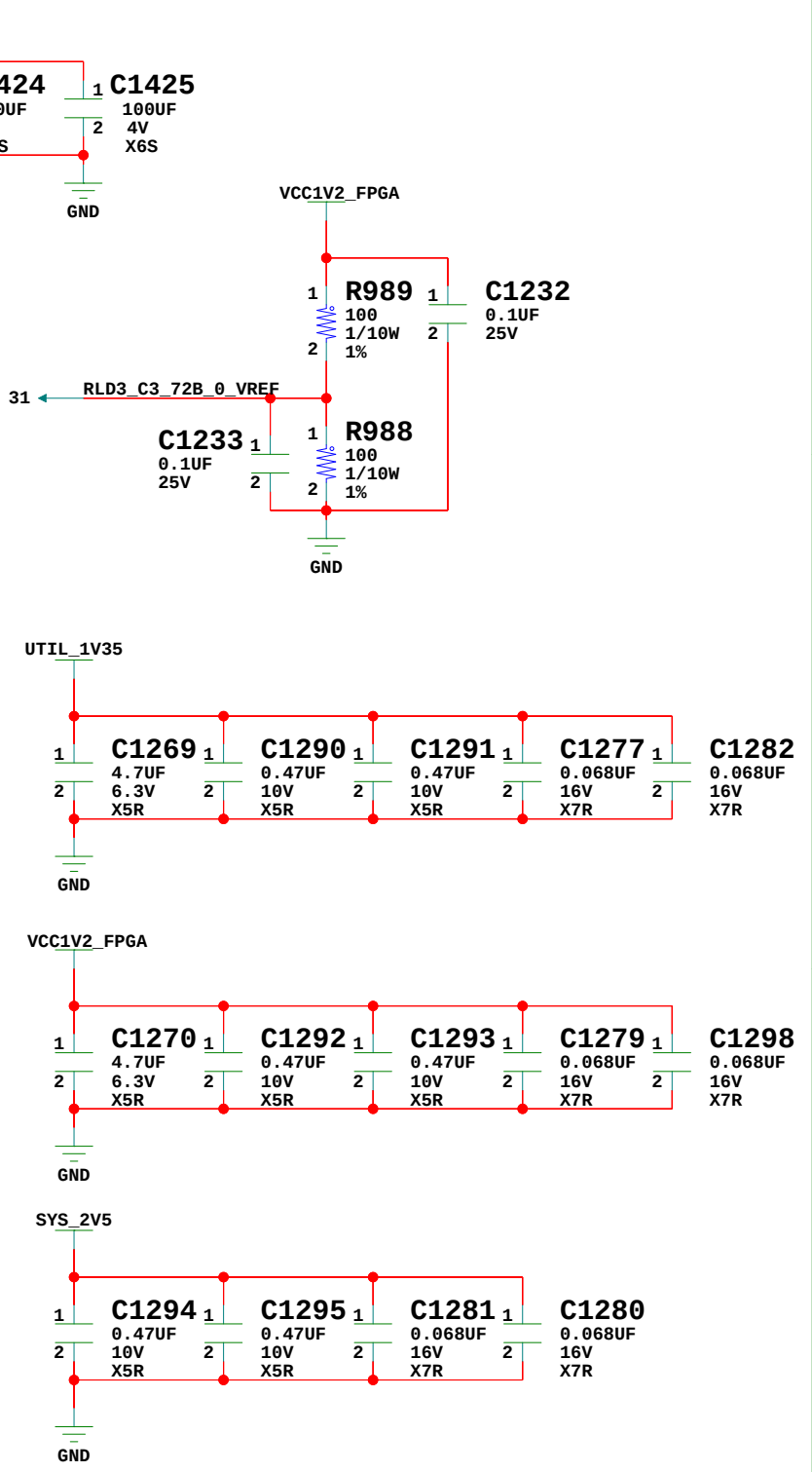
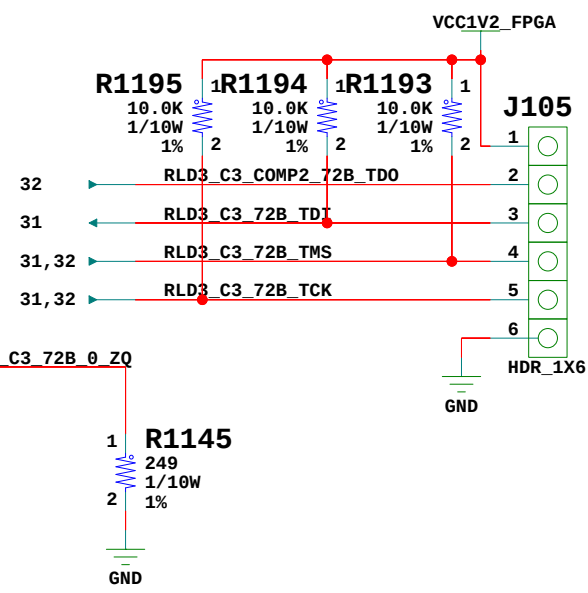
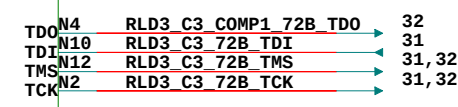
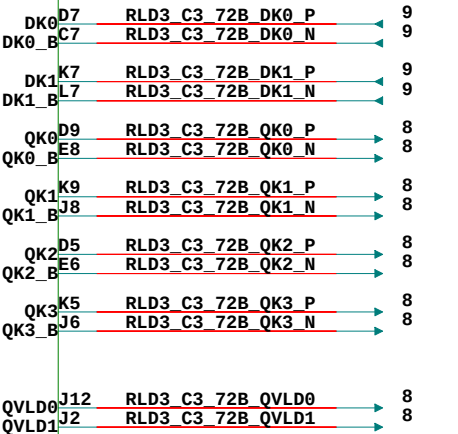




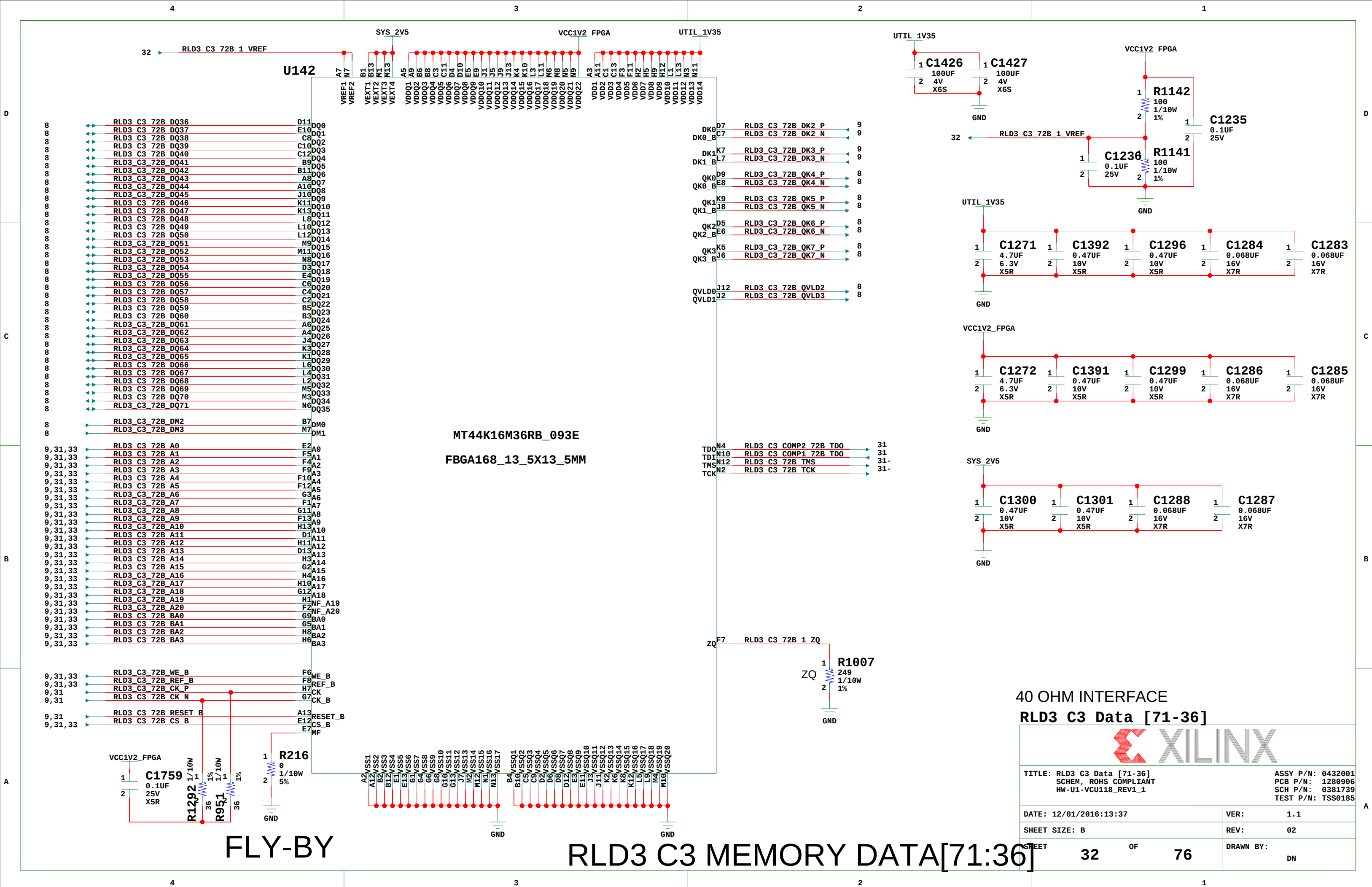


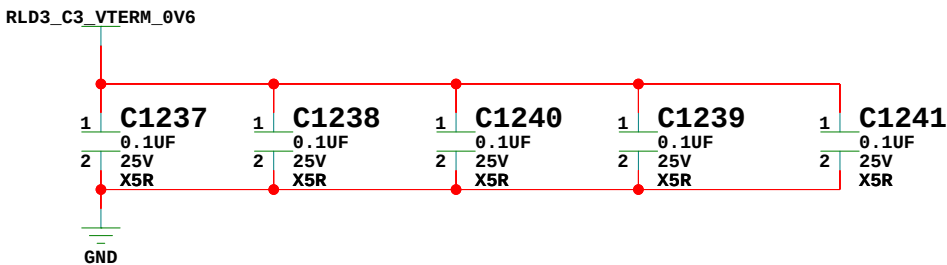
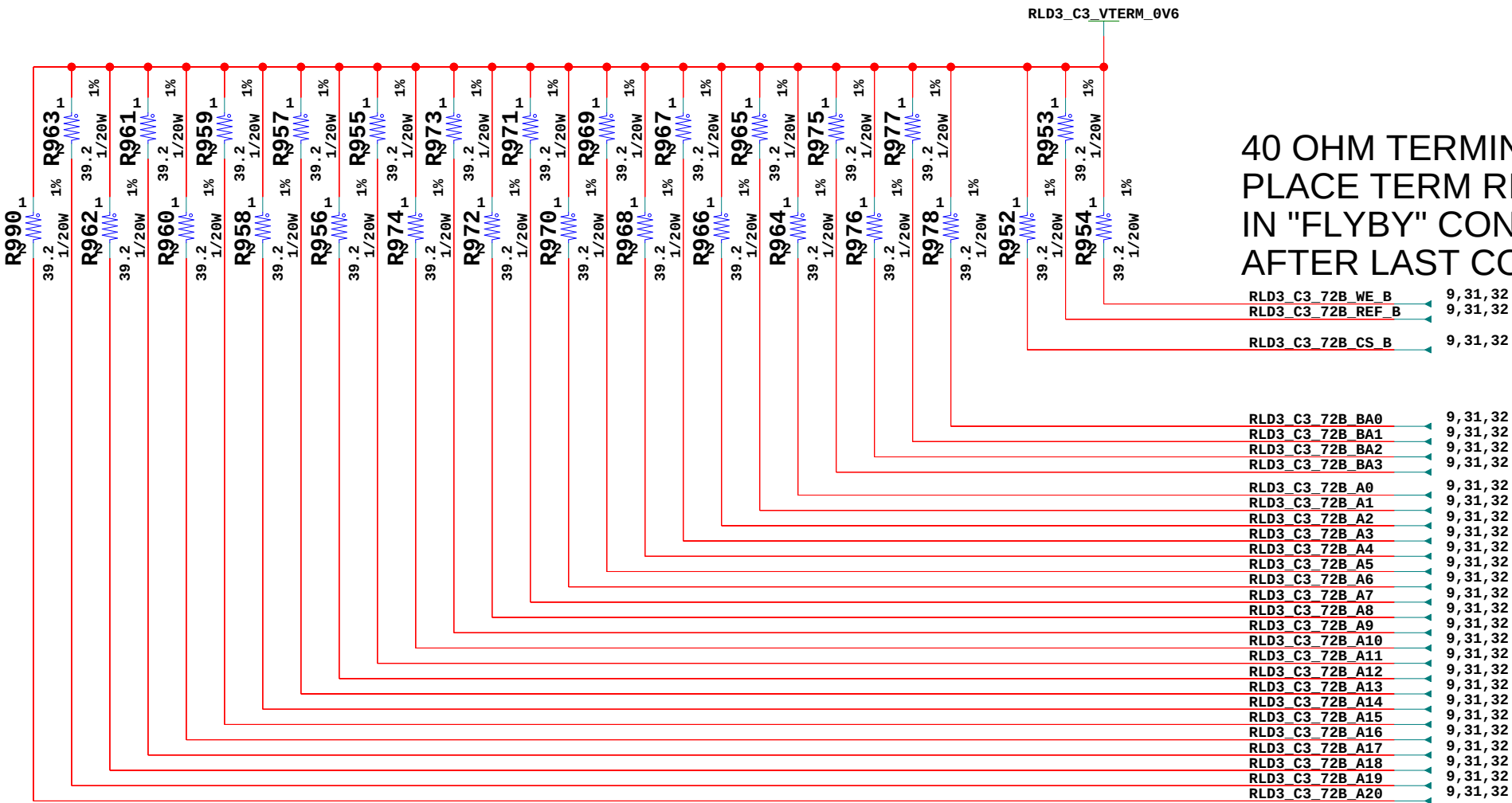
MT44K16M36RB_093E
FBGA168_13_5X13_5MM

RLD3 C3 MEMORY DATA[35:0]

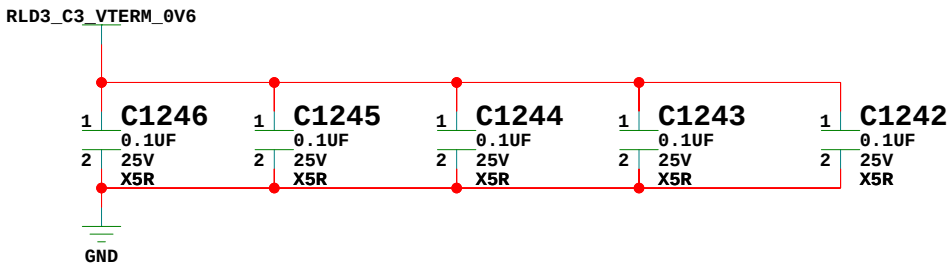


40 OHM INTERFACE			
RLD3 C3 Data [35-0]			
		TITLE: RLD3 C3 Data [35-0] SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
DATE: 12/01/2016:13:37		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
SHEET SIZE: B		VER: 1.1	
SHEET 31 OF 76		REV: 02	
		DRAWN BY: DN	



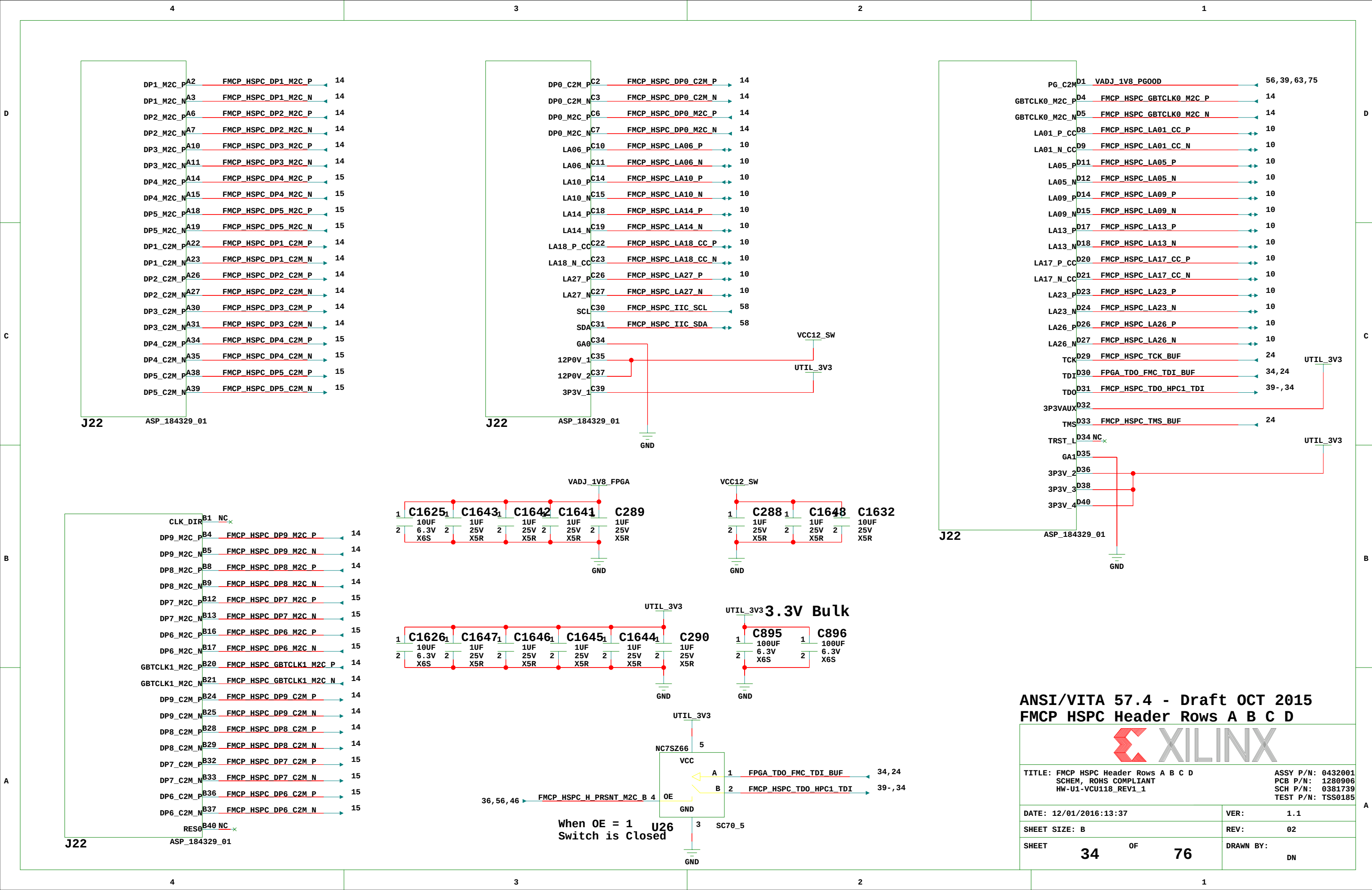


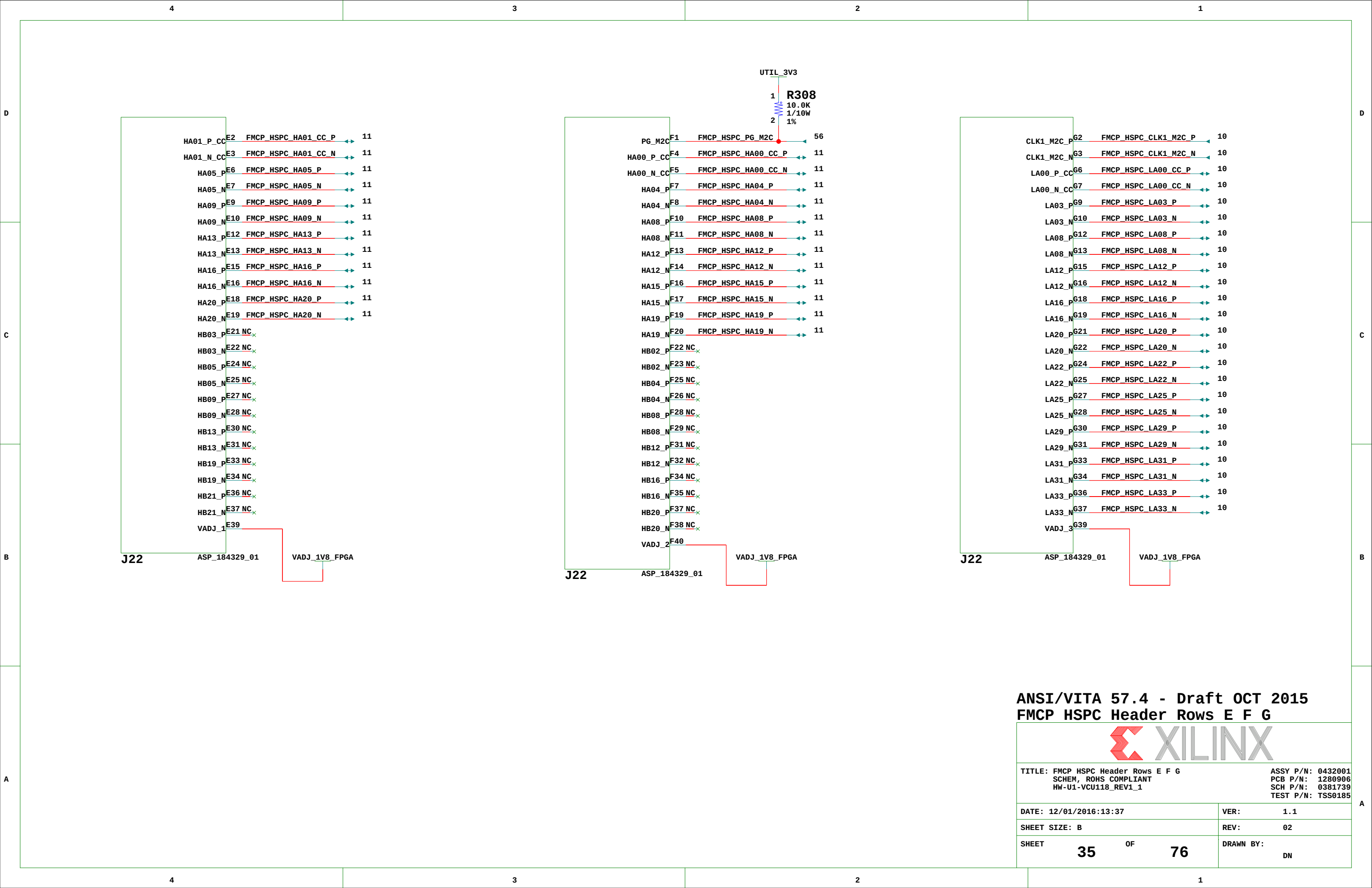
RLD3 C3 72-BIT RTERM DECOUPLING



40 OHM INTERFACE
RLD3 C3 Termination/Decoupling

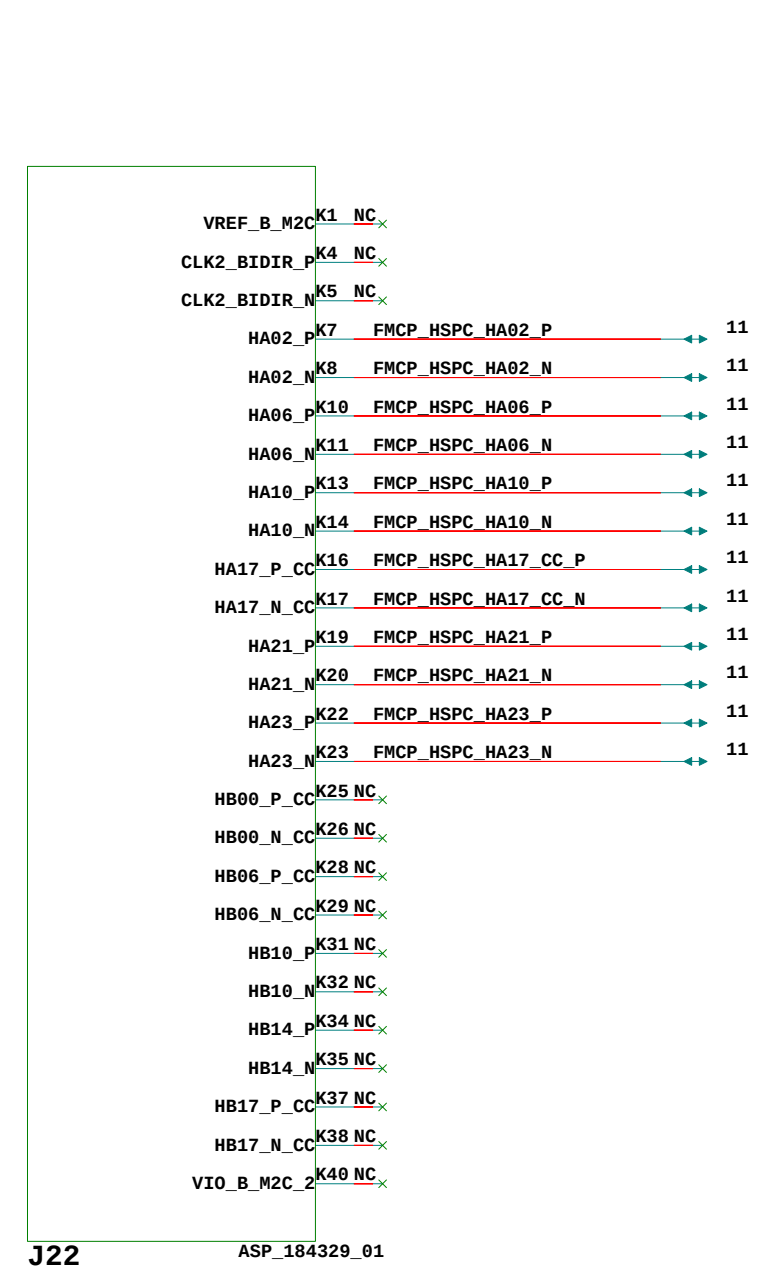
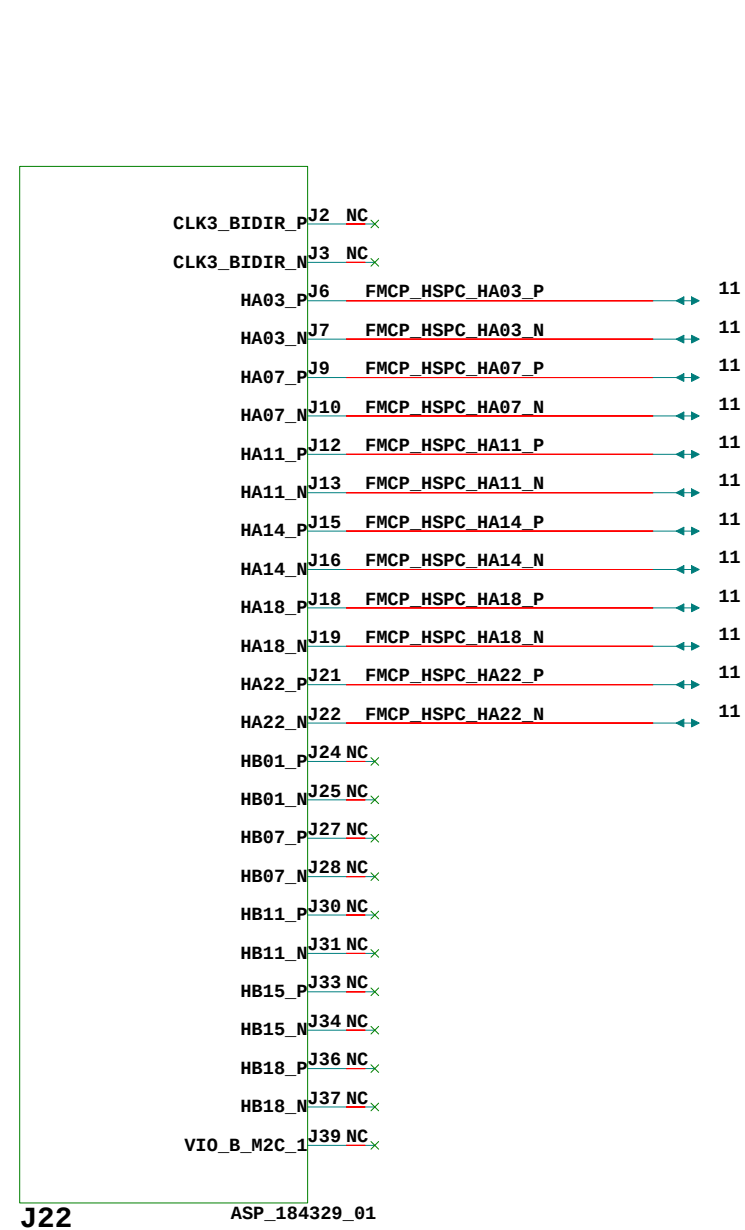
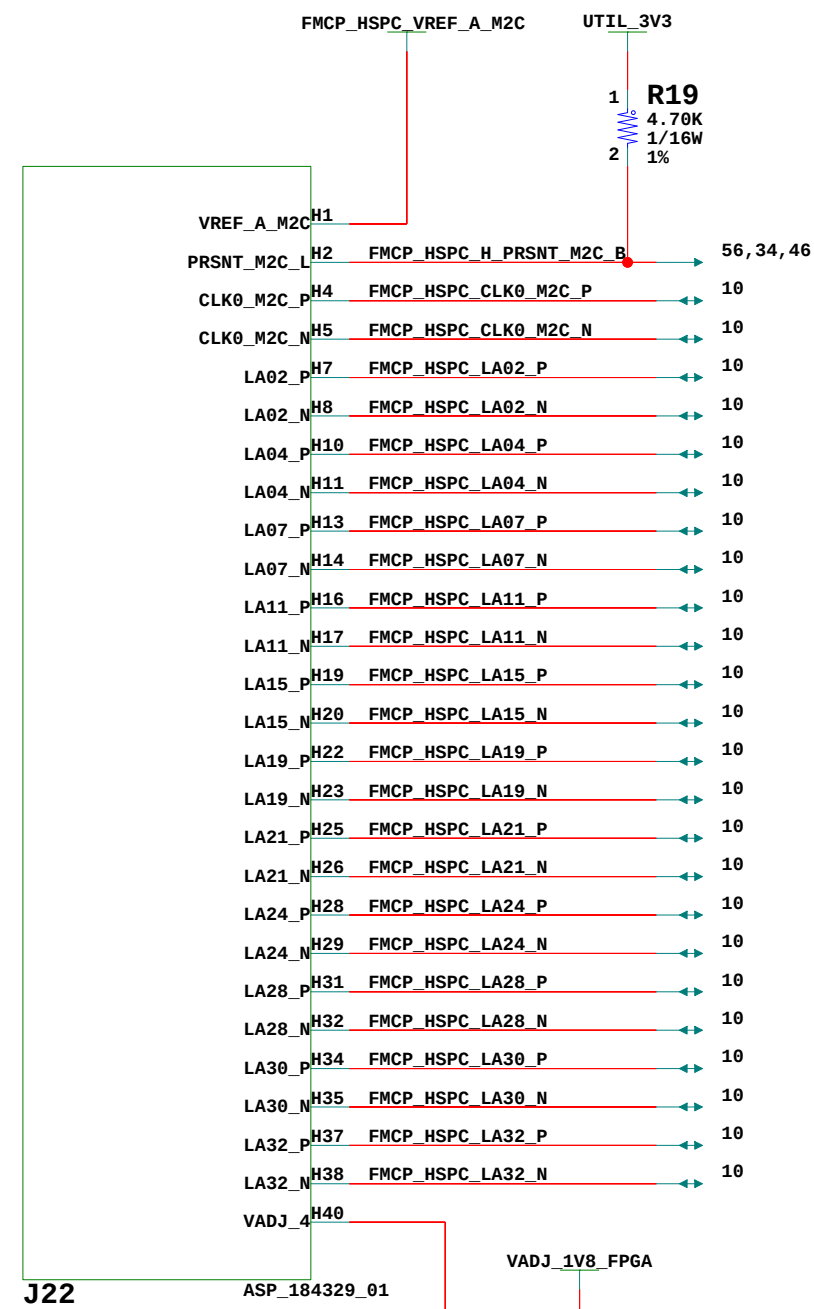
	
TITLE: RLD3 C3 Termination/Decoupling SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 33 OF 76	DRAWN BY: DN





ANSI/VITA 57.4 - Draft OCT 2015
FMCP HSPC Header Rows E F G

	
TITLE: FMCP HSPC Header Rows E F G SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 35 OF 76	DRAWN BY: DN



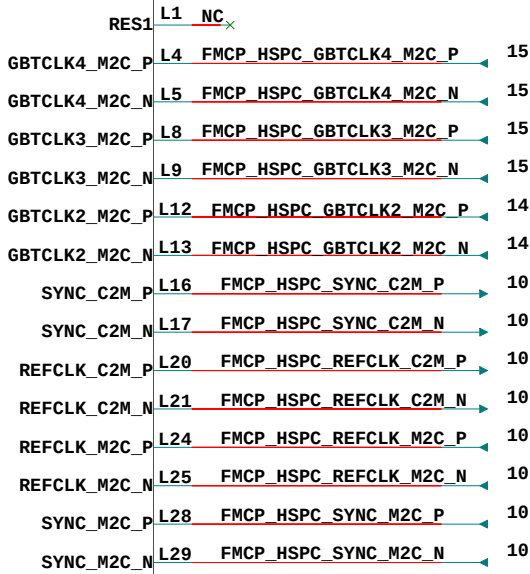
ANSI/VITA 57.4 - Draft OCT 2015
FMCP HSPC Header Rows H J K



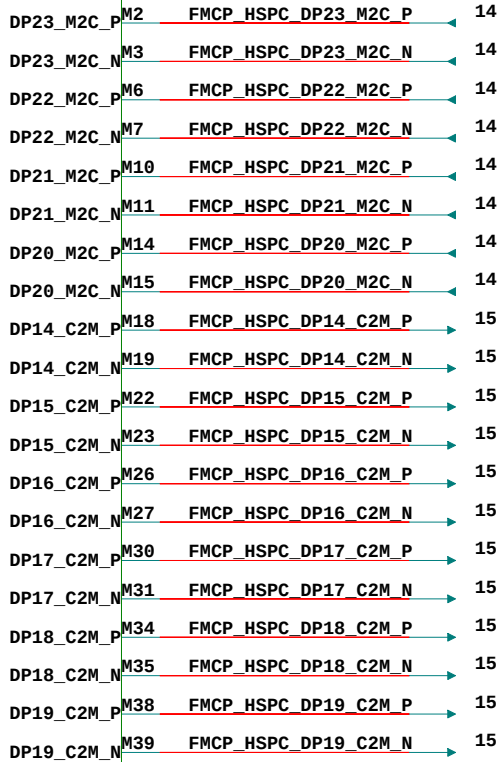
TITLE: FMCP HSPC Header Rows H J K
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

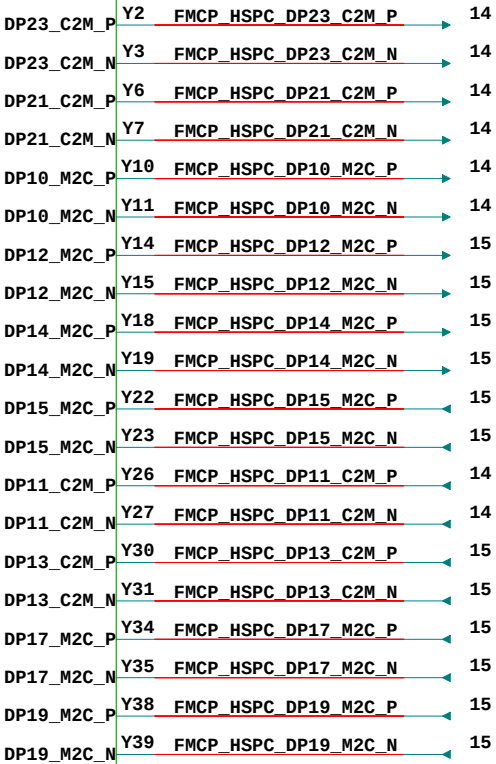
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 36 OF 76	DRAWN BY: DN



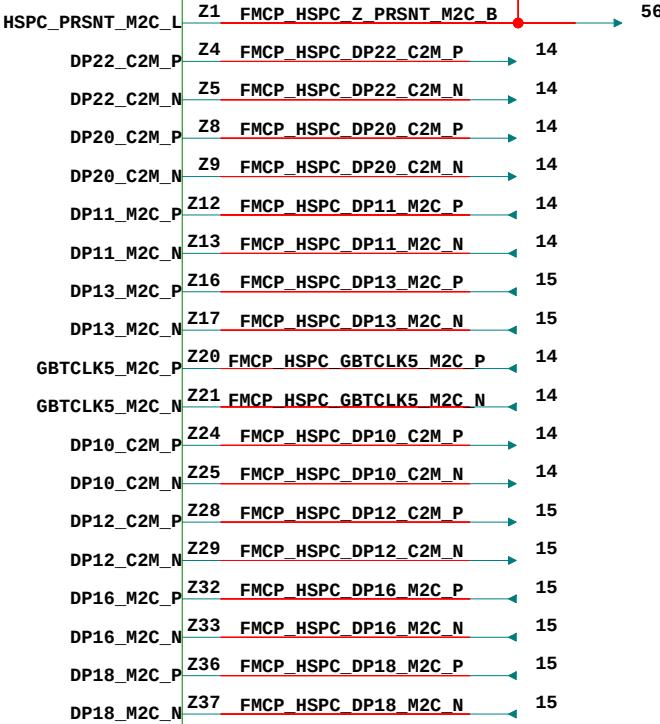
J22 ASP_184329_01



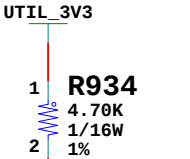
J22 ASP_184329_01



J22 ASP_184329_01



J22 ASP_184329_01



ANSI/VITA 57.4 - Draft OCT 2015
FMCP HSPC Header Rows L M Y Z

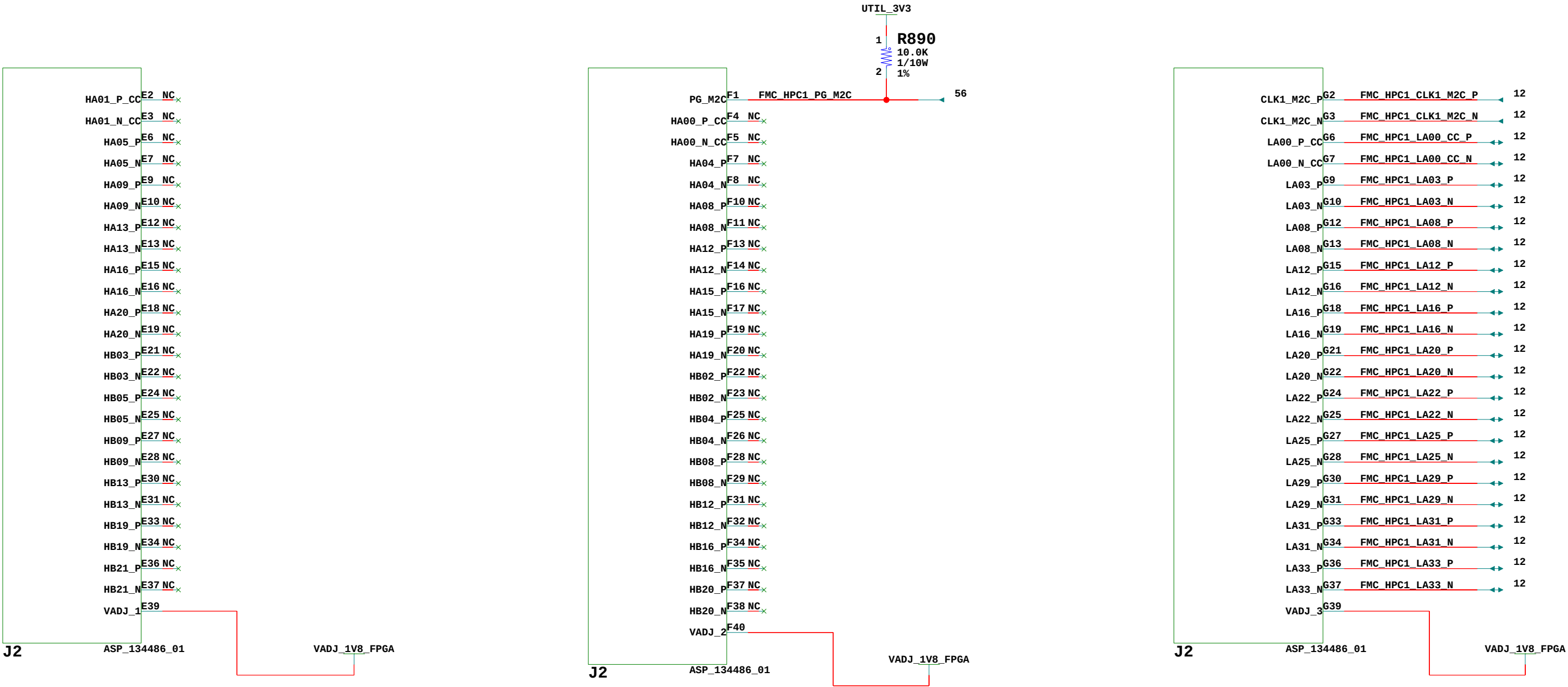


TITLE: FMCP HSPC Header Rows L M Y Z
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

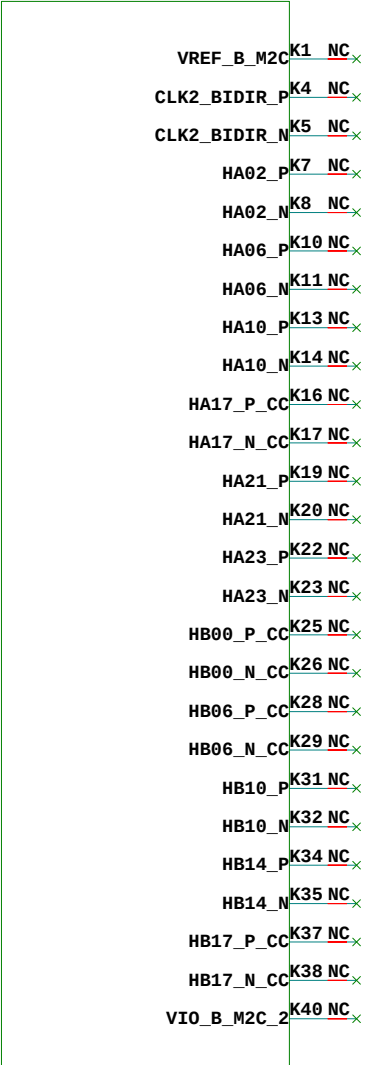
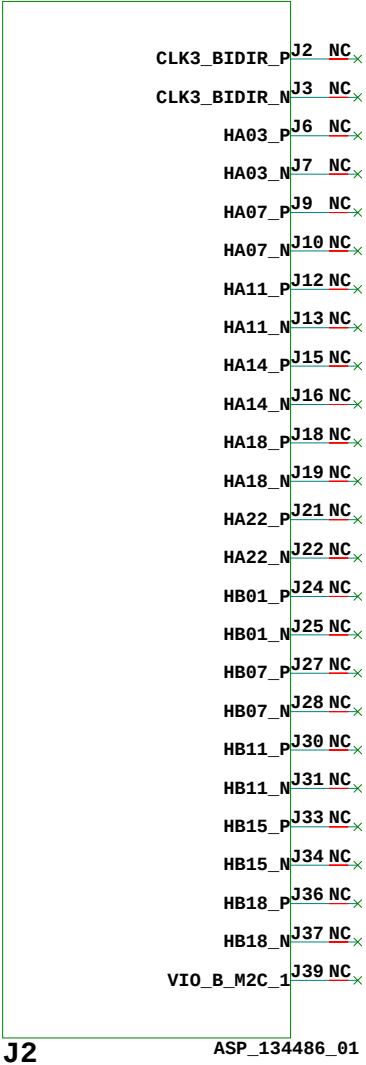
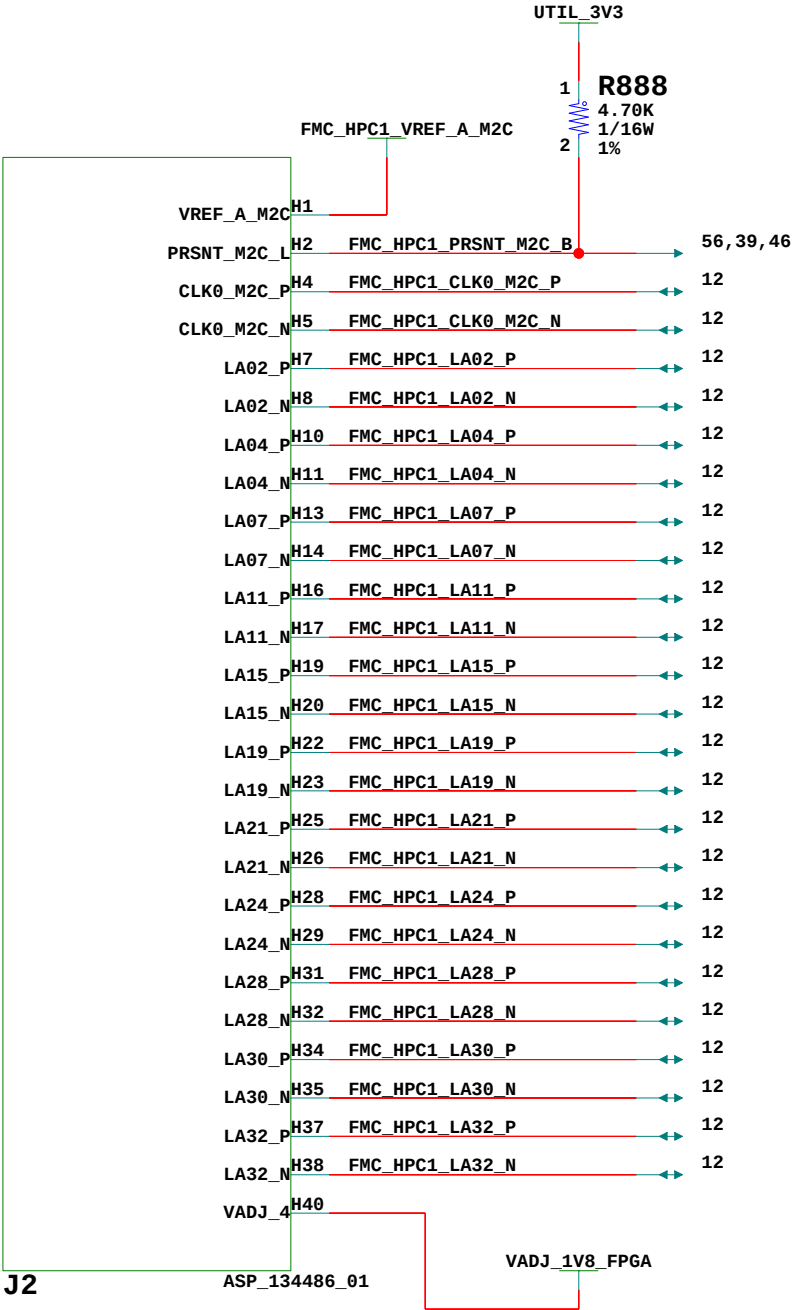
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 37 OF 76	DRAWN BY: DN





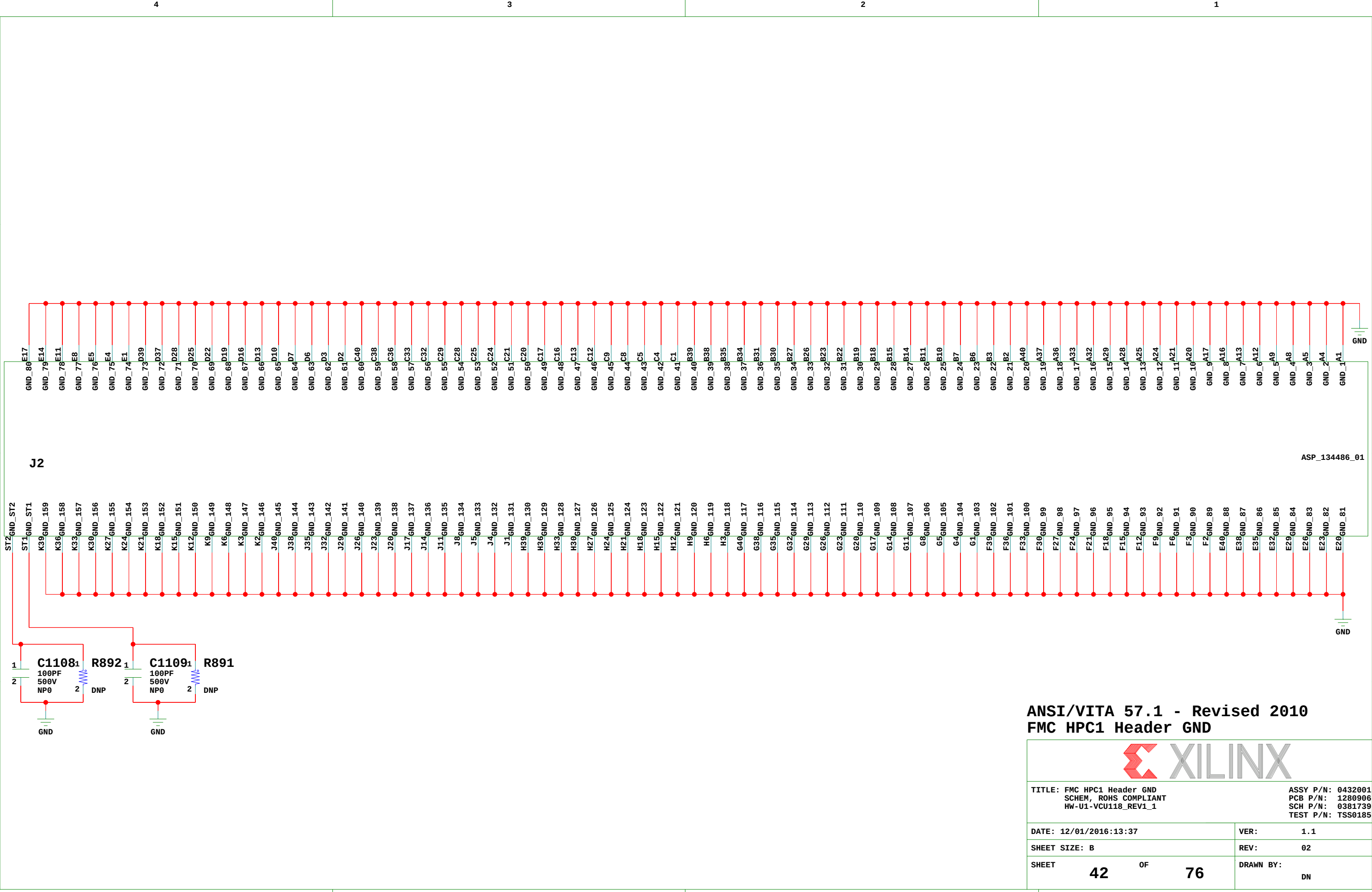
ANSI/VITA 57.1 - Revised 2010
FMC HPC1 Header Rows E F G

TITLE: FMC HPC1 Header Rows E F G SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 40 OF 76	DRAWN BY: DN



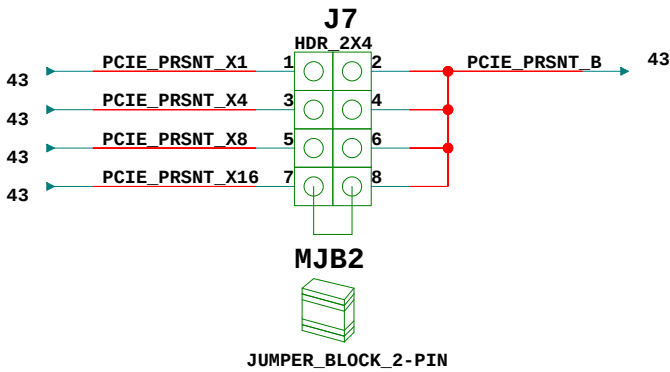
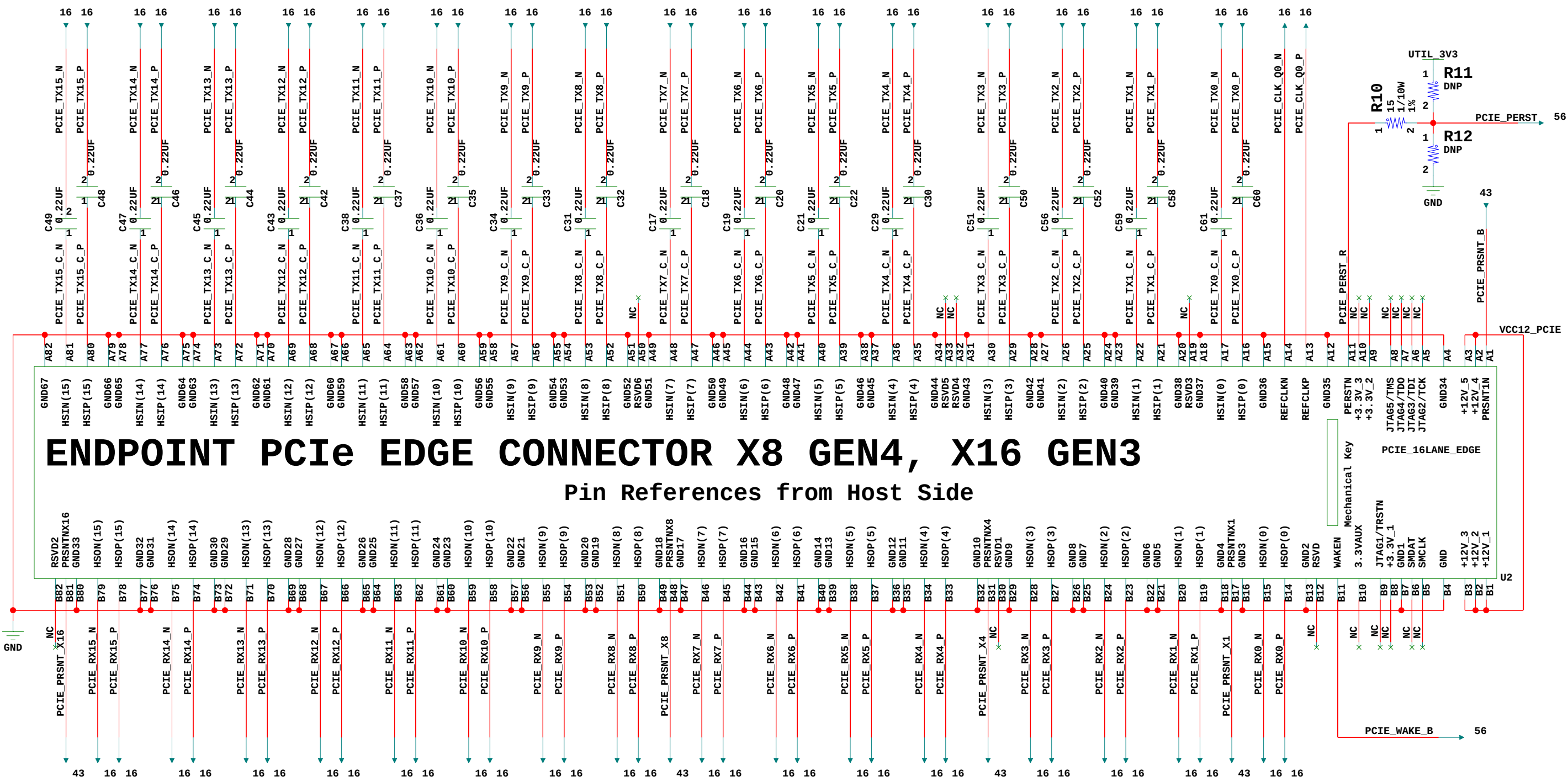
ANSI/VITA 57.1 - Revised 2010
FMC HPC1 Header Rows H J K

TITLE: FMC HPC1 Header Rows H J K SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 41 OF 76	DRAWN BY: DN



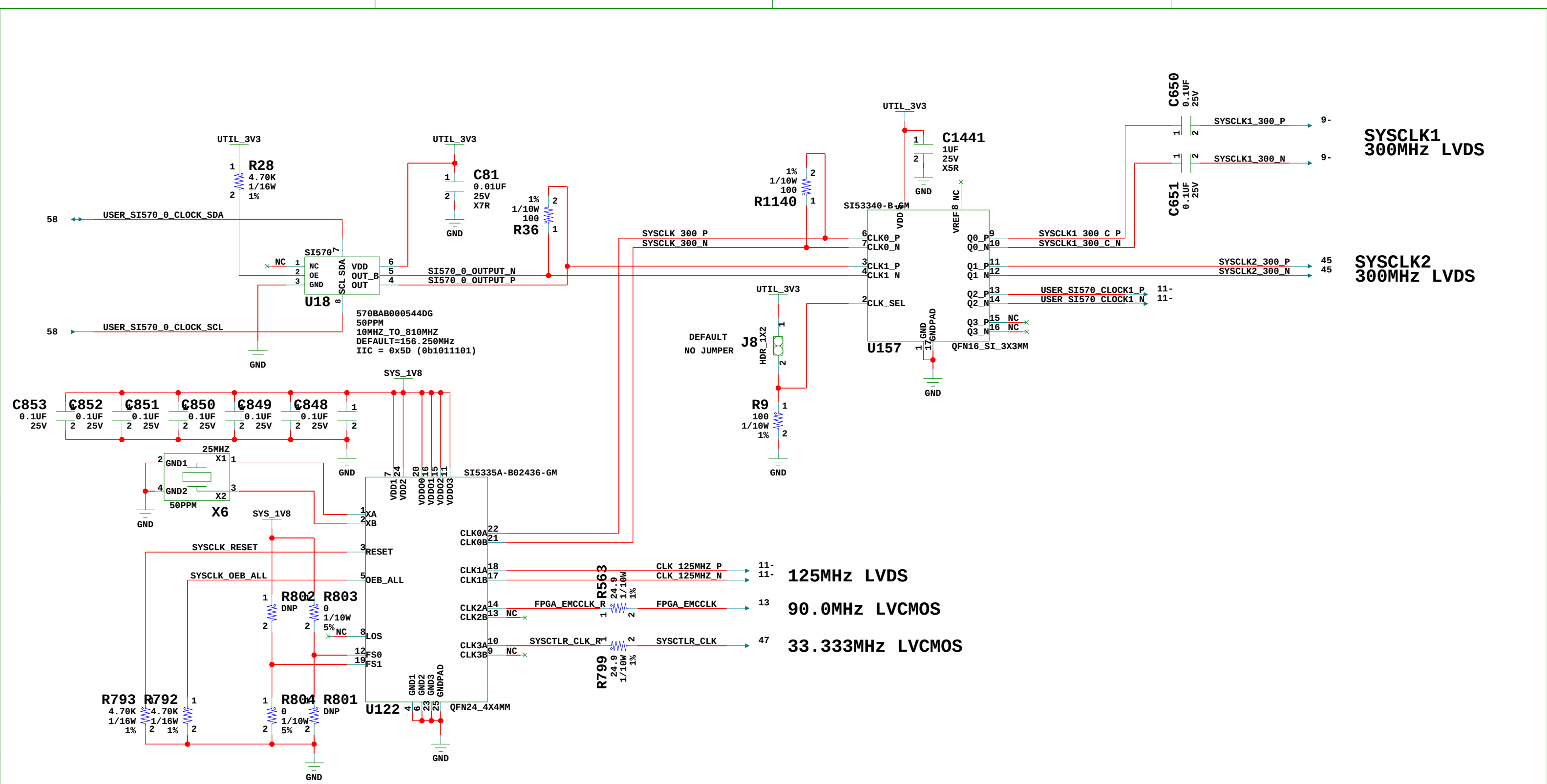
ENDPOINT PCIe EDGE CONNECTOR X8 GEN4, X16 GEN3

Pin References from Host Side



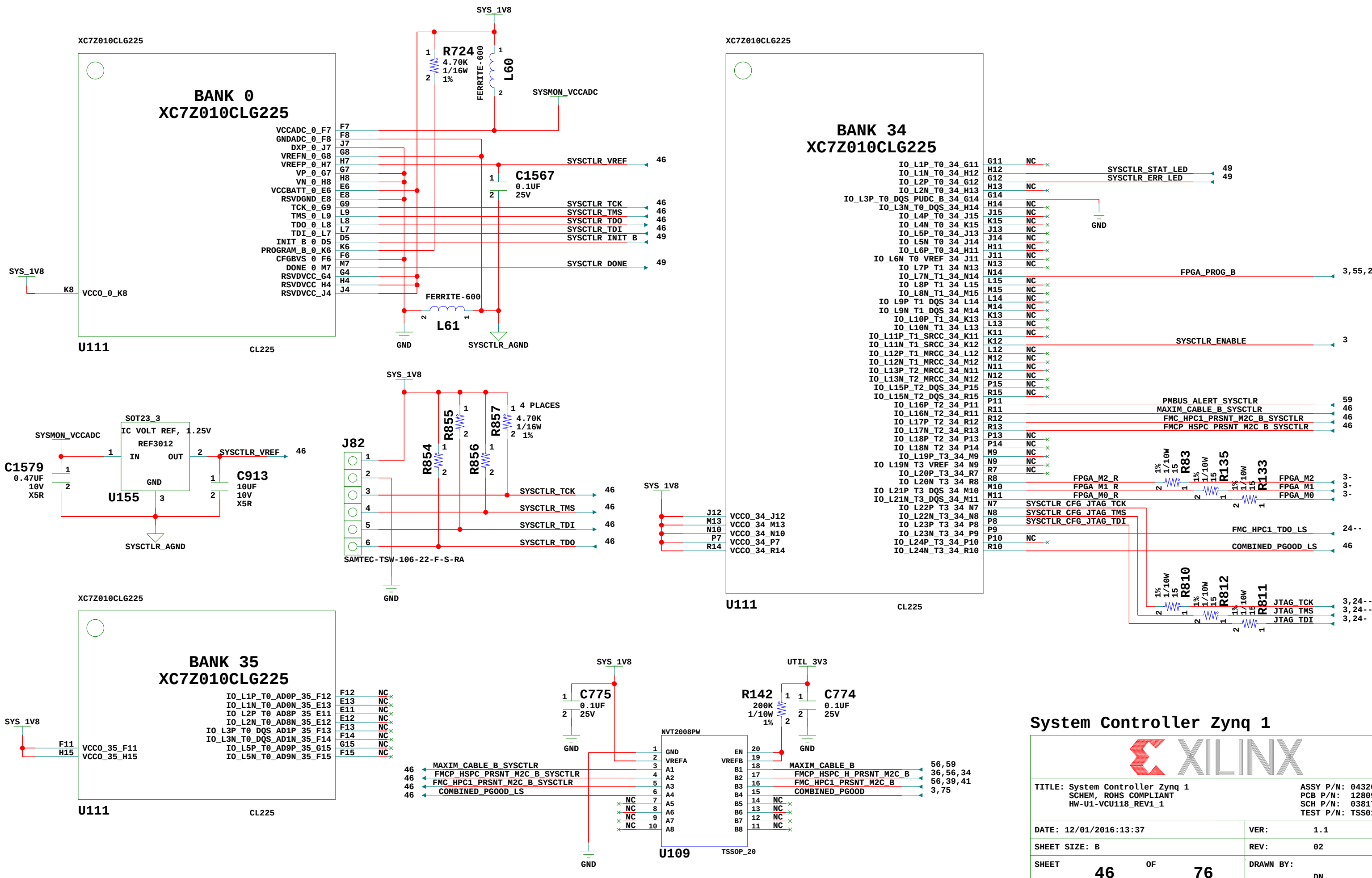
PCIe x16 Endpoint Edge Connector

TITLE: PCIe x16 Endpoint Edge Connector SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 43 OF 76	DRAWN BY: DN



EMCCLK and System Clocks

TITLE: EMCCLK and System Clocks SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 44 OF 76	DRAWN BY: DN



System Controller Zynq 1

TITLE: System Controller Zynq 1 SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 46 OF 76	DRAWN BY: DN

XC7Z010CLG225

BANK 500
XC7Z010CLG225

PS_CLK_500_C7
PS_POR_B_500_C9
PS_MIO15_500_D10
PS_MIO14_500_B9
PS_MIO13_500_C6
PS_MIO12_500_B7
PS_MIO11_500_B10
PS_MIO10_500_D6
PS_MIO9_500_B5
PS_MIO8_500_B6
PS_MIO7_500_D9
PS_MIO6_500_A10
PS_MIO5_500_A9
PS_MIO4_500_C8
PS_MIO3_500_A7
PS_MIO2_500_A8
PS_MIO1_500_A5
PS_MIO0_500_D8

U111

CL225

SYS_1V8

R1197 20K 1/10W 1%
R1196 20K 1/10W 1%
R776 20K 1/10W 1%

SYSCTLR_CLK 44
SYSCTLR_POR_B 59, 49
SDIO_DATA3 49
SDIO_DATA2 49
SDIO_DATA1 49
SDIO_CLK 49
SDIO_CMD 49
SDIO_DATA0 49

SYSCTLR_QSPI_SCLK 49
SYSCTLR_QSPI_IO3 49
SYSCTLR_QSPI_IO2 49
SYSCTLR_QSPI_IO1 49
SYSCTLR_QSPI_IO0 49
SYSCTLR_QSPI_SS_B 49
SDIO_CD 49

R1124 4.70K 1/16W 1%
J110 HDR_1X2
R775 20K 1/10W 1%
R772 20K 1/10W 1%
R773 20K 1/10W 1%
R774 20K 1/10W 1%
GND

XC7Z010CLG225

BANK 501
XC7Z010CLG225

PS_MIO29_501_D11
PS_MIO31_501_C12
PS_MIO33_501_C11
PS_MIO35_501_A14
PS_MIO38_501_A13
PS_MIO48_501_B12
PS_MIO52_501_A12
PS_SRST_B_501_B11
PS_MIO28_501_A15
PS_MIO30_501_B15
PS_MIO32_501_E15
PS_MIO34_501_D15
PS_MIO36_501_B14
PS_MIO37_501_C14
PS_MIO39_501_D14
PS_MIO49_501_D13
PS_MIO53_501_C13

U111

CL225

SYS_1V8

R826 4.70K 1/16W 1%

IIC_MAIN_SDA 58, 11
SYSCTLR_UART_TX 53
SYSCTLR_MODE1 49
SYSCTLR_MODE3 49
GPIO_SW_N 55, 11
GPIO_SW_C 55, 11
SYSCTLR_GPIO_6 11
IIC_MAIN_SCL 58, 11
SYSCTLR_UART_RX 53
SYSCTLR_MODE0 49
SYSCTLR_MODE2 49
GPIO_SW_E 55, 11
GPIO_SW_S 55, 11
GPIO_SW_W 55, 11
SYSCTLR_GPIO_5 11
SYSCTLR_GPIO_7 11

XC7Z010CLG225

BANK 502
XC7Z010CLG225

PS_DDR_DRST_B_502_L4 L4 NC
PS_DDR_DQ0_502_D4 D4 NC
PS_DDR_DQ1_502_A2 A2 NC
PS_DDR_DQ2_502_C4 C4 NC
PS_DDR_DQ3_502_C1 B1 NC
PS_DDR_DM0_502_B1 C2 NC
PS_DDR_DQS_P0_502_C2 B2 NC
PS_DDR_DQS_N0_502_B2 B4 NC
PS_DDR_DQ4_502_B4 A4 NC
PS_DDR_DQ5_502_A4 C3 NC
PS_DDR_DQ6_502_C3 A3 NC
PS_DDR_DQ7_502_A3 E1 NC
PS_DDR_DQ8_502_E1 D1 NC
PS_DDR_DQ9_502_D1 E2 NC
PS_DDR_DQ10_502_E2 E3 NC
PS_DDR_DQ11_502_E3 D3 NC
PS_DDR_DM1_502_D3 G2 NC
PS_DDR_DQS_P1_502_G2 F2 NC
PS_DDR_DQS_N1_502_F2 F3 NC
PS_DDR_DQ12_502_F3 G1 NC
PS_DDR_DQ13_502_G1 H1 NC
PS_DDR_DQ14_502_H1 H2 NC
PS_DDR_DQ15_502_H2 K1 NC
PS_DDR_A14_502_K1 K2 NC
PS_DDR_A13_502_K2 M2 NC
PS_DDR_A12_502_M2 L2 NC
PS_DDR_A11_502_L2 J1 NC
PS_DDR_A10_502_J1 N4 NC
PS_DDR_A9_502_N4 P6 NC
PS_DDR_A8_502_P6 M5 NC
PS_DDR_A7_502_M5 P5 NC
PS_DDR_A6_502_P5 P4 NC
PS_DDR_A5_502_P4 P3 NC
PS_DDR_A4_502_P3 M4 NC
PS_DDR_A3_502_M4 J3 NC
PS_DDR_VRN_502_J3 H3 NC
PS_DDR_VRP_502_H3 N3 NC
PS_DDR_CKP_502_N3 N2 NC
PS_DDR_CKN_502_N2 M1 NC
PS_DDR_A2_502_M1 N1 NC
PS_DDR_A1_502_N1 P1 NC
PS_DDR_A0_502_P1 N6 NC
PS_DDR_BA2_502_N6 R1 NC
PS_DDR_BA1_502_R1 M6 NC
PS_DDR_BA0_502_M6 K3 NC
PS_DDR_ODT_502_K3 R2 NC
PS_DDR_CS_B_502_R2 L3 NC
PS_DDR_CKE_502_L3 R3 NC
PS_DDR_WE_B_502_R3 R5 NC
PS_DDR_CAS_B_502_R5 R6 NC
PS_DDR_RAS_B_502_R6 F4 NC
PS_DDR_VREF0_502_F4

U111

CL225

SYS_1V8

B3 VCCO_DDR_502_B3
E4 VCCO_DDR_502_E4
F1 VCCO_DDR_502_F1
J2 VCCO_DDR_502_J2
M3 VCCO_DDR_502_M3
R4 VCCO_DDR_502_R4

System Controller Zynq 2



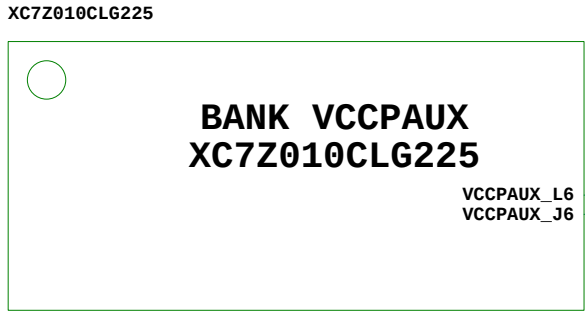
TITLE: System Controller Zynq 2
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 47 OF 76	DRAWN BY: DN



U111

CL225



U111

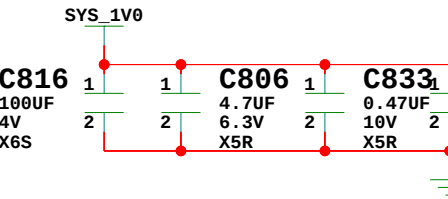
CL225



U111

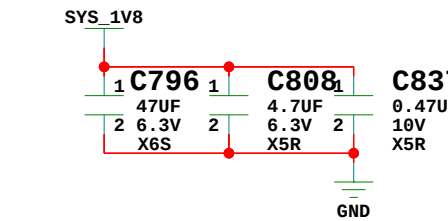
CL225

VCCINT/VCCBRAM

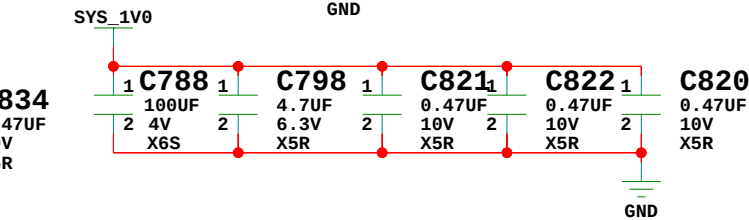


Bank 0

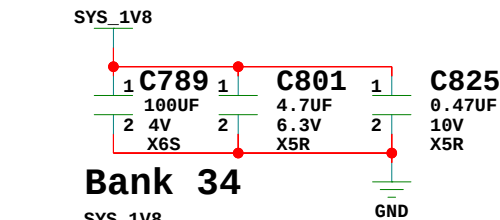
VCCAUX



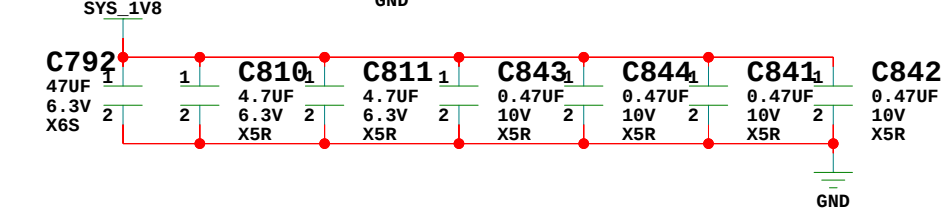
VCCPINT



VCCPAUX

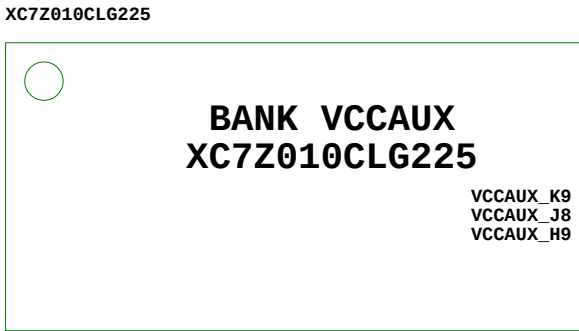


Bank 34



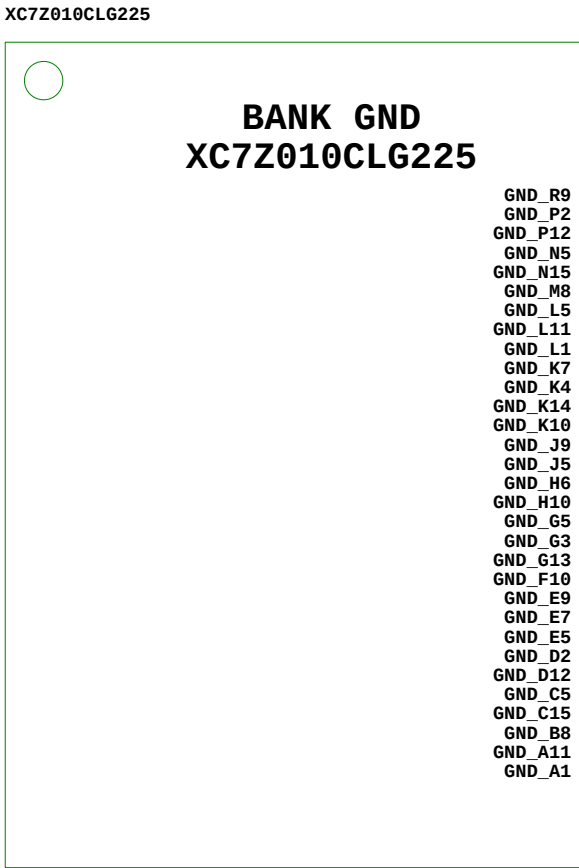
U111

CL225



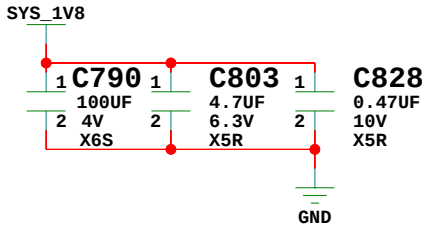
U111

CL225

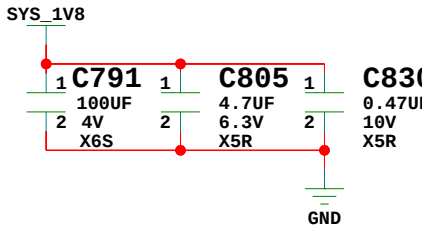


U111

VCCO_MI00_BANK500



VCCO_MI01_BANK501



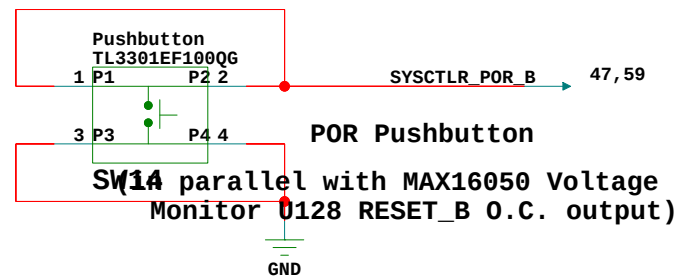
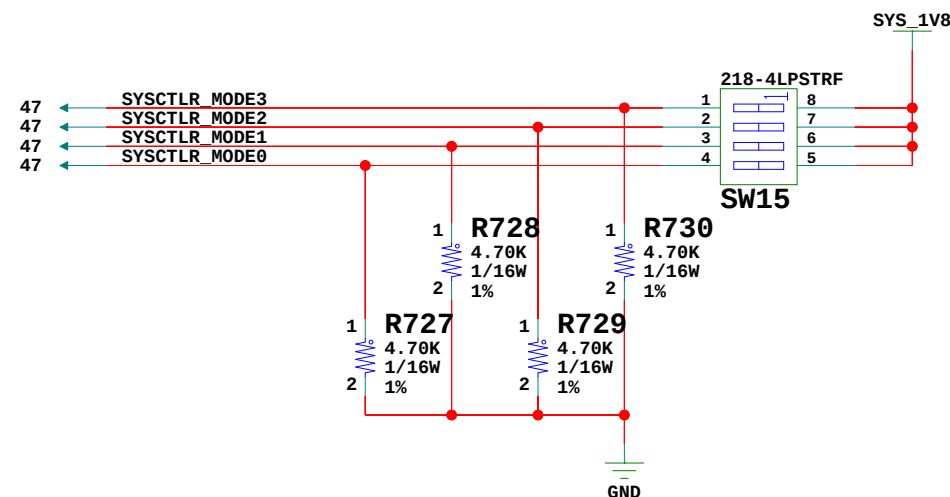
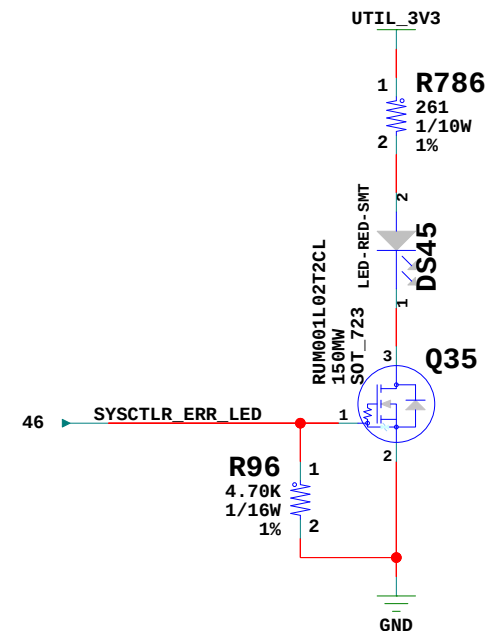
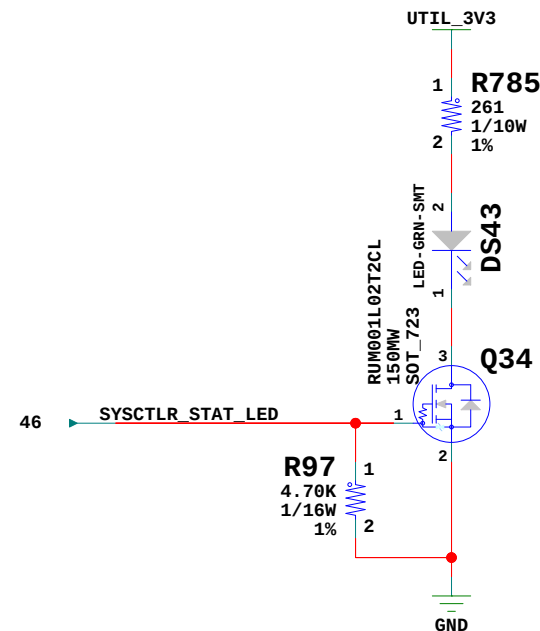
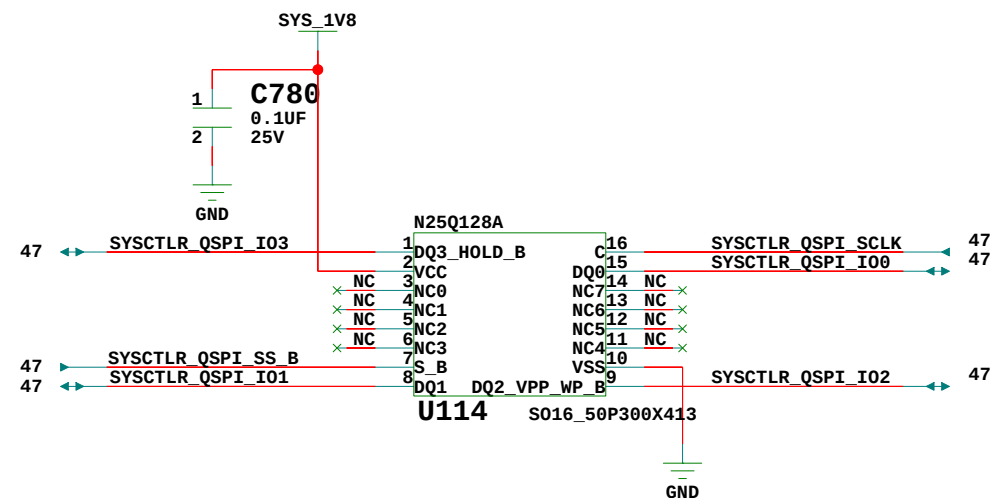
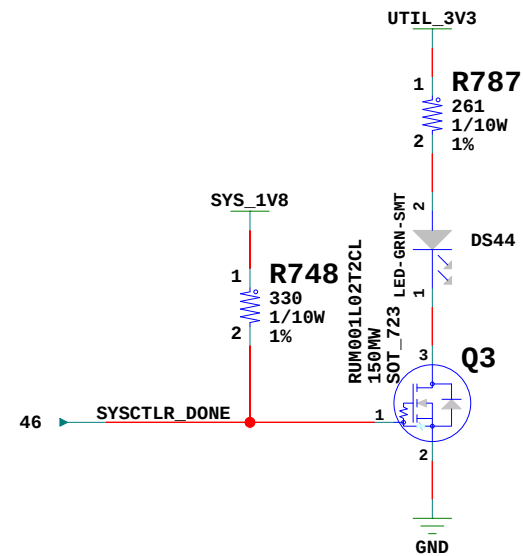
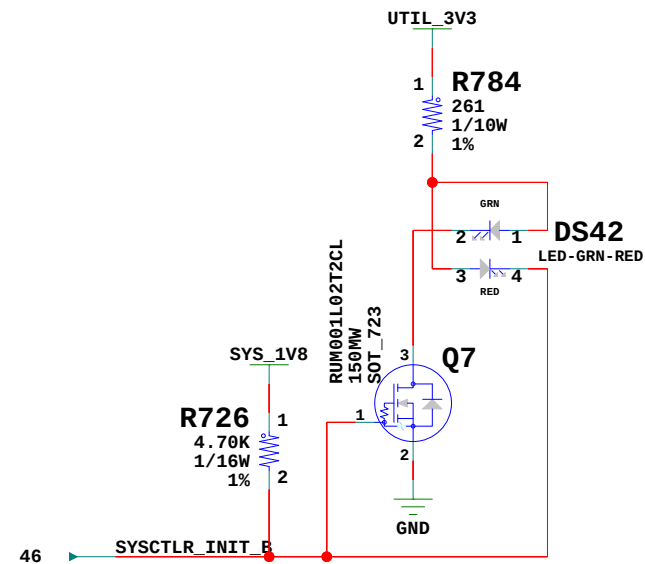
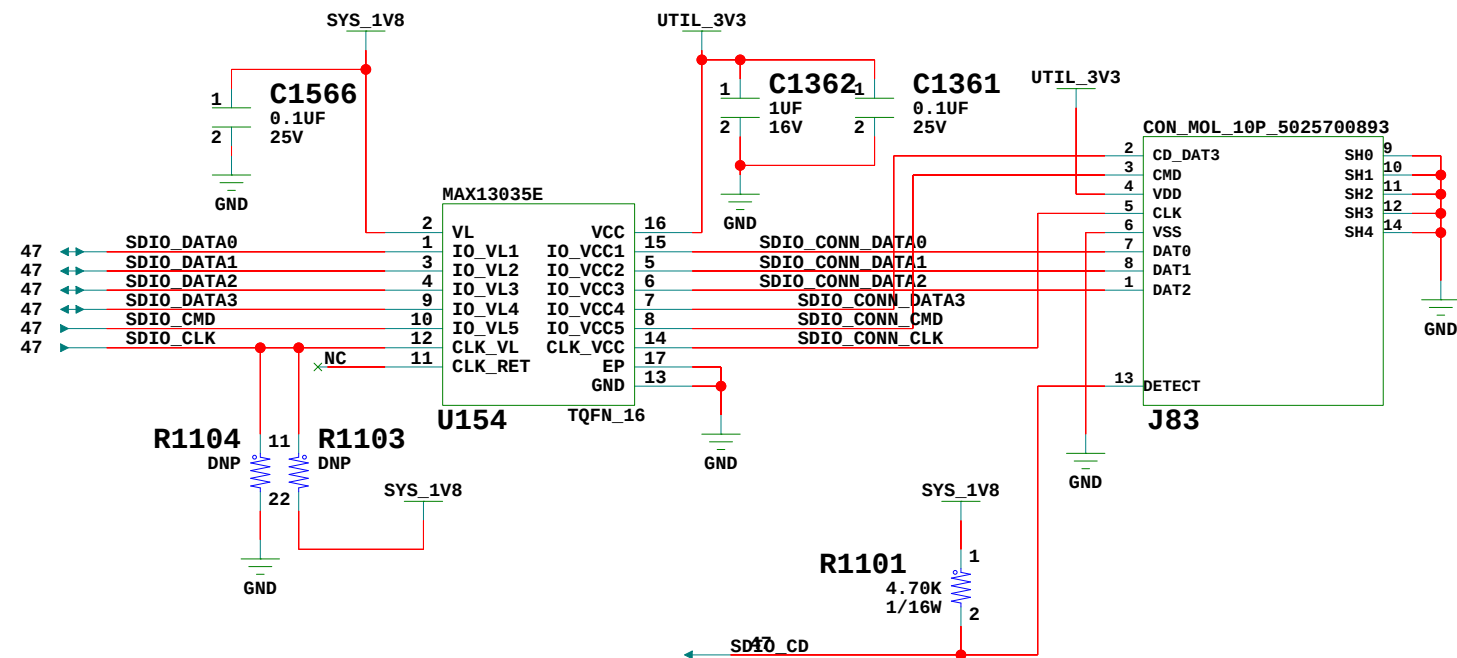
System Controller Zynq 3



TITLE: System Controller Zynq 3
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 48 OF 76	DRAWN BY: DN



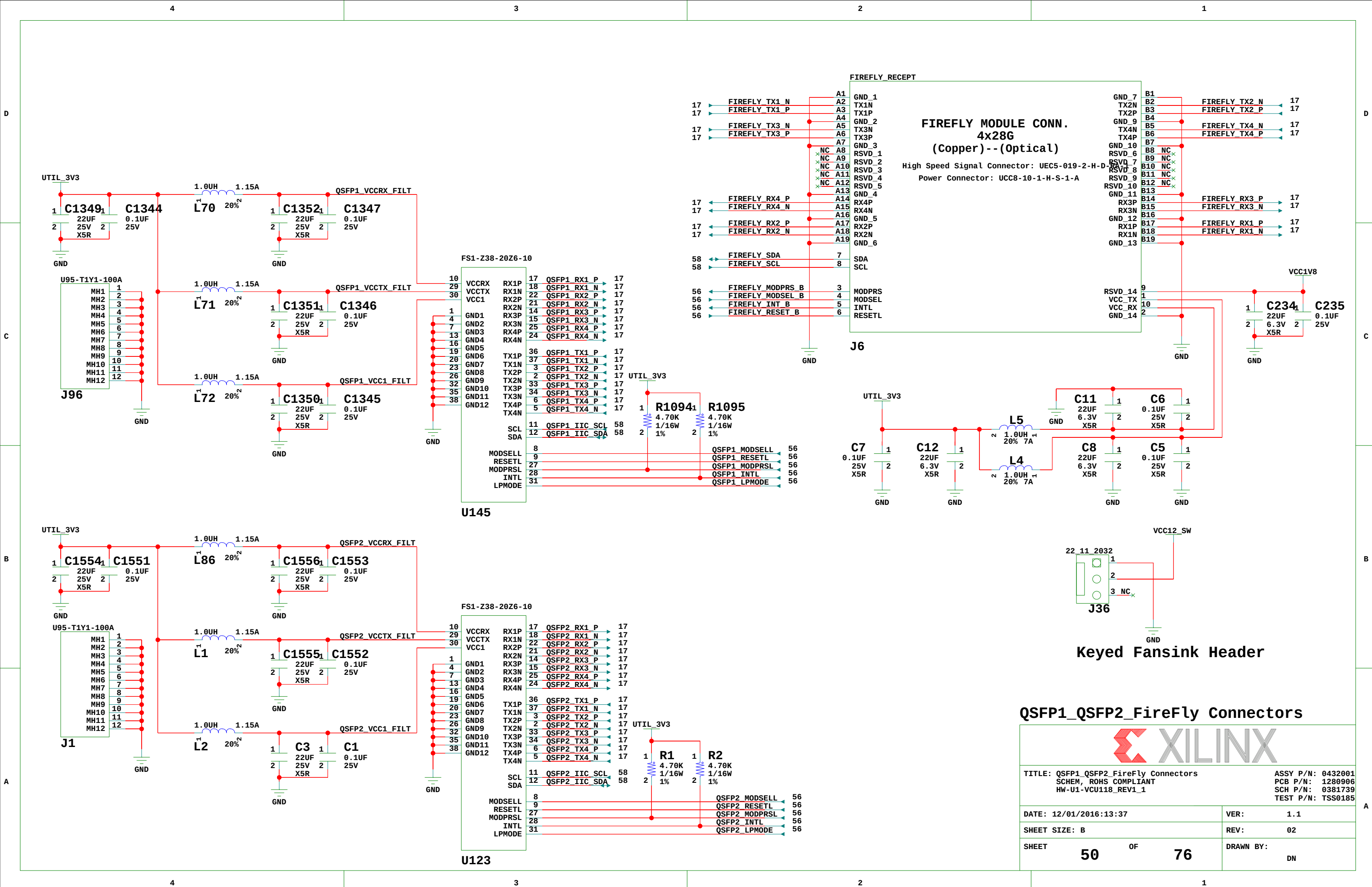
System Controller Zynq 4

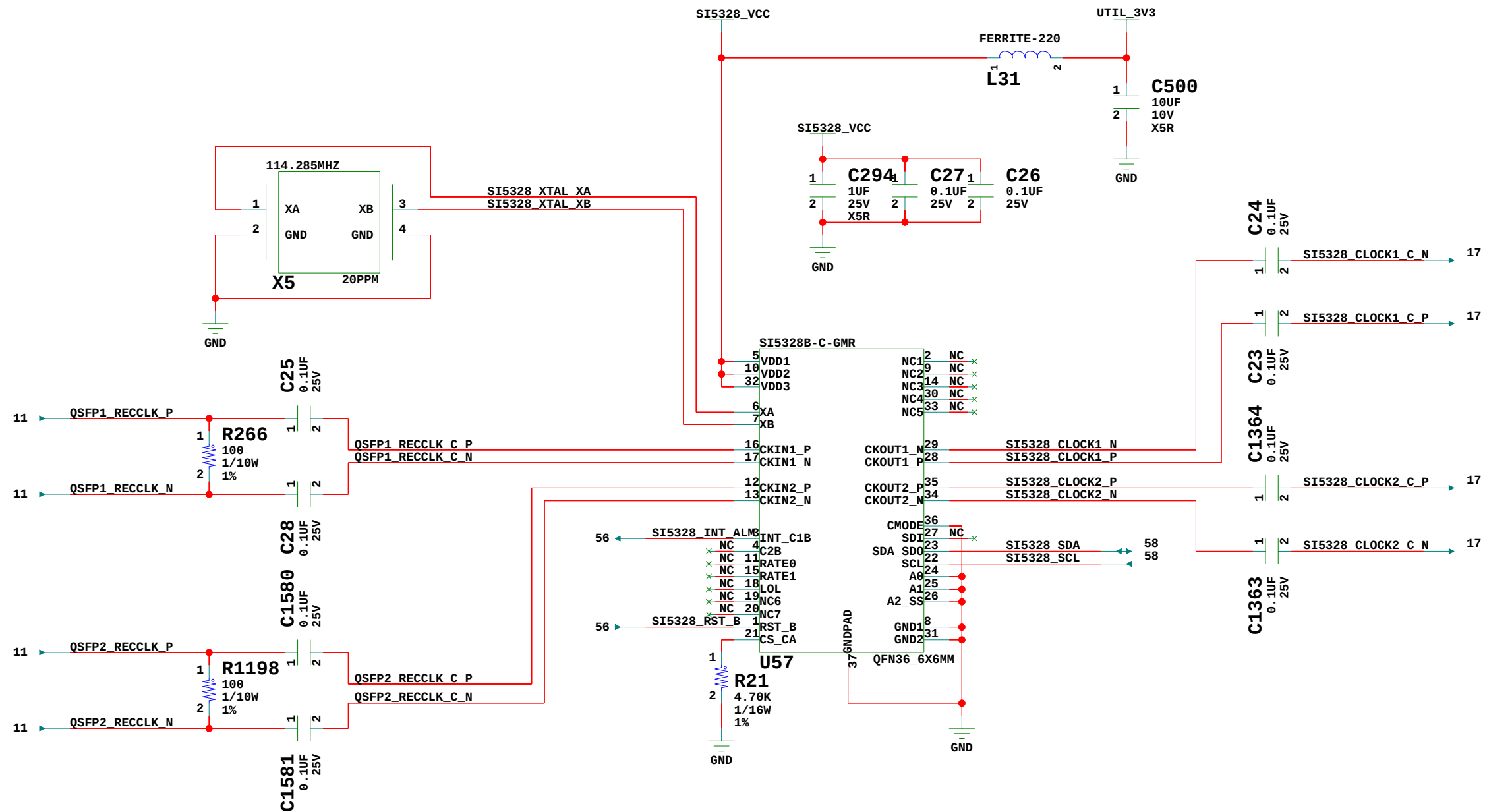


TITLE: System Controller Zynq 4
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 49 OF 76	DRAWN BY: DN





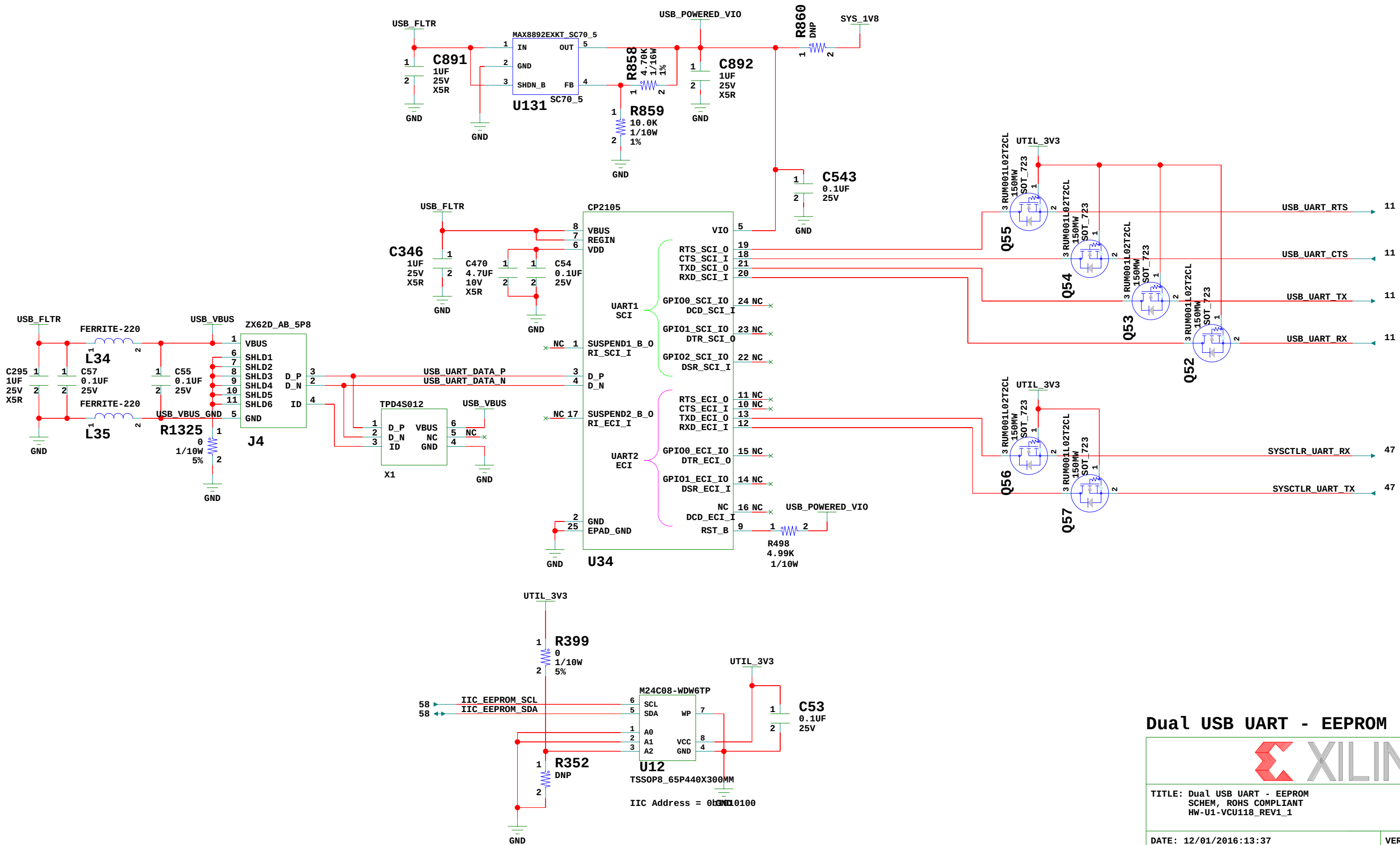
QSF+ 5328 Jitter Recovery



TITLE: QSF+ 5328 Jitter Recovery SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185
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DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 51 OF 76	DRAWN BY: DN

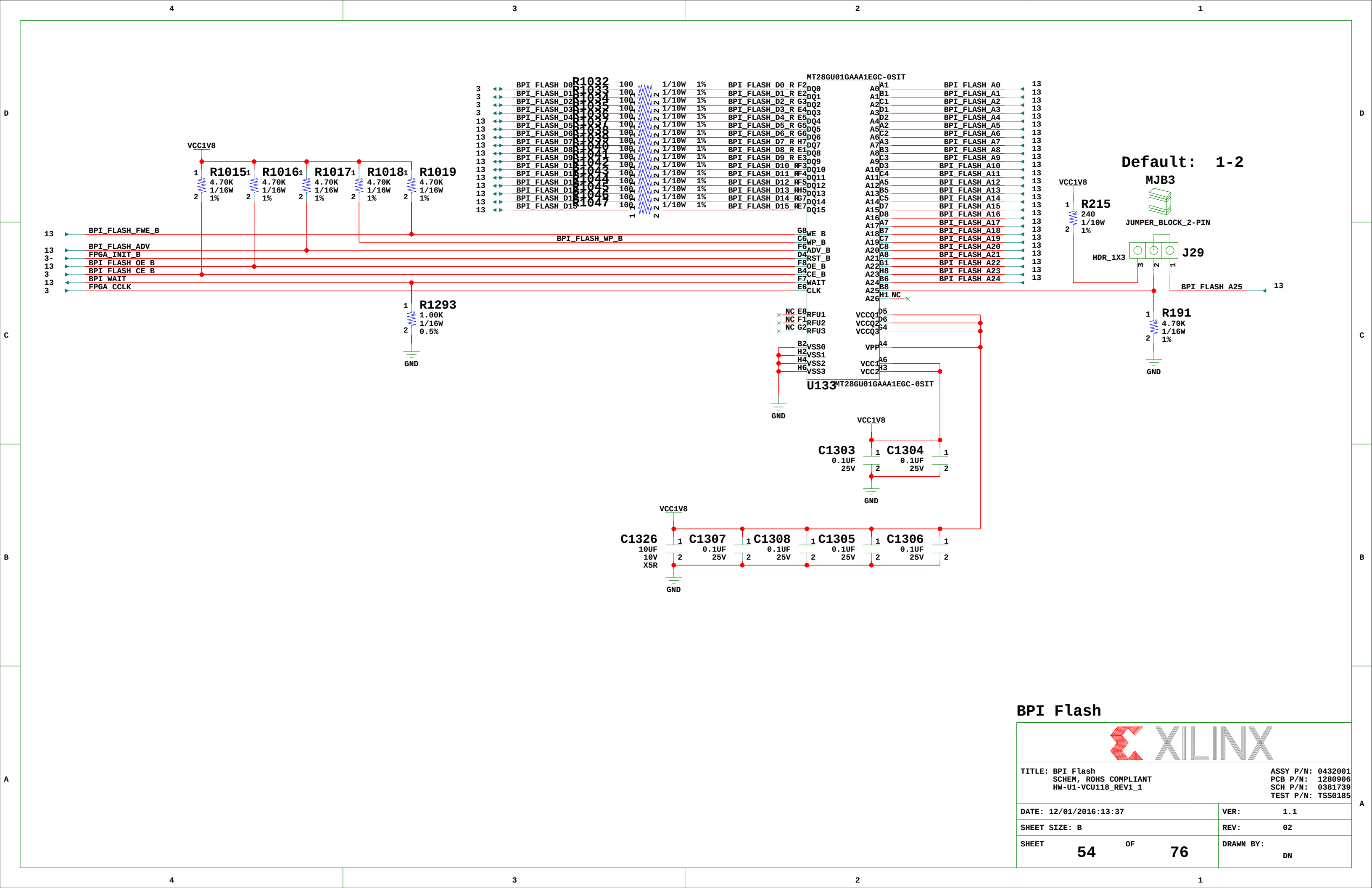
	
TITLE: 10/100/1000 Ethernet PHY - SGMII SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET	DRAWN BY: DN

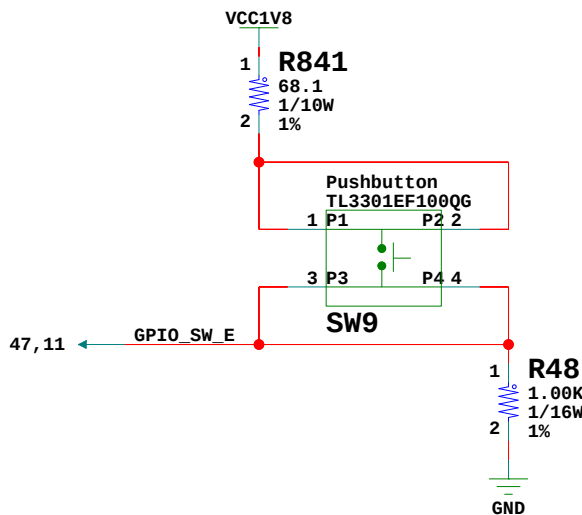
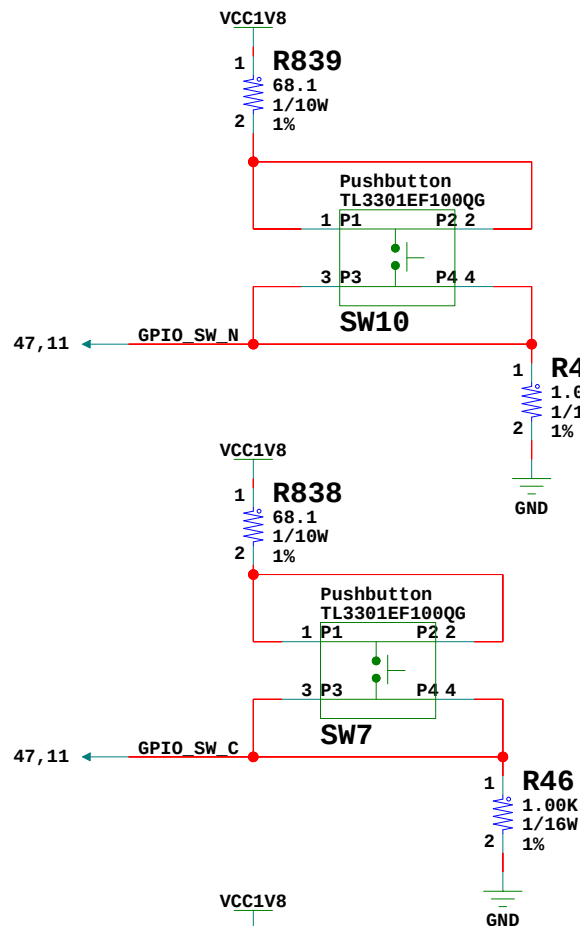
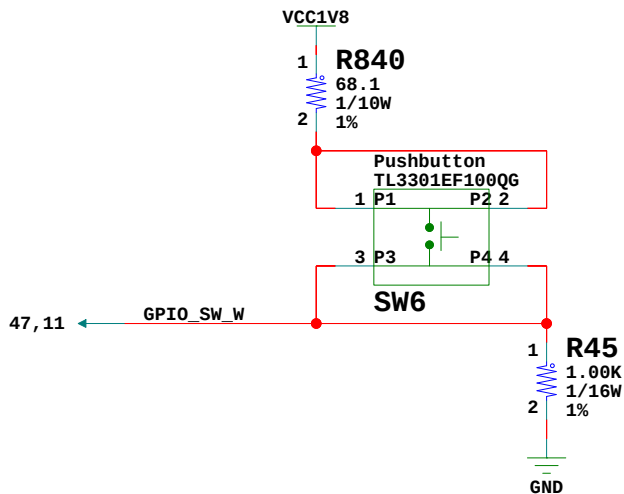
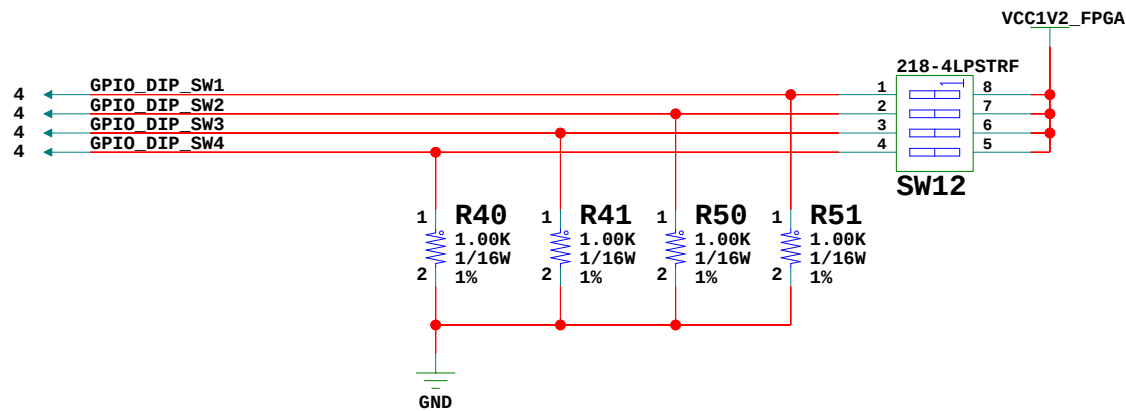


TITLE: Dual USB UART - EEPROM
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

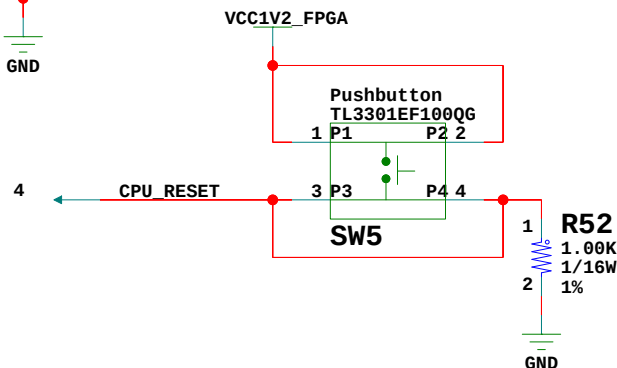
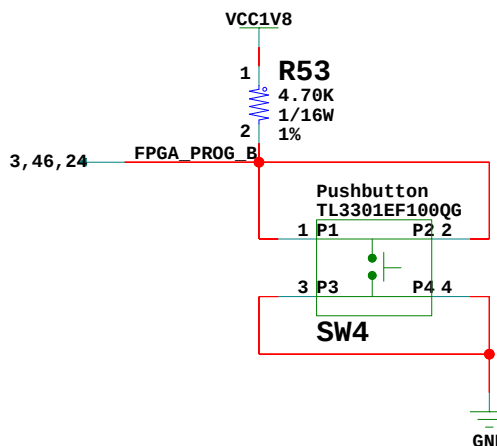
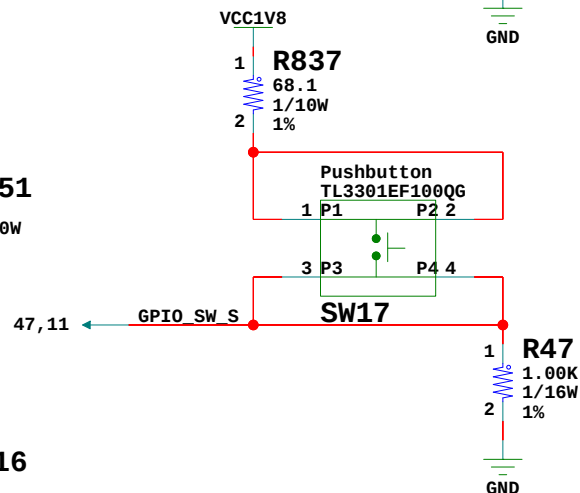
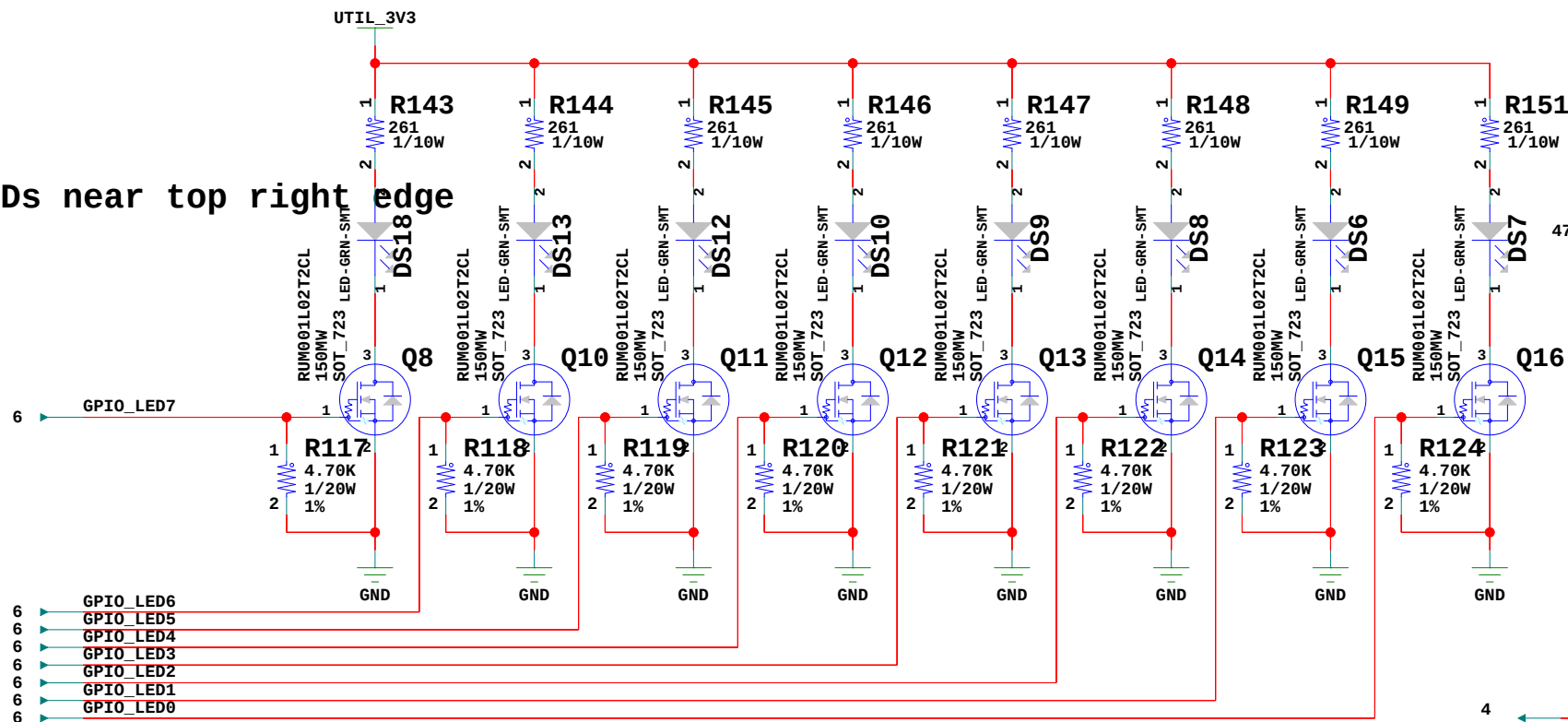
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 53 OF 76	DRAWN BY: DN





LEDs near top right edge



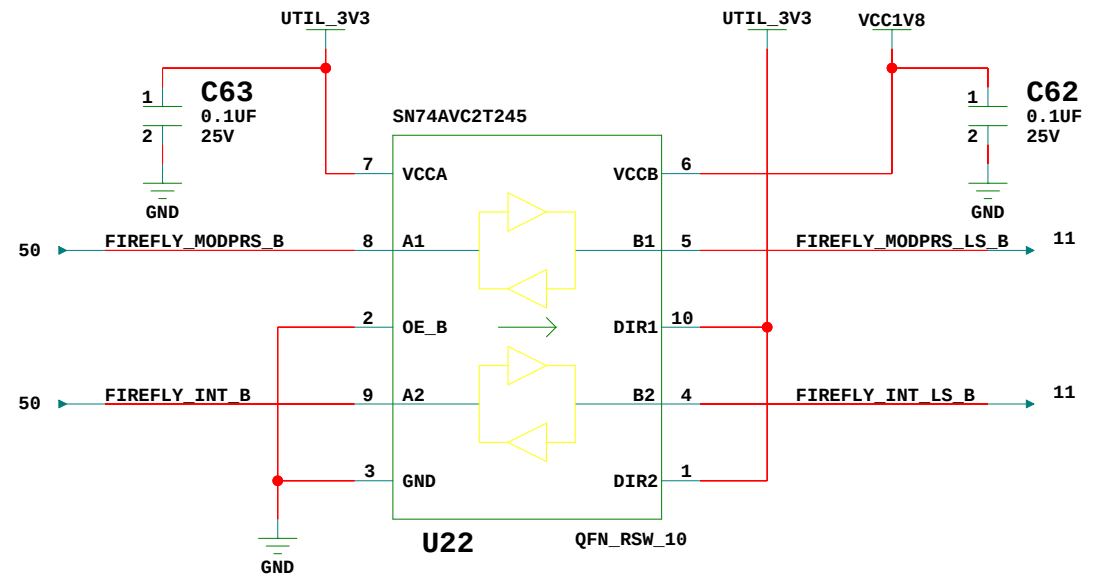
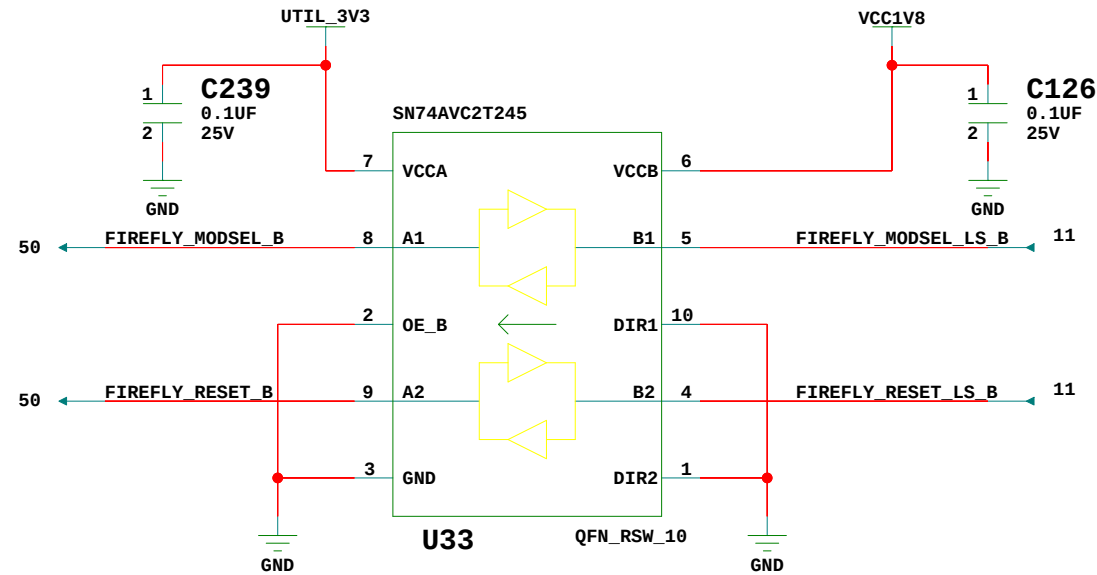
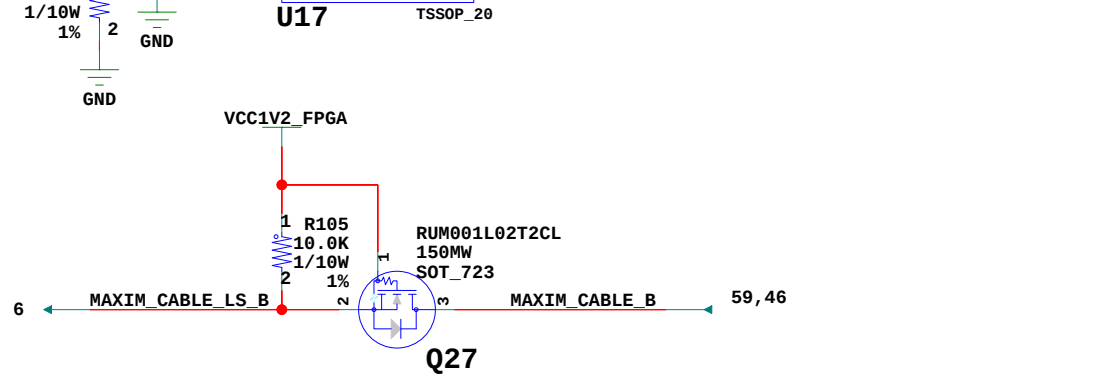
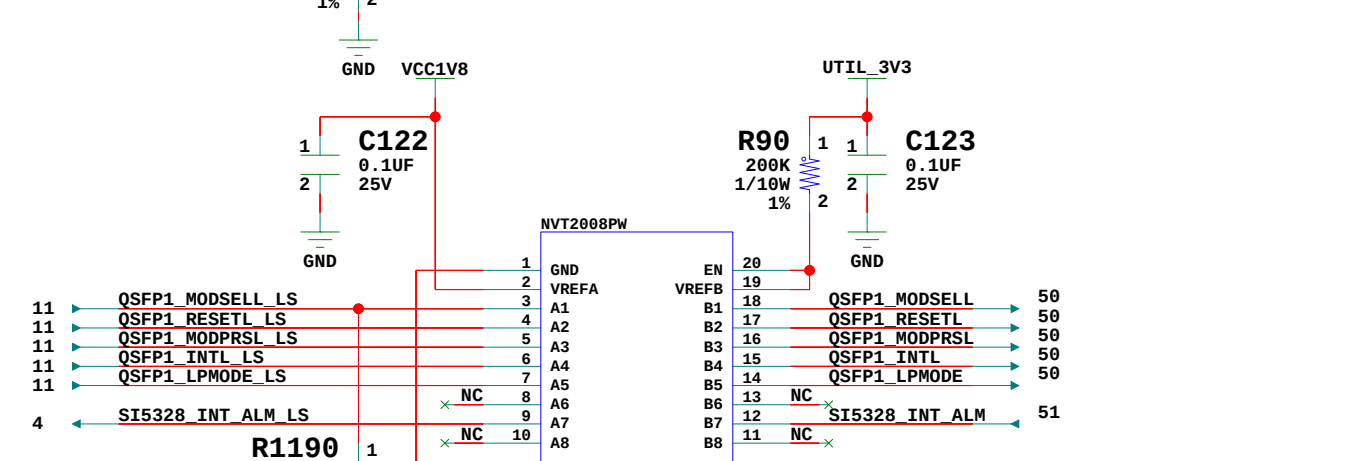
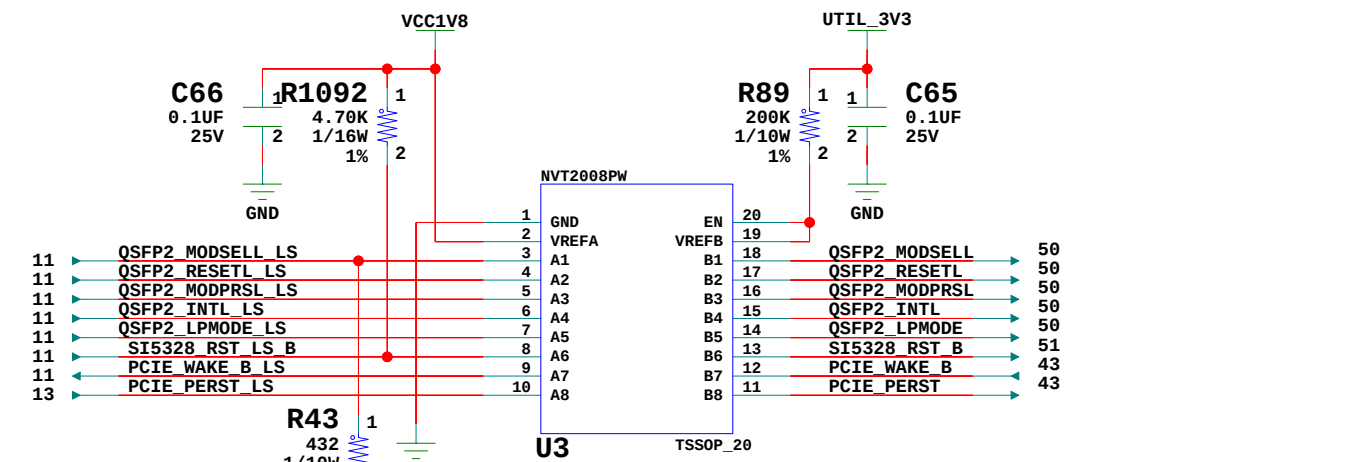
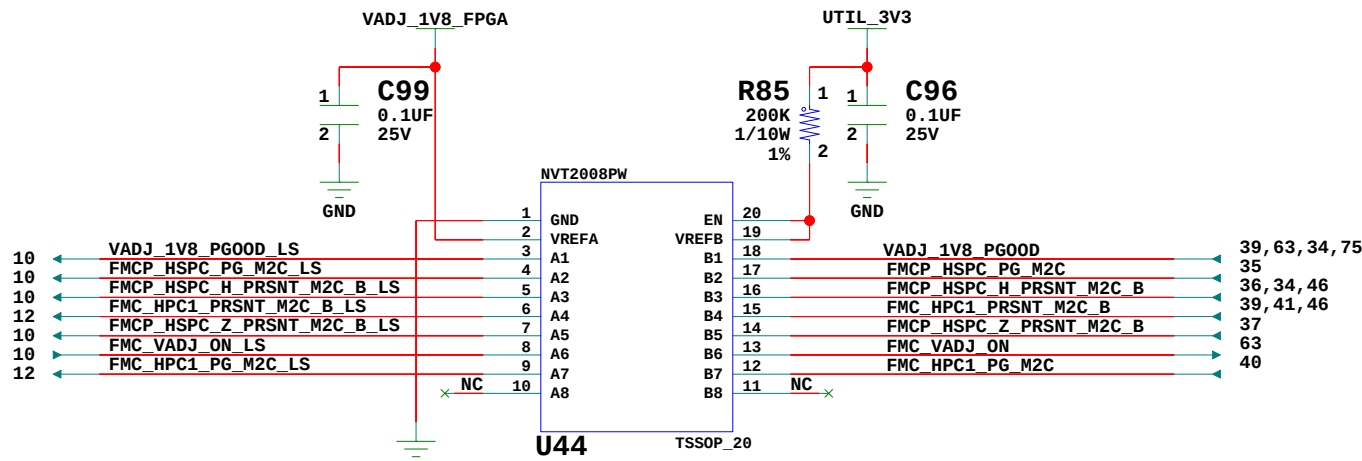
Buttons - Switches - LEDs



TITLE: Buttons - Switches - LEDs
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 55 OF 76	DRAWN BY: DN



Level Shifters

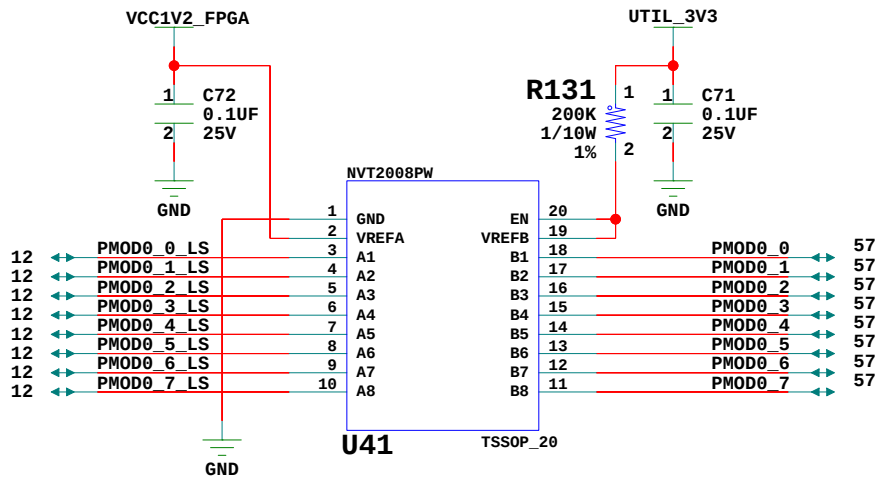
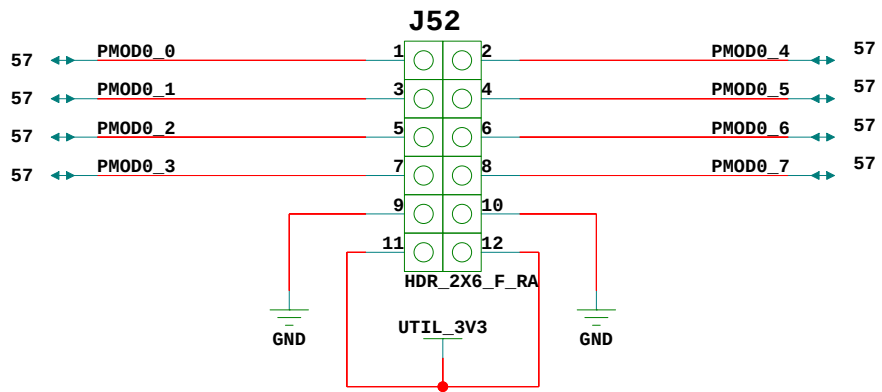


TITLE: Level Shifters
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

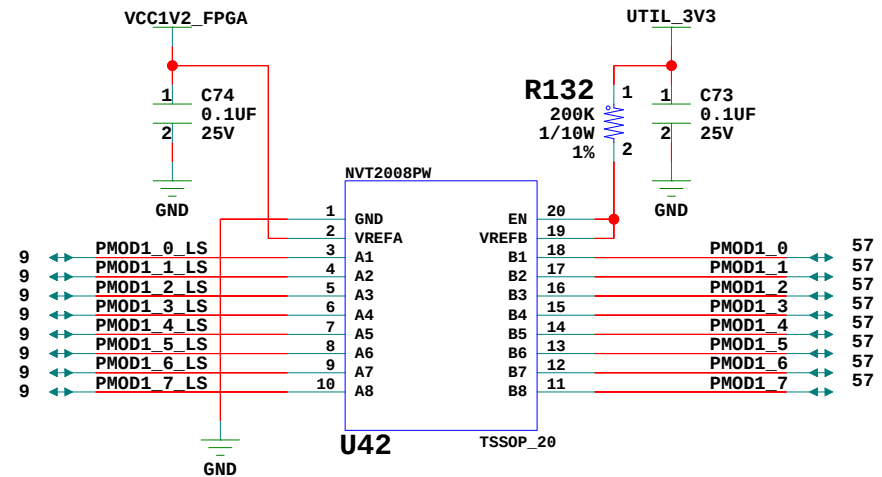
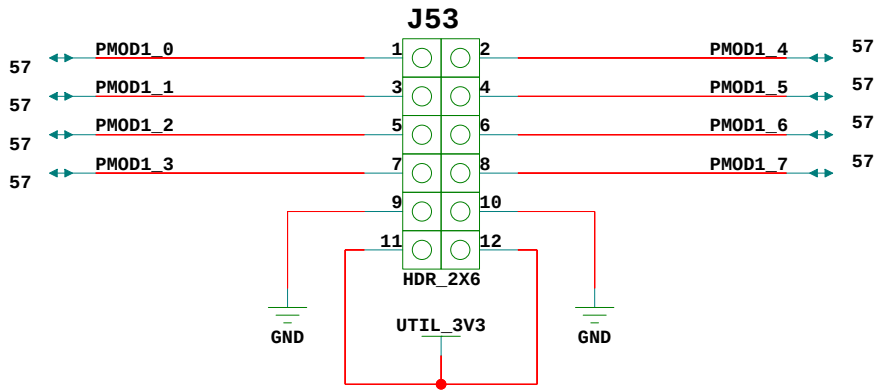
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 56 OF 76	DRAWN BY: DN

PMOD Rt-Angle Female Receptacle



PMOD Male Pin Header

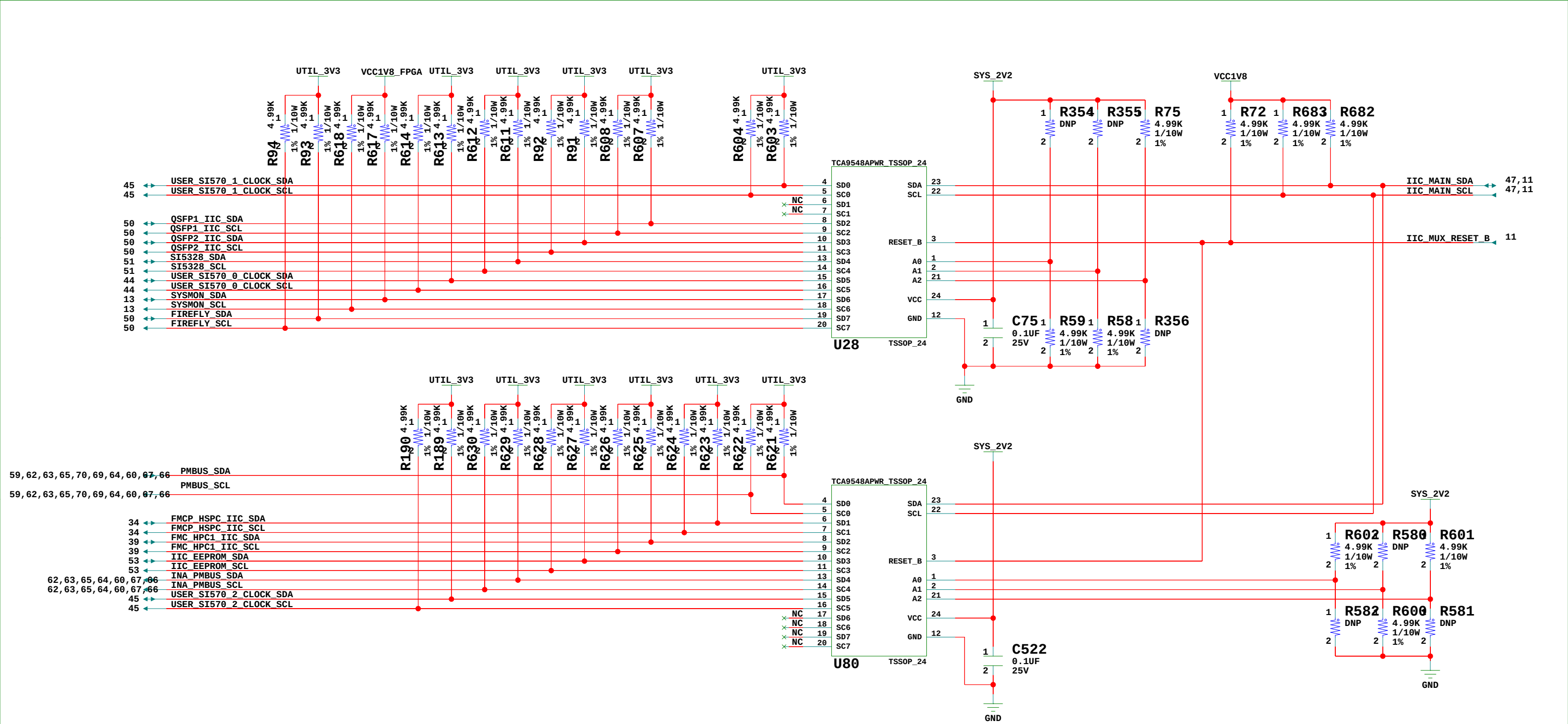


PMODs and Level Shifters

TITLE: PMODs and Level Shifters
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

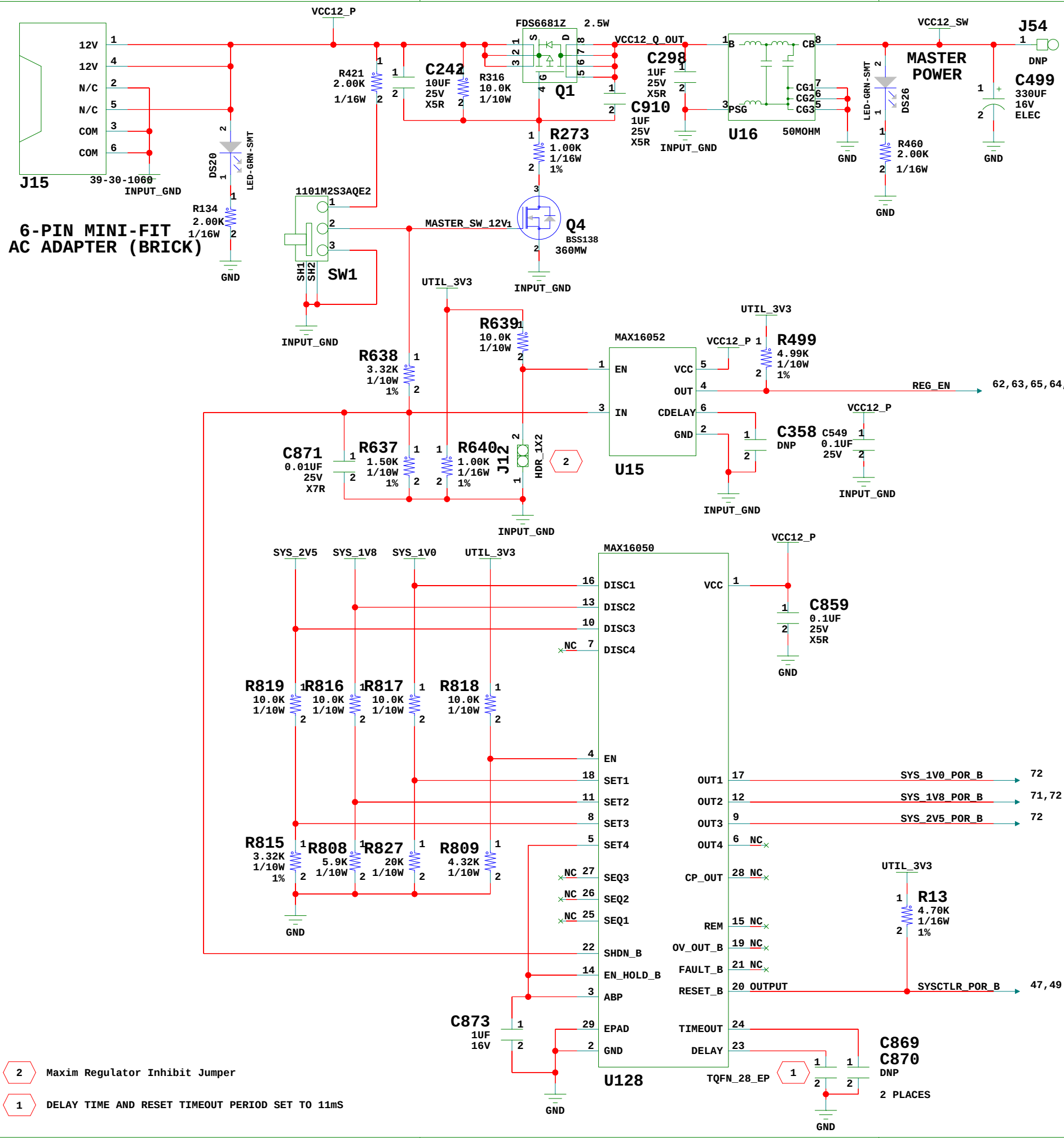
DATE: 12/01/2016:13:39	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 57 OF 76	DRAWN BY: DN



TITLE: IIC MUXs
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

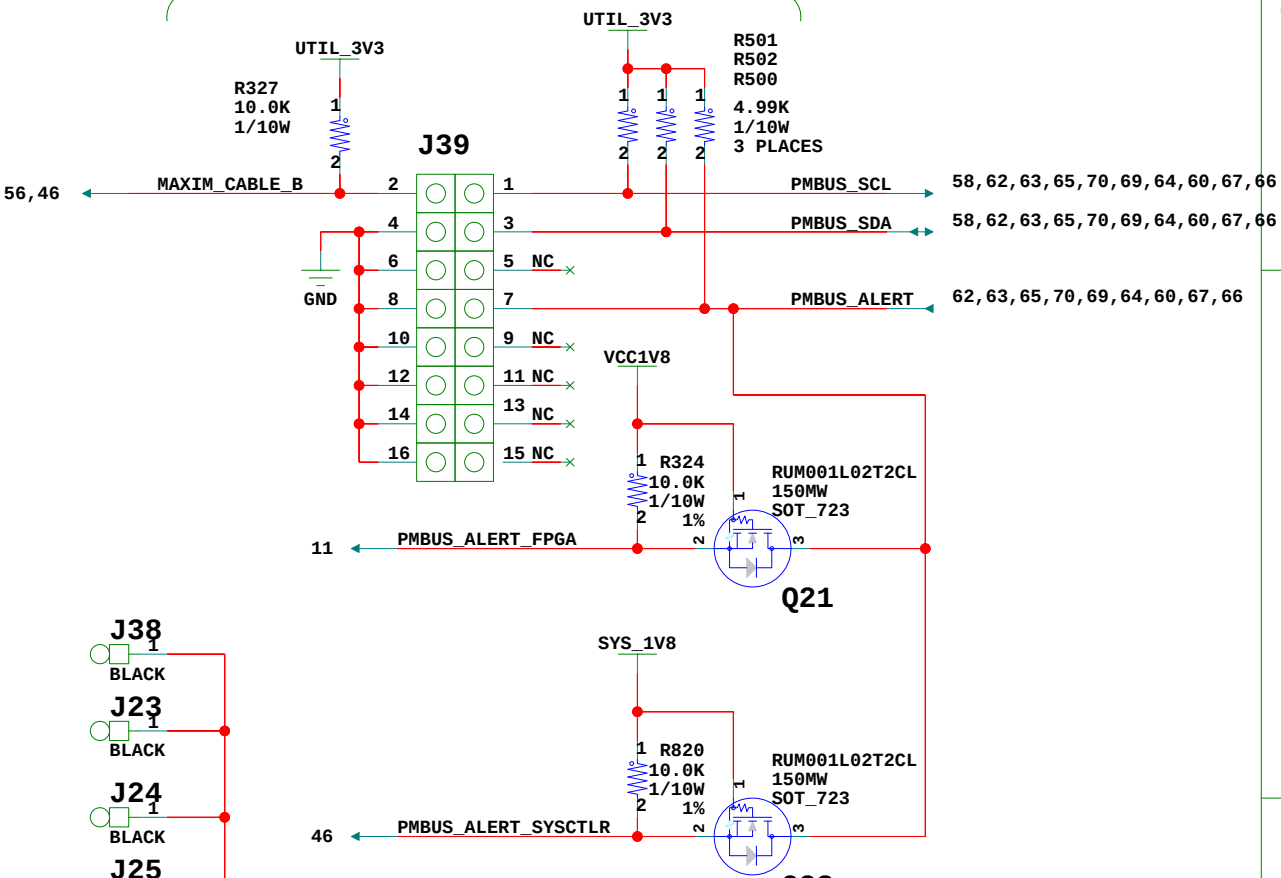
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 58 OF 76	DRAWN BY: DN



Fan Control circuit on page 13

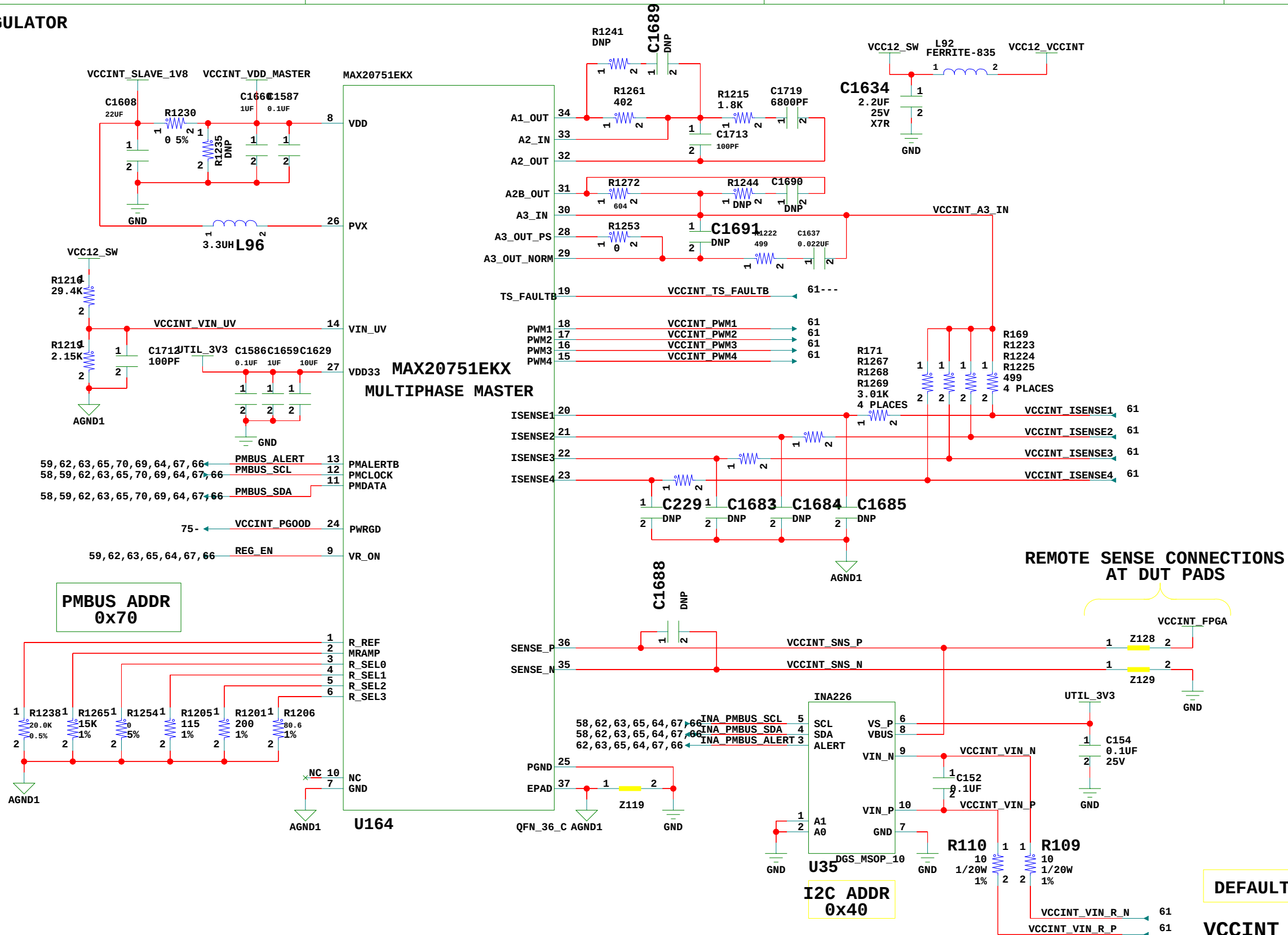
MAXIM PMBUS PROGRAMMING CABLE



Pwr Connector - Switch - Pwr Supervisor

TITLE: Pwr Connector - Switch - Pwr Supervisor SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 59 OF 76	DRAWN BY: DN

VCCINT REGULATOR



DEFAULT = 0.72VDC

VCCINT 80A Regulator Master



**TITLE: VCCINT 80A Regulator Master
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1**

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER:	1.1
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SHEET SIZE: B

REV: 02

SHEET

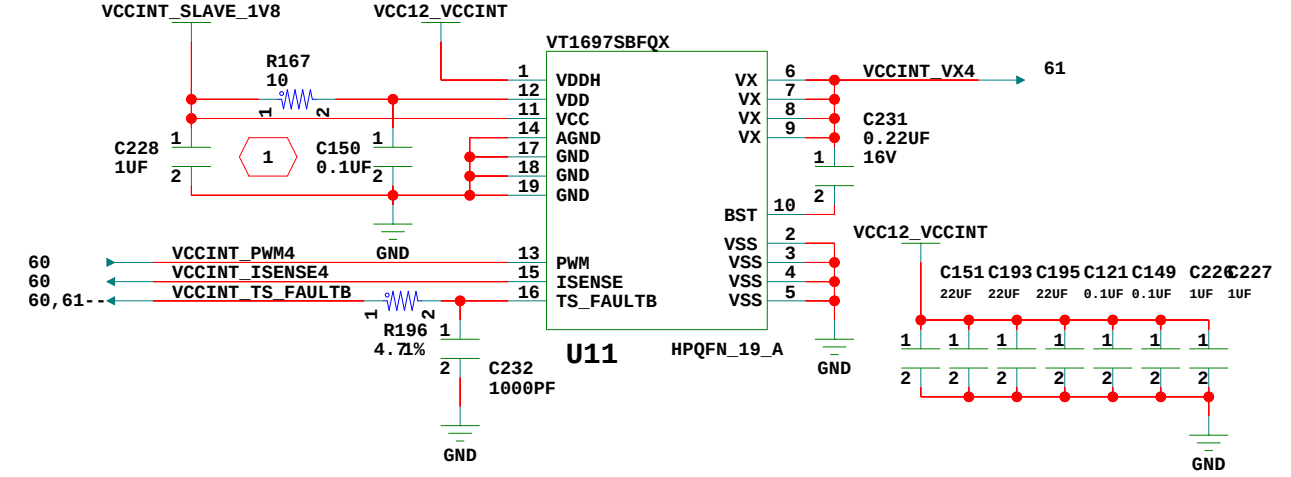
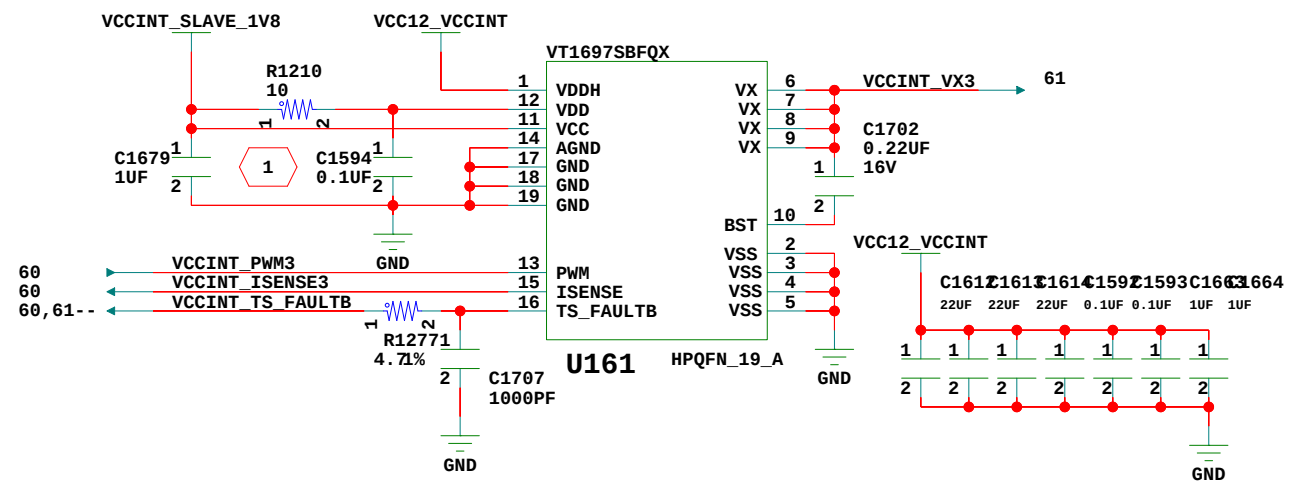
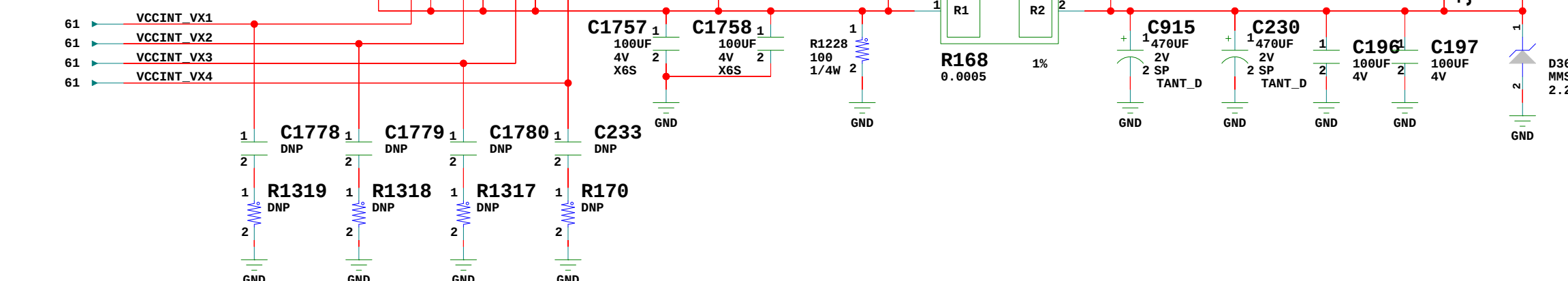
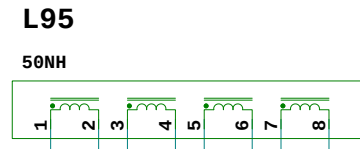
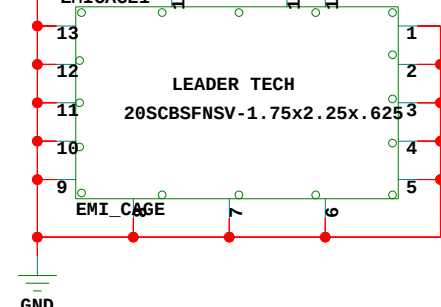
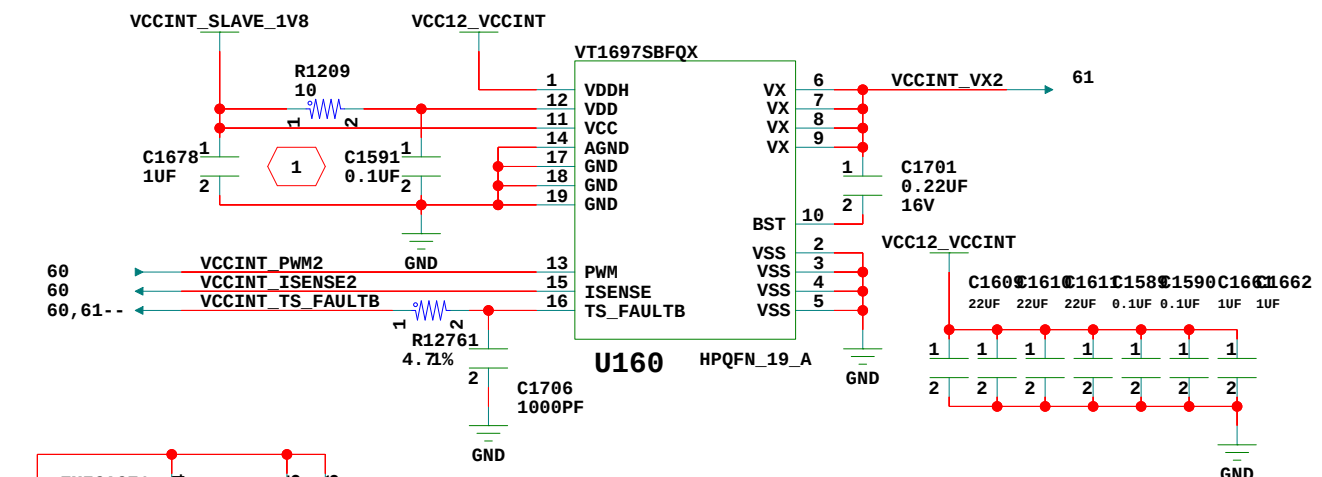
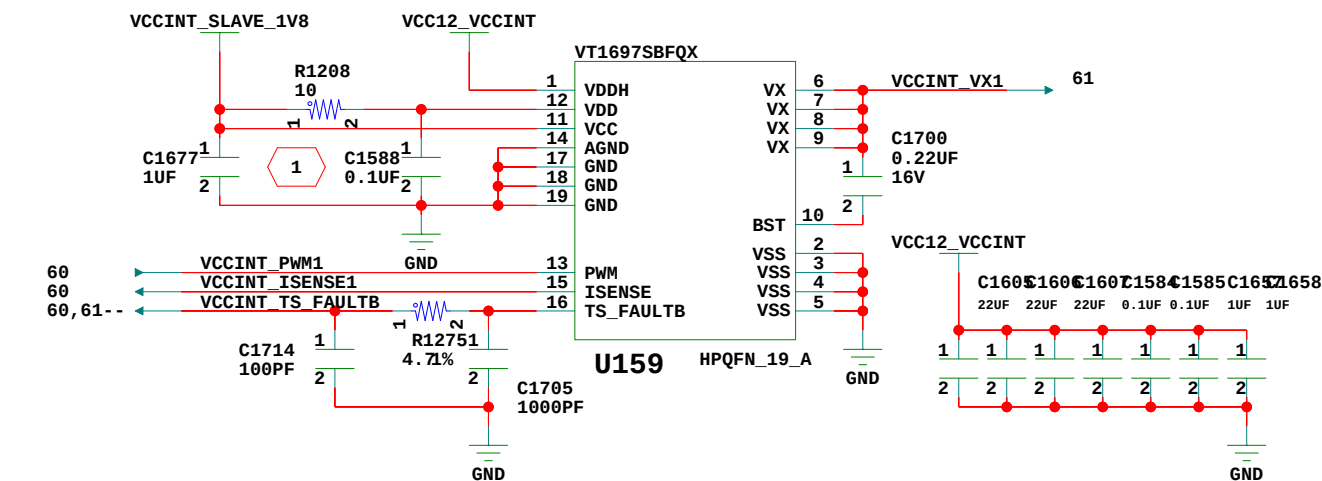
60

OF

76

DRAWN BY:

DN



NOTES

- 1 Capacitors should be placed as close as possible to Pin12 and 14

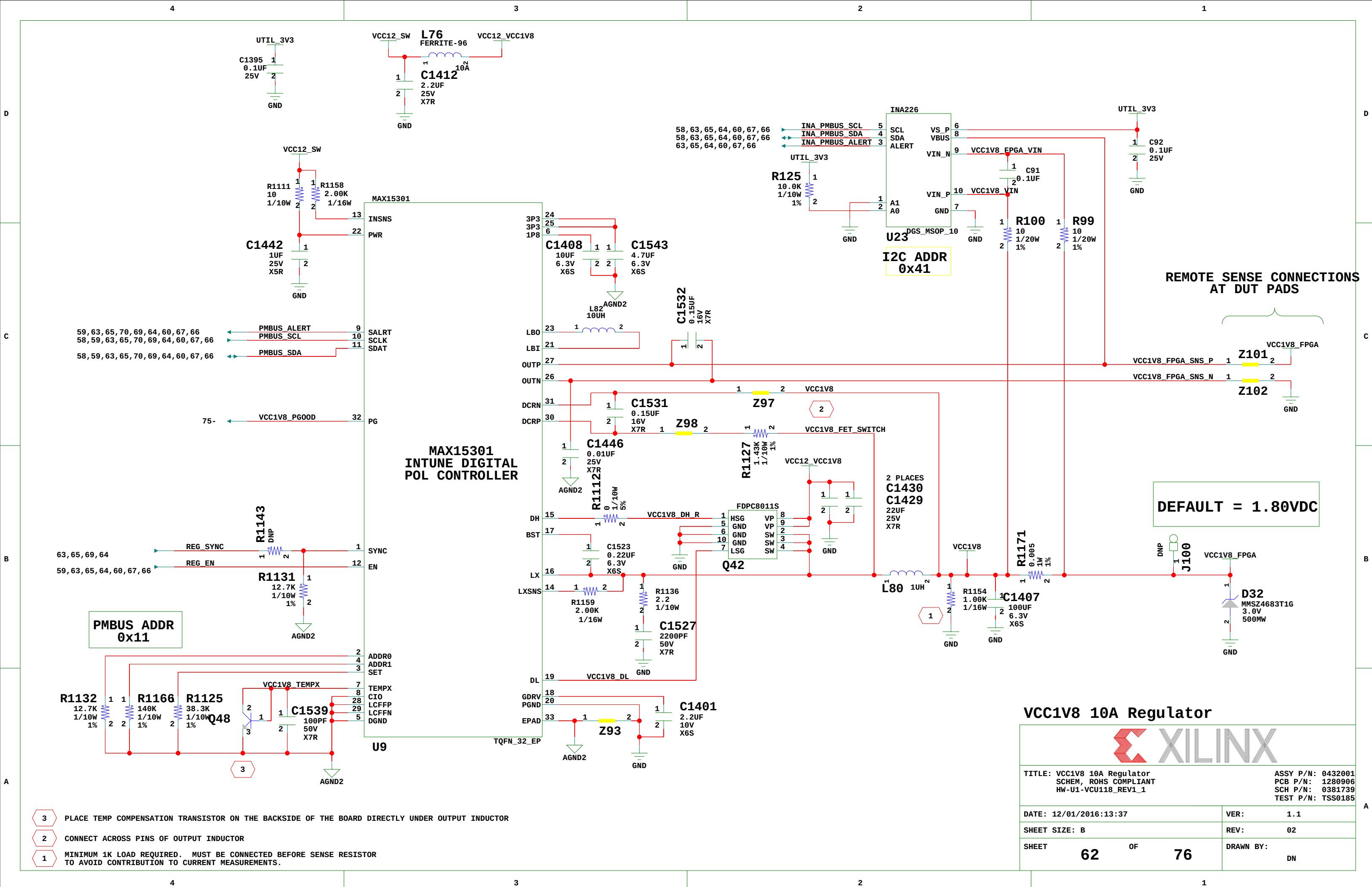
VCCINT 80A Regulator Slave

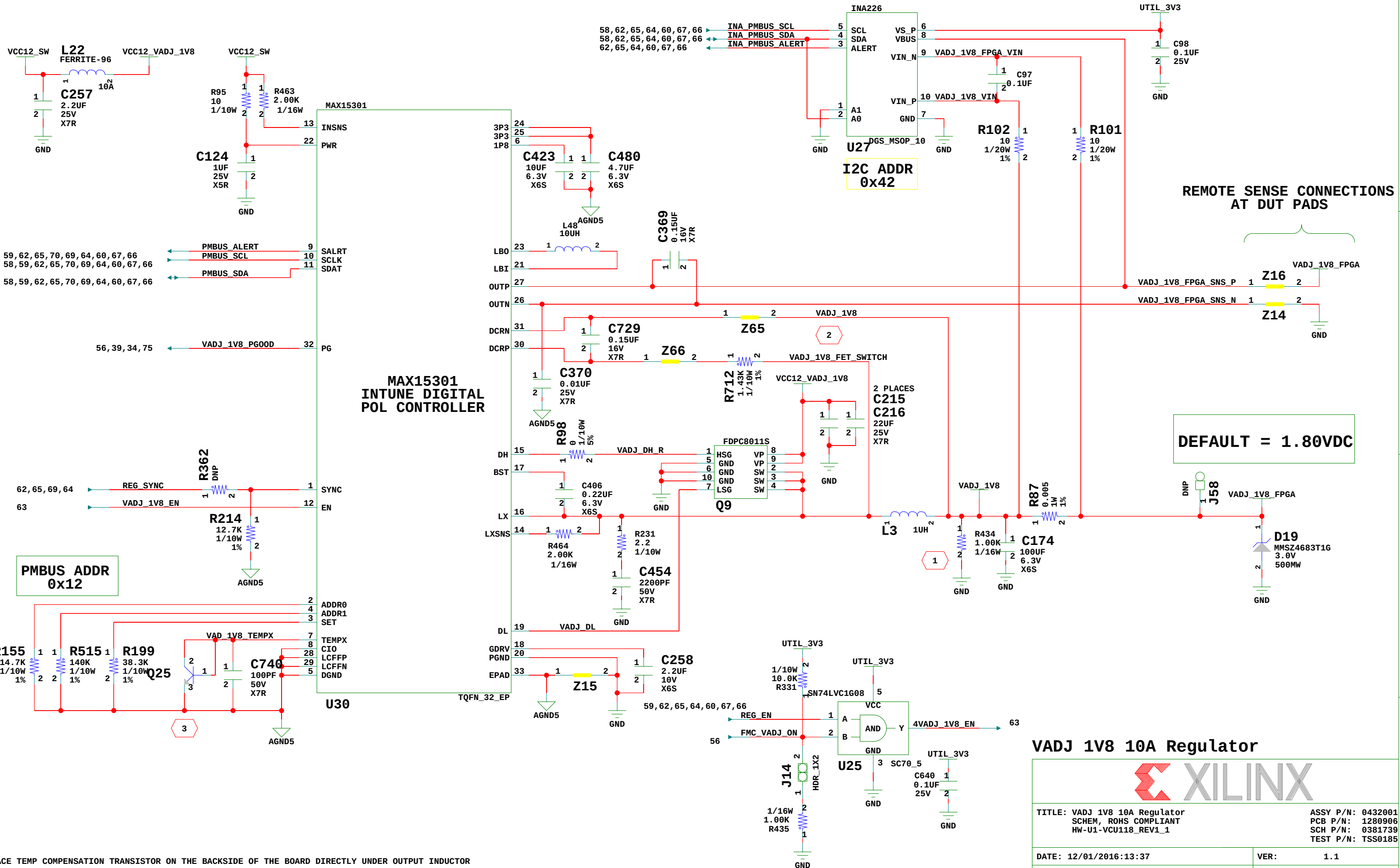


TITLE: VCCINT 80A Regulator Slave
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 61 OF 76	DRAWN BY: DN





- 3 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR
- 2 CONNECT ACROSS PINS OF OUTPUT INDUCTOR
- 1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

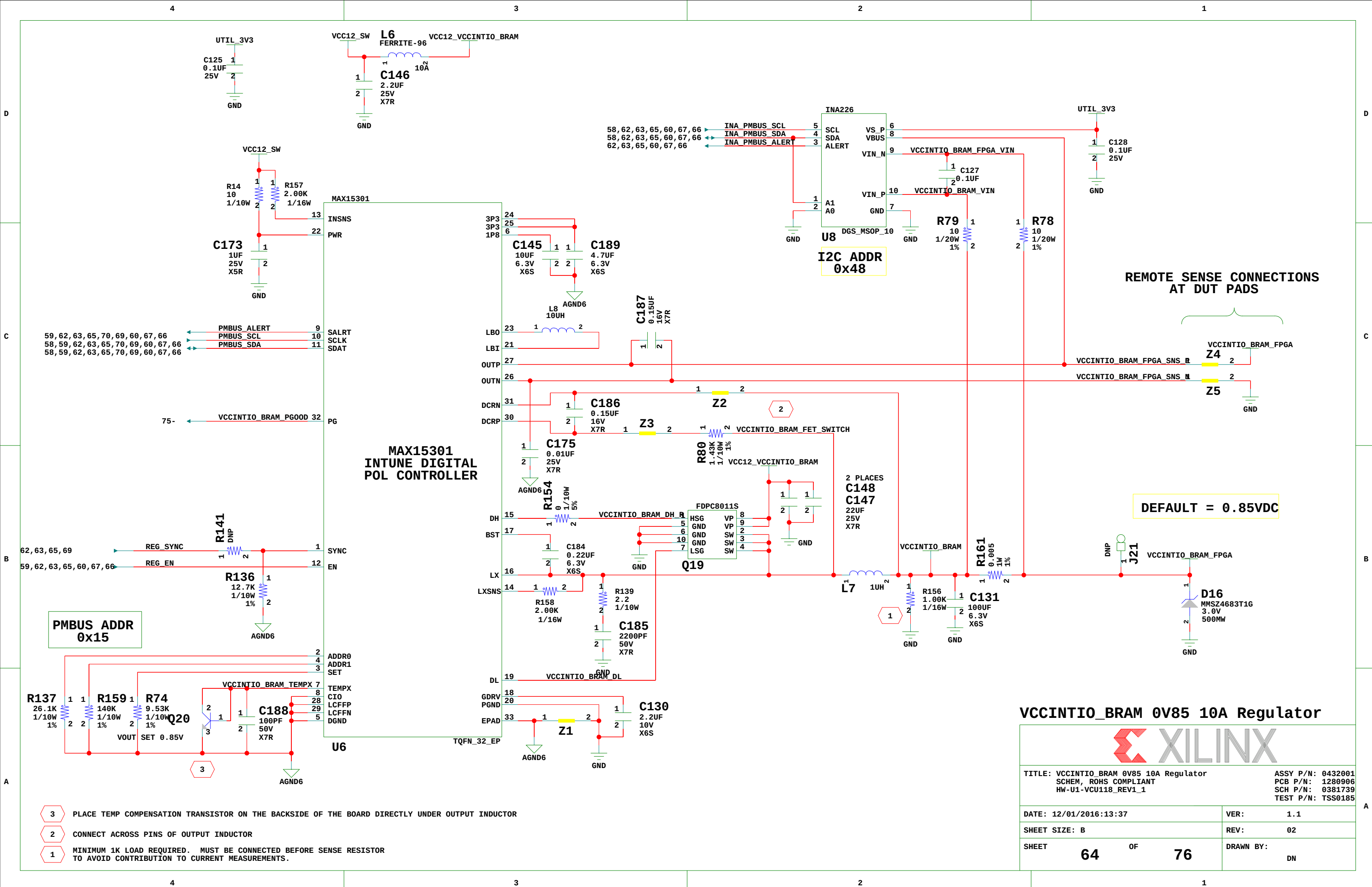
VADJ 1V8 10A Regulator

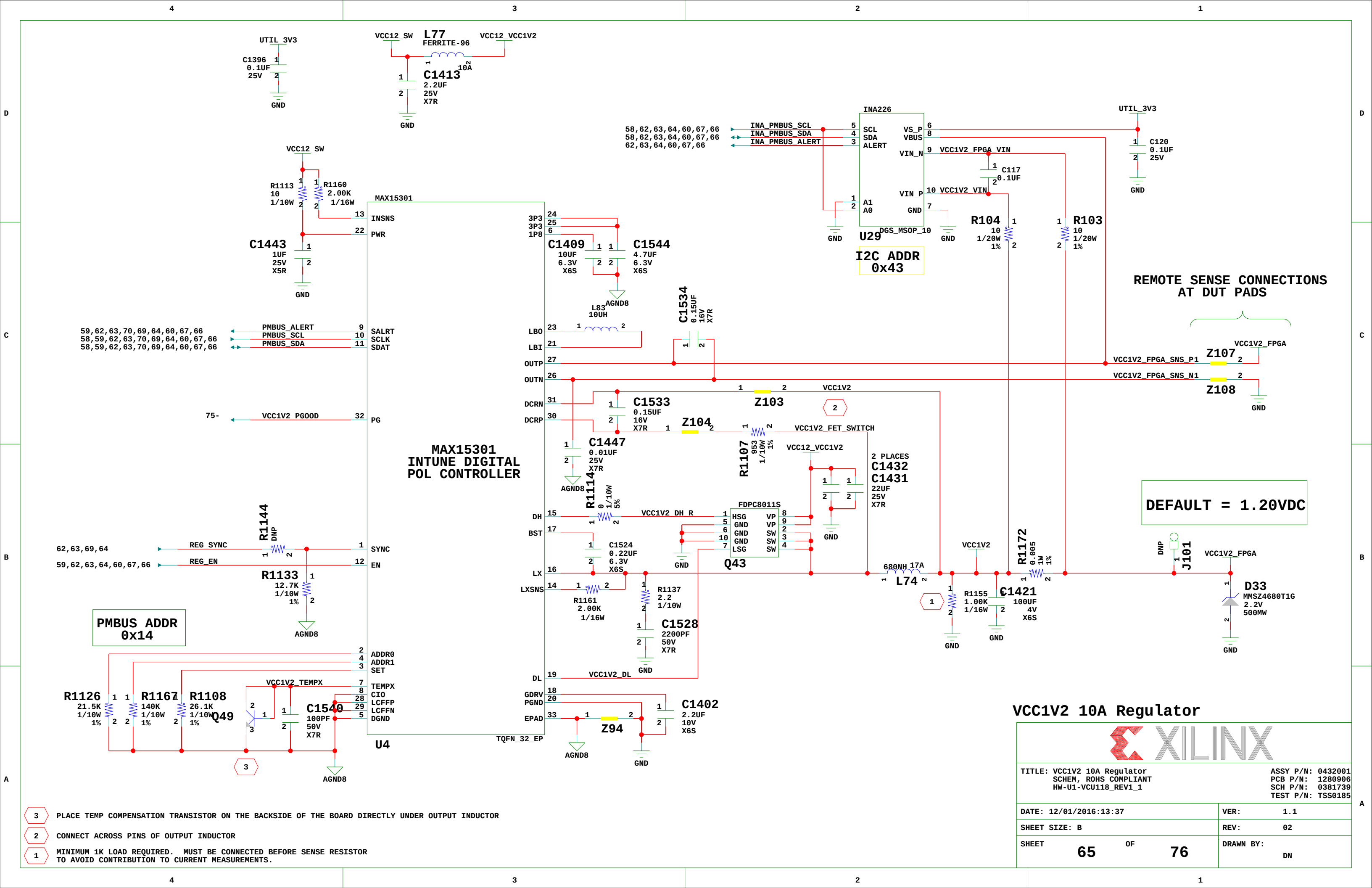


TITLE: VADJ 1V8 10A Regulator
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 63 OF 76	DRAWN BY: DN





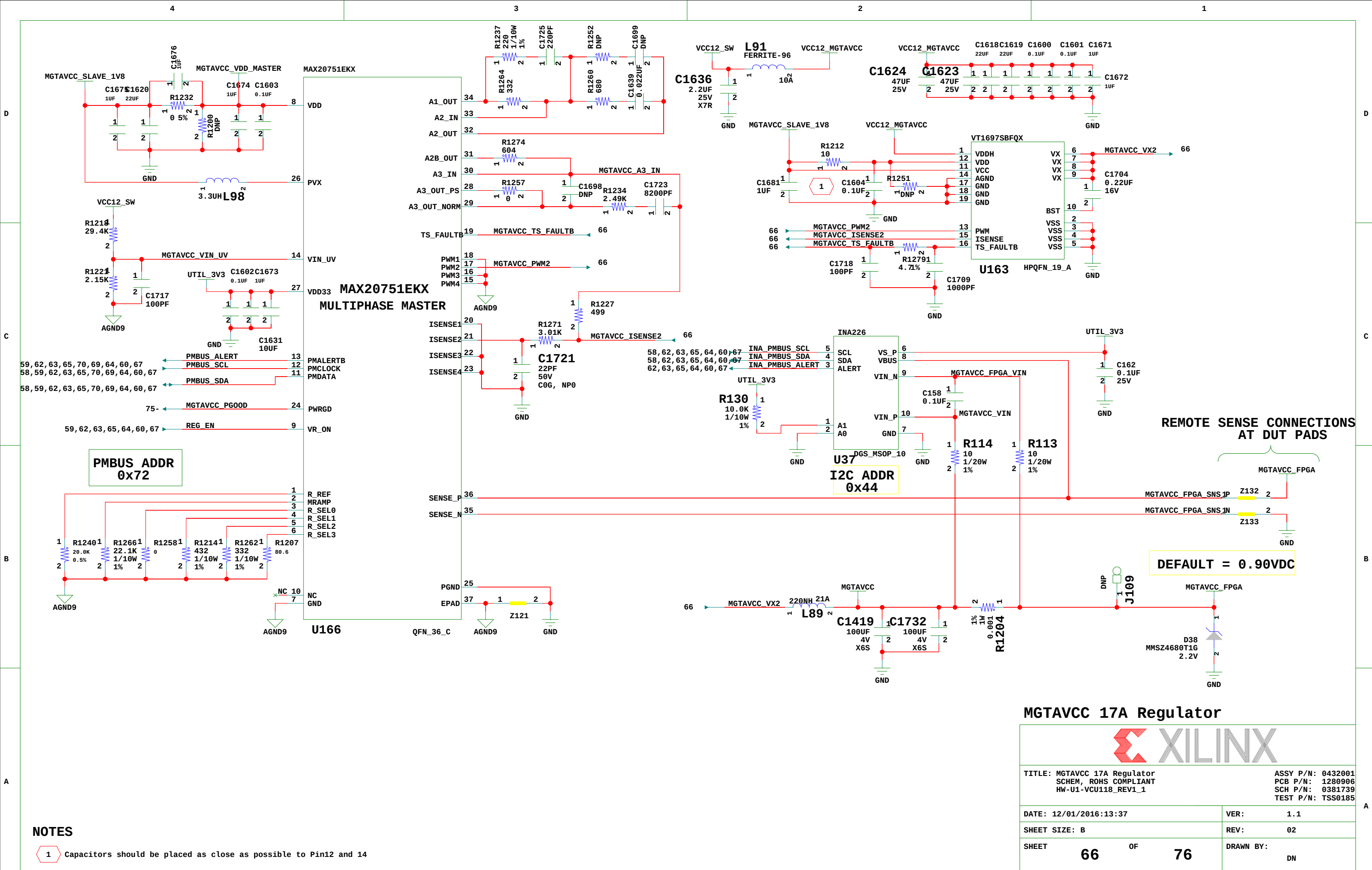
- 3 PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR
- 2 CONNECT ACROSS PINS OF OUTPUT INDUCTOR
- 1 MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.

VCC1V2 10A Regulator

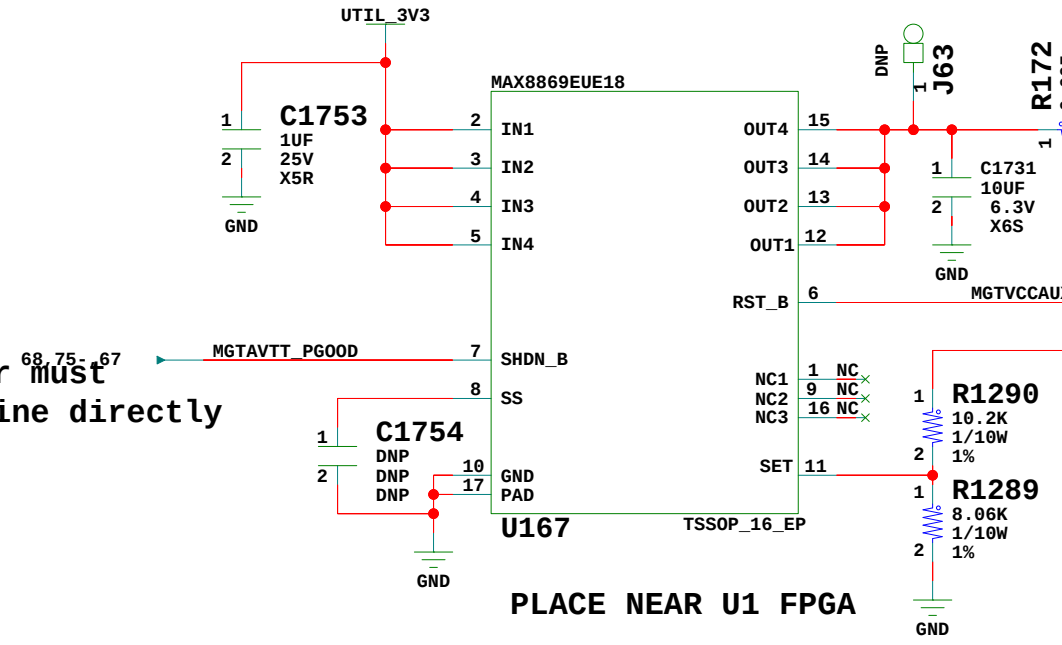
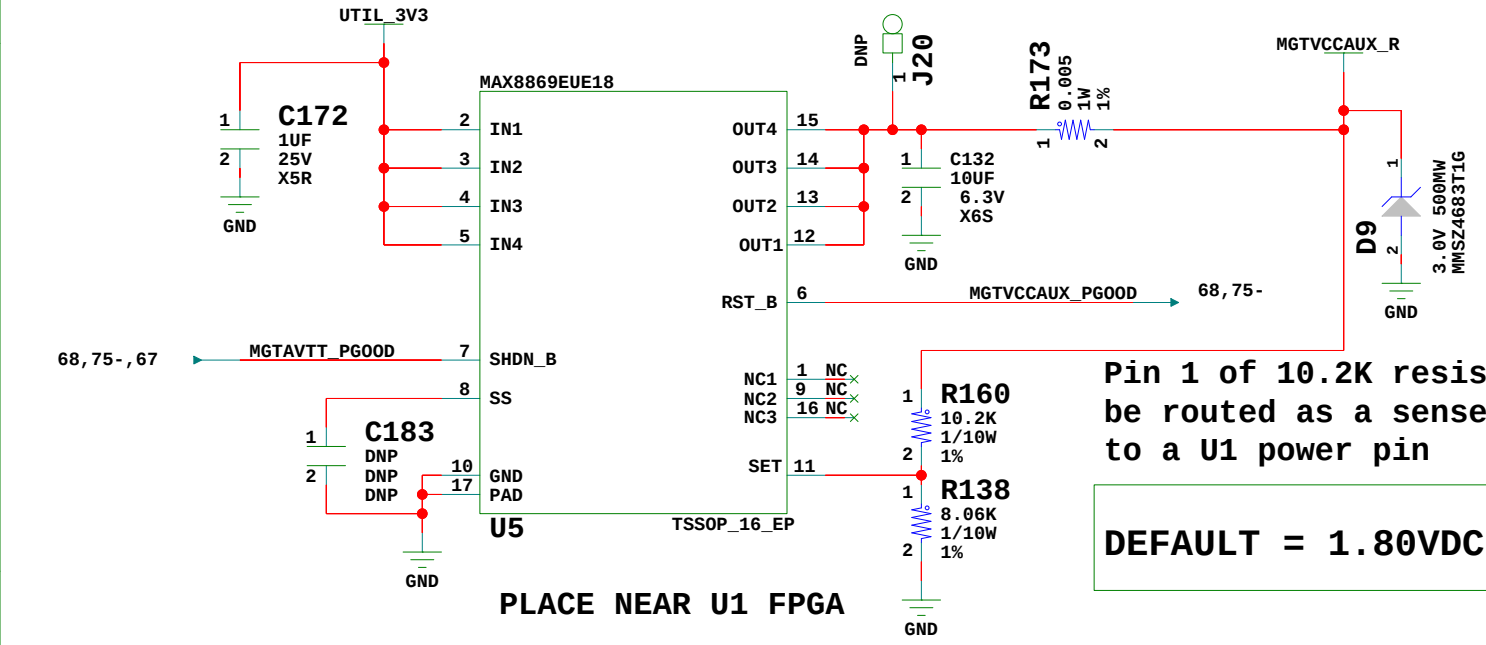
TITLE: VCC1V2 10A Regulator
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 65 OF 76	DRAWN BY: DN

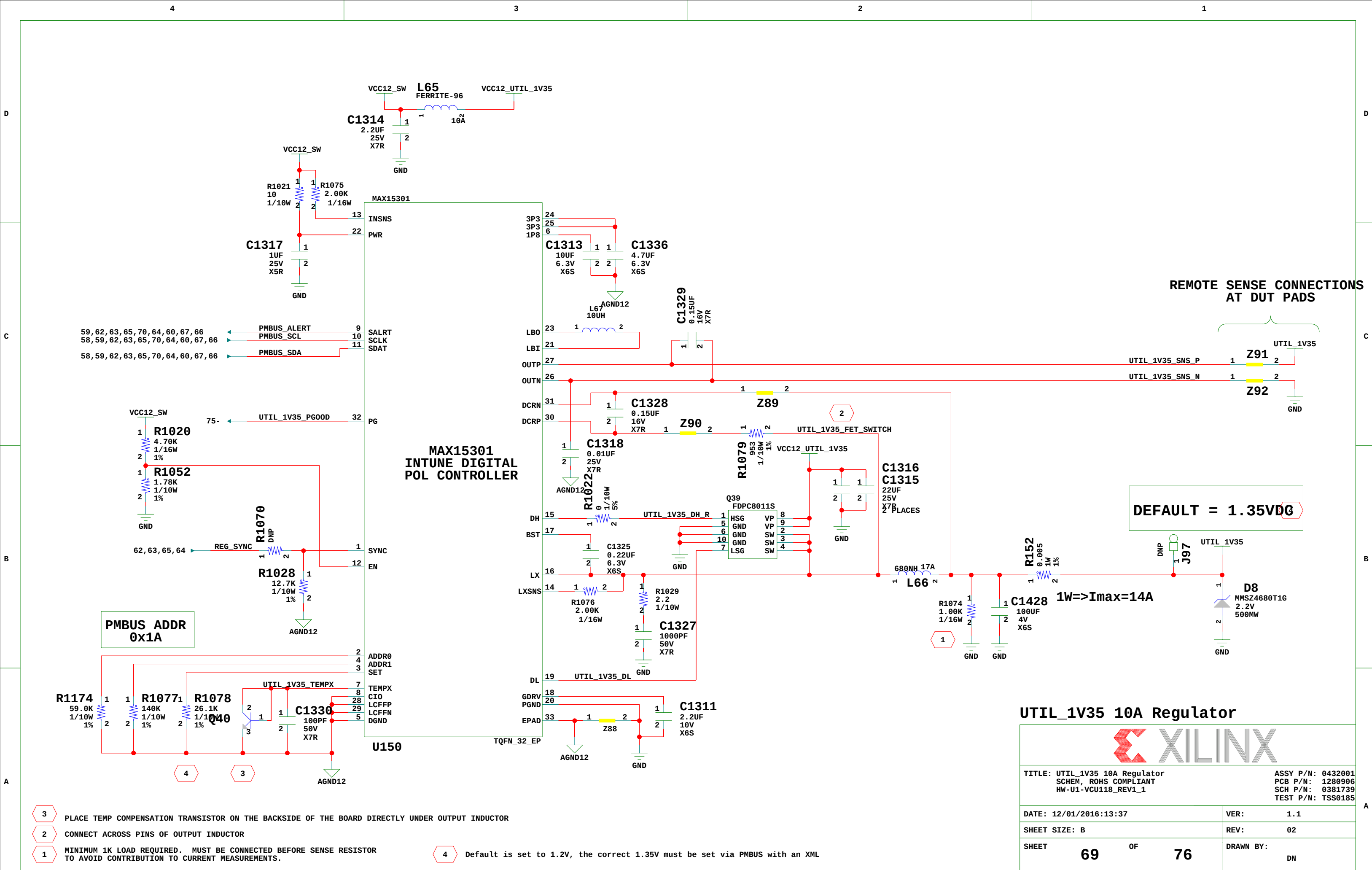


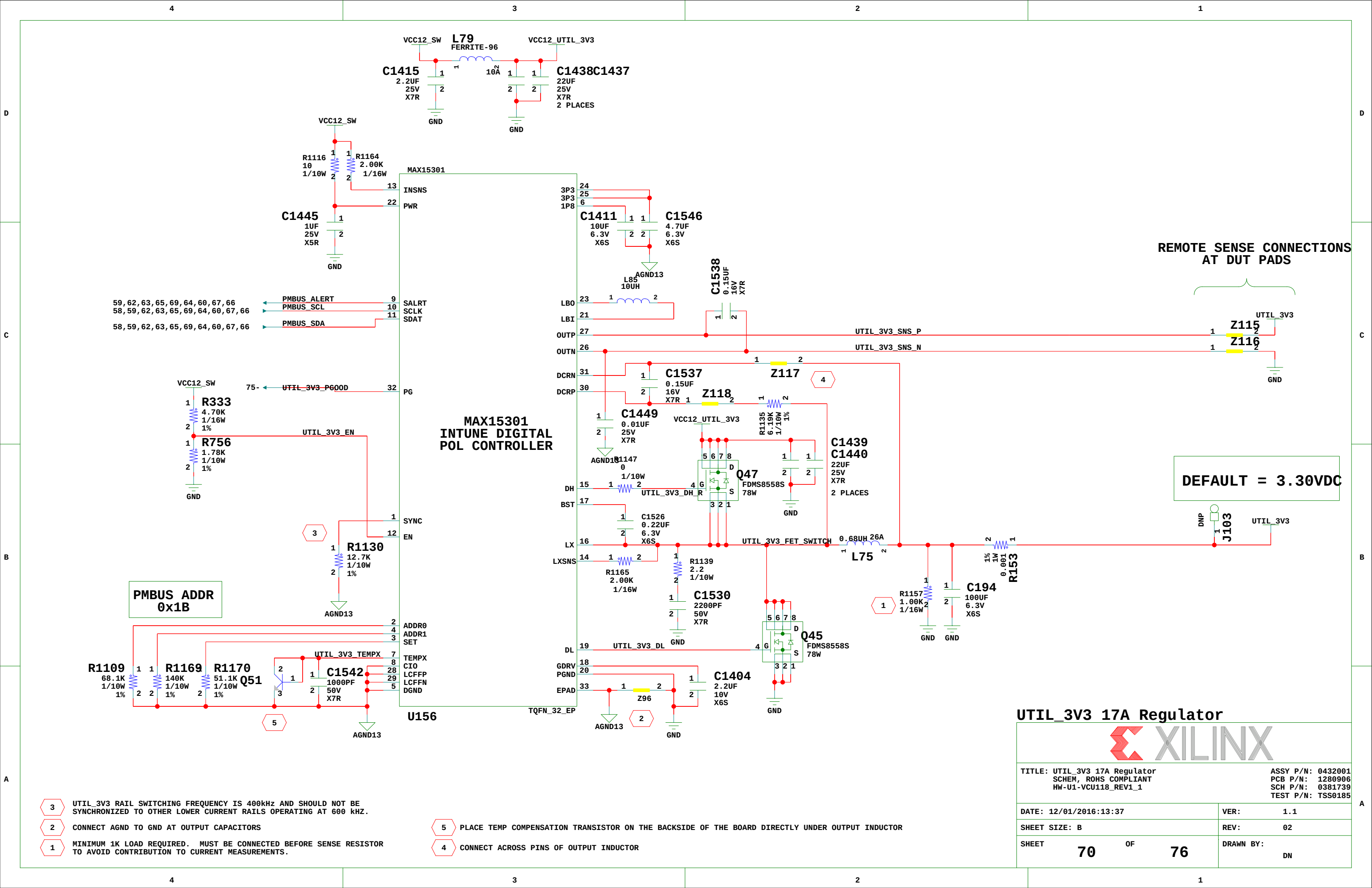
DATE: 12/01/2016:13:39	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 67 OF 76	DRAWN BY: DN



MGTVCCAUX 1A Regulator

TITLE: MGTVCCAUX 1A Regulator SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 68 OF 76	DRAWN BY: DN





REMOTE SENSE CONNECTIONS
AT DUT PADS

DEFAULT = 3.30VDC

UTIL_3V3 17A Regulator



TITLE: UTIL_3V3 17A Regulator
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 70 OF 76	DRAWN BY: DN

- 3

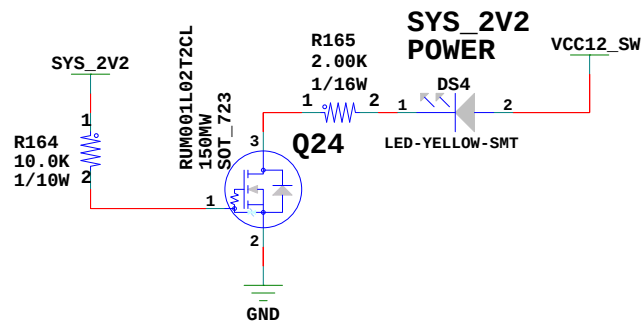
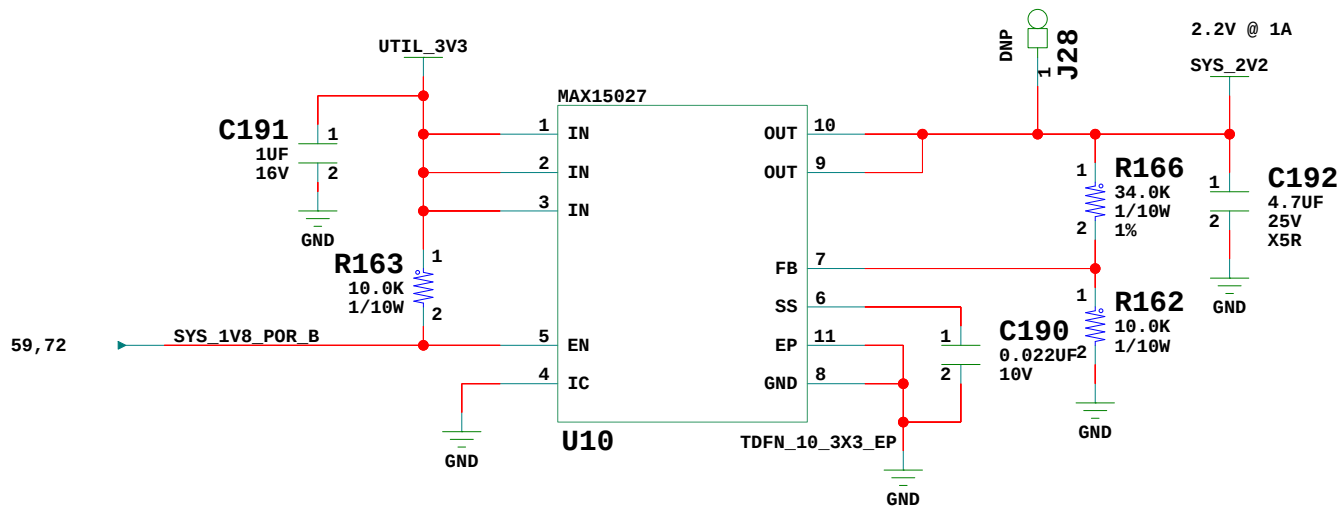
UTIL_3V3 RAIL SWITCHING FREQUENCY IS 400kHz AND SHOULD NOT BE SYNCHRONIZED TO OTHER LOWER CURRENT RAILS OPERATING AT 600 KHZ.
- 2

CONNECT AGND TO GND AT OUTPUT CAPACITORS
- 1

MINIMUM 1K LOAD REQUIRED. MUST BE CONNECTED BEFORE SENSE RESISTOR TO AVOID CONTRIBUTION TO CURRENT MEASUREMENTS.
- 5

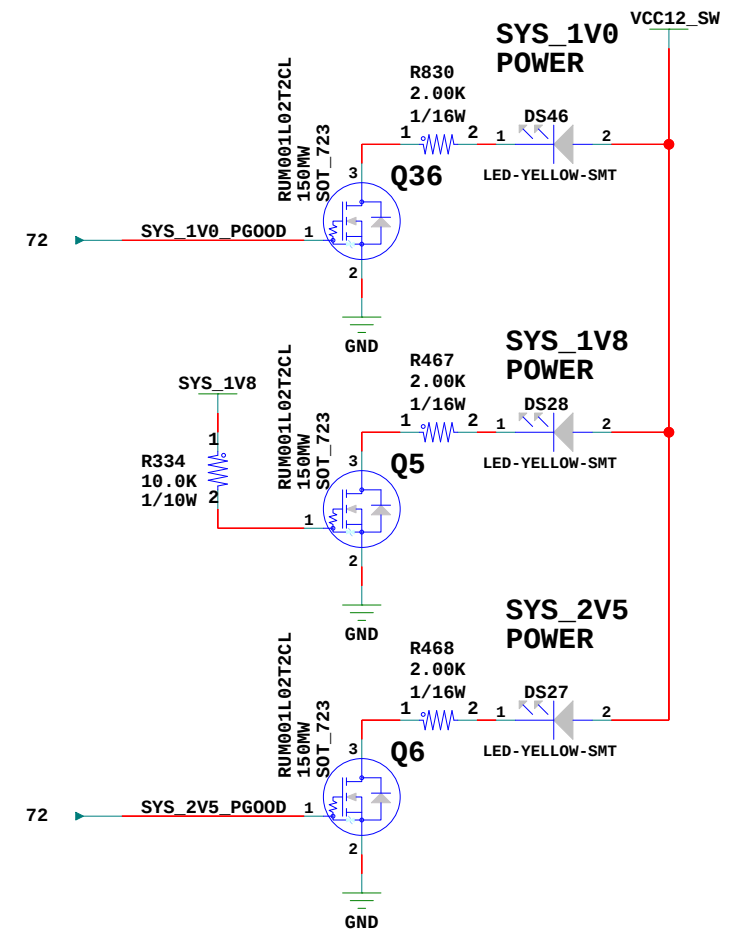
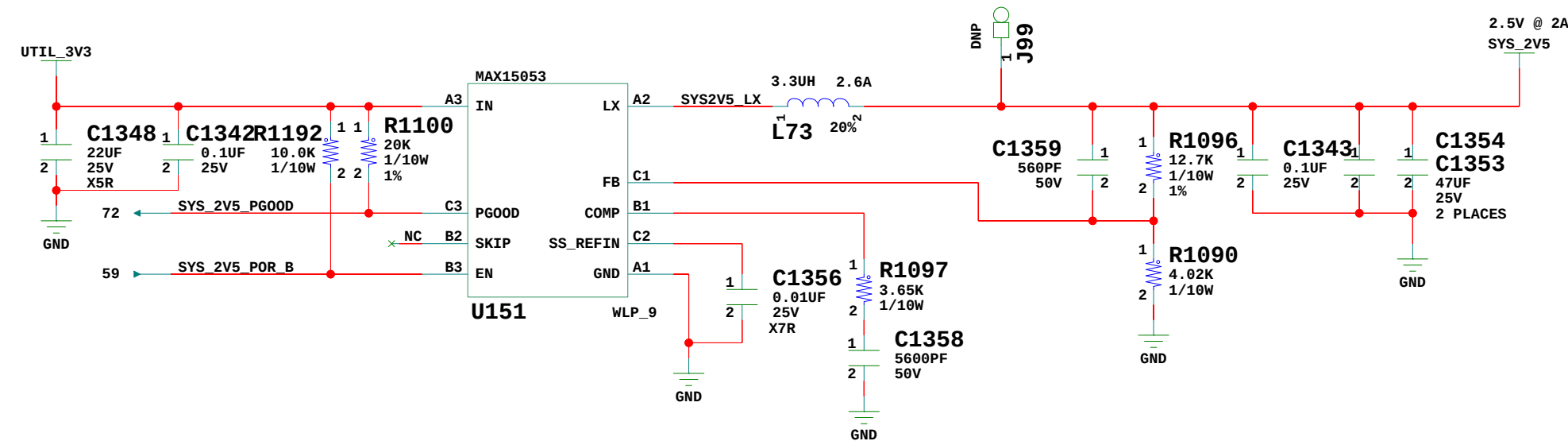
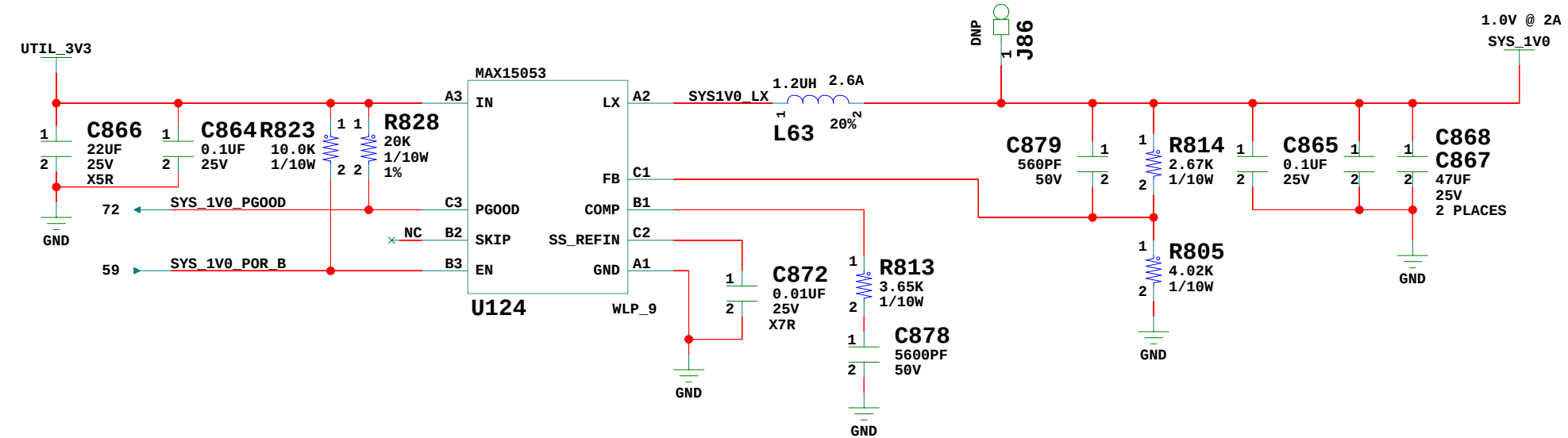
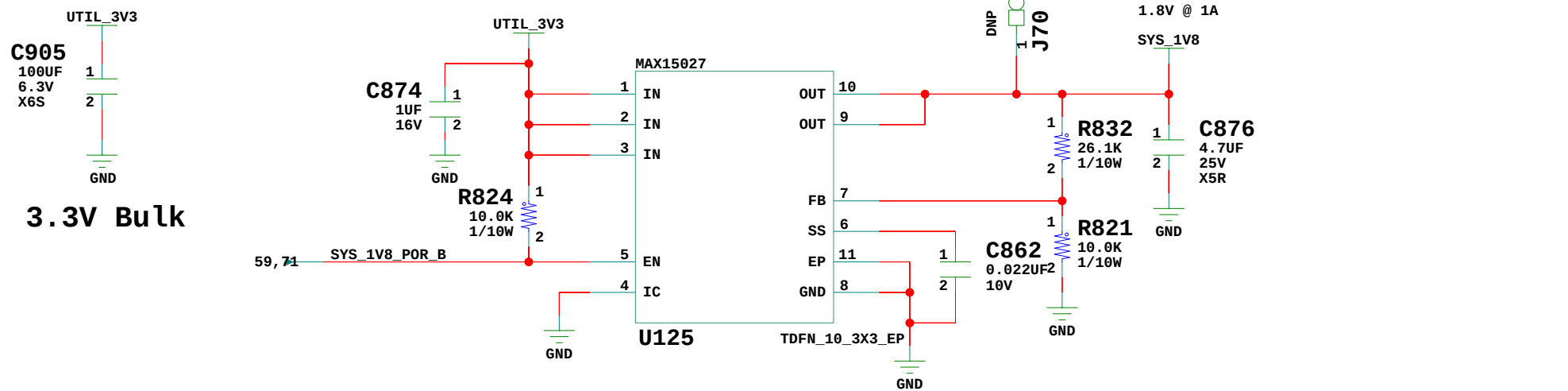
PLACE TEMP COMPENSATION TRANSISTOR ON THE BACKSIDE OF THE BOARD DIRECTLY UNDER OUTPUT INDUCTOR
- 4

CONNECT ACROSS PINS OF OUTPUT INDUCTOR



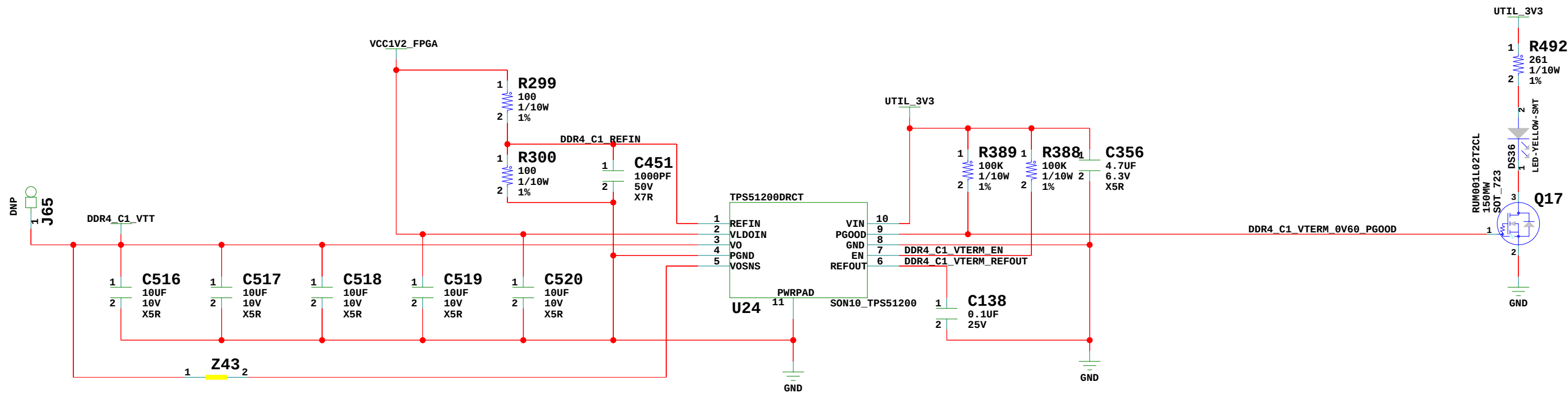
SYS_ 2V2 Regulator

TITLE: SYS_ 2V2 Regulator SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37		VER:	1.1
SHEET SIZE: B		REV:	02
SHEET	71	OF	76
		DRAWN BY:	DN

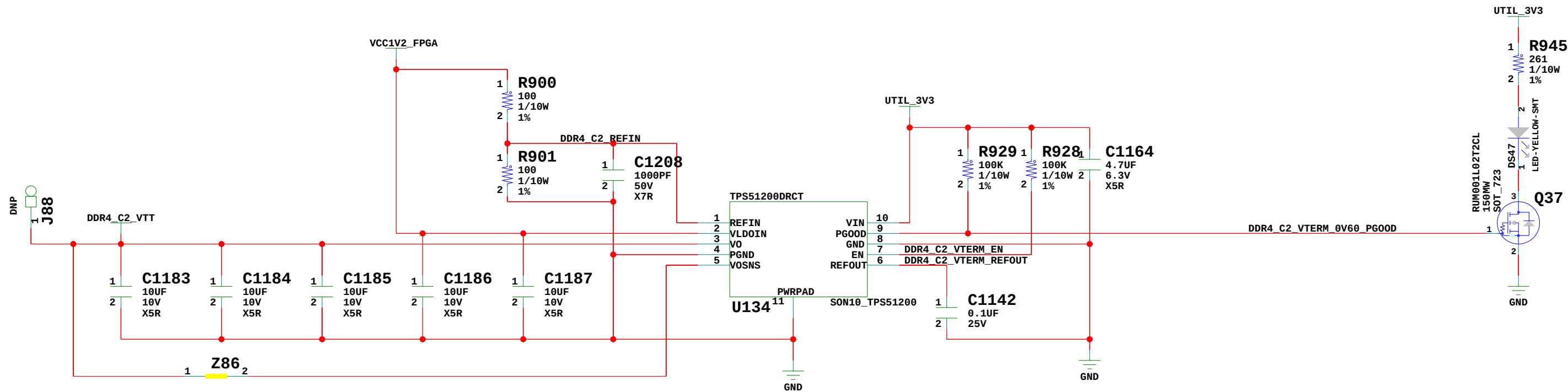


SYS_2V5 SYS_1V8 SYS_1V0 Regulators

TITLE: SYS_2V5 SYS_1V8 SYS_1V0 Regulators SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 72 OF 76	DRAWN BY: DN



Place sense near DDR4 C1 components

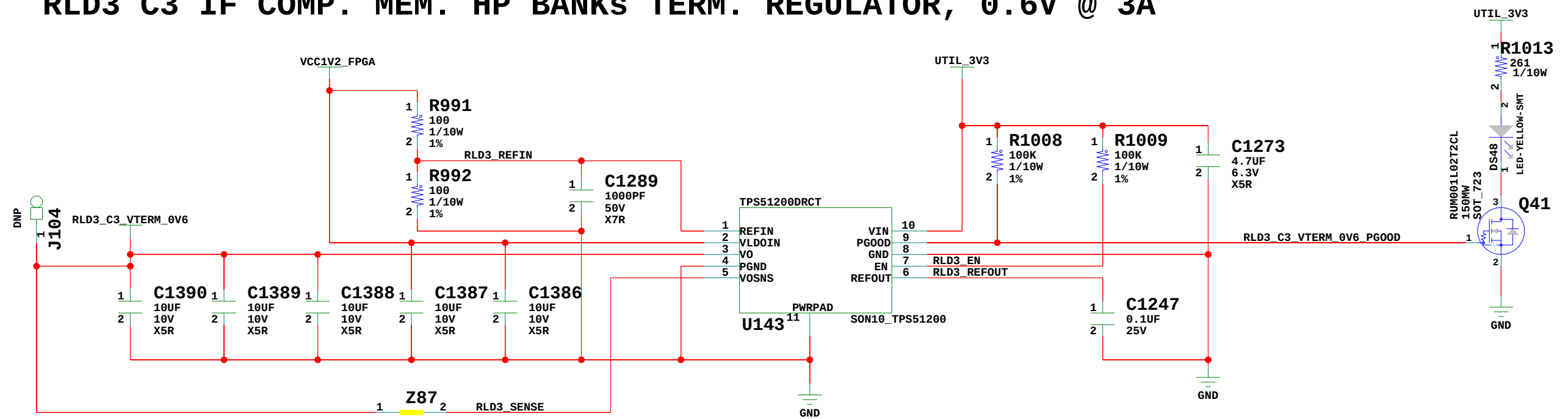


Place sense near DDR4 C2 components

DDR4 Termination Supplies

TITLE: DDR4 Termination Supplies SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 73 OF 76	DRAWN BY: DN

RLD3 C3 IF COMP. MEM. HP BANKs TERM. REGULATOR, 0.6V @ 3A



Place sense near RLD3 C3 components

RLD3 Termination Supply



**TITLE: RLD3 Termination Supply
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1**

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER:	1.1
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SHEET SIZE: B

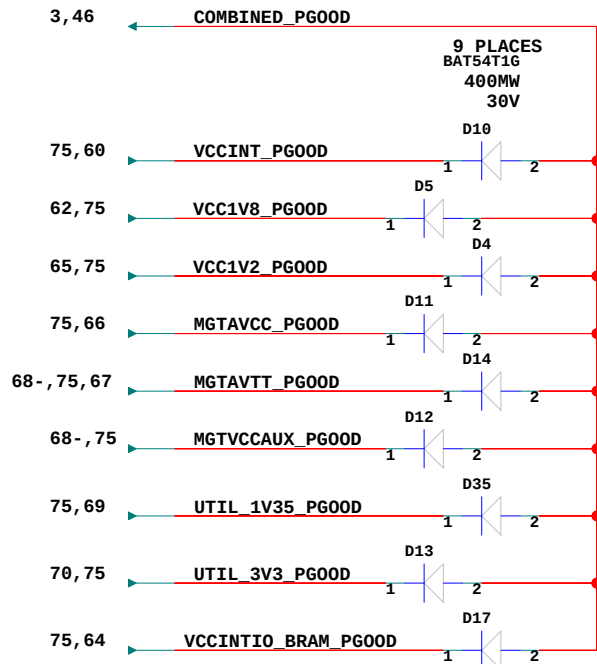
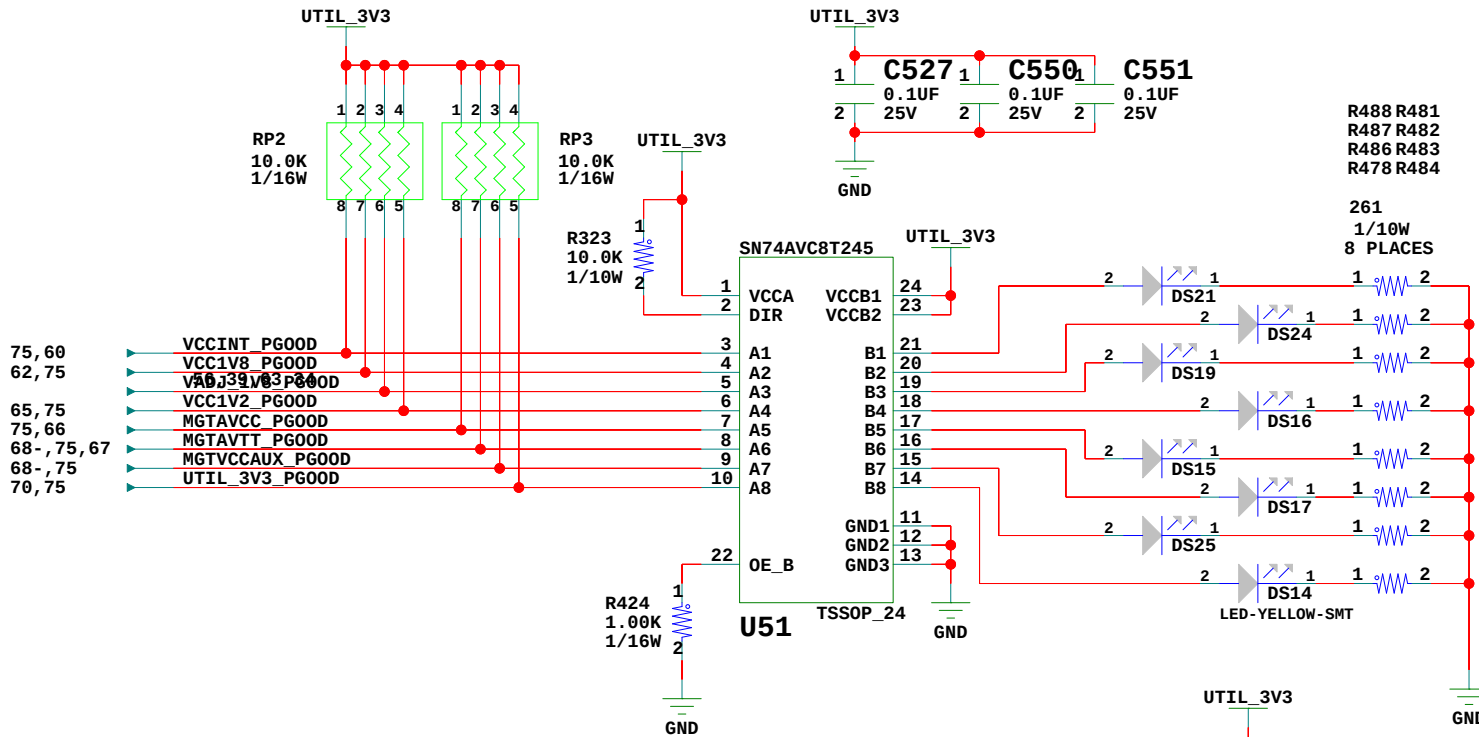
REV:	02
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SHEET **74** OF **76**

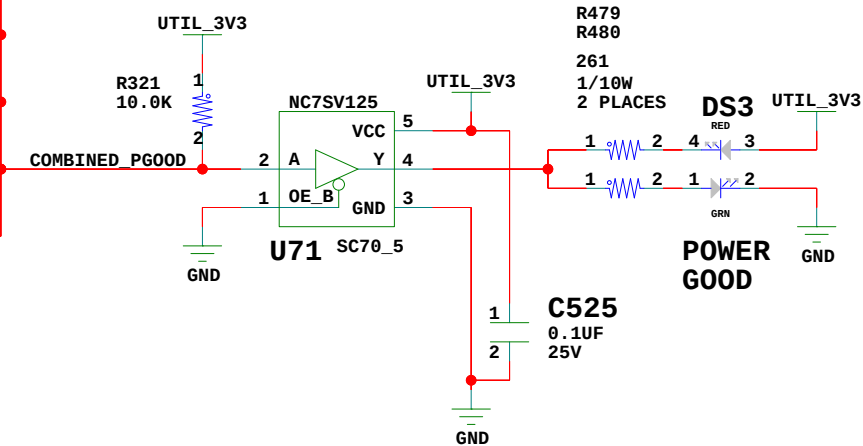
DRAWN BY:

DN

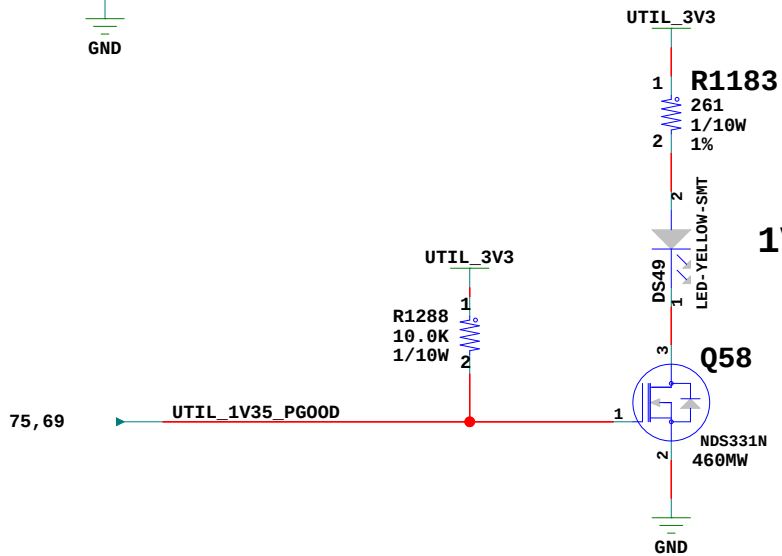
DS19 must be labeled VADJ Power Good Place LEDs near top right corner of board



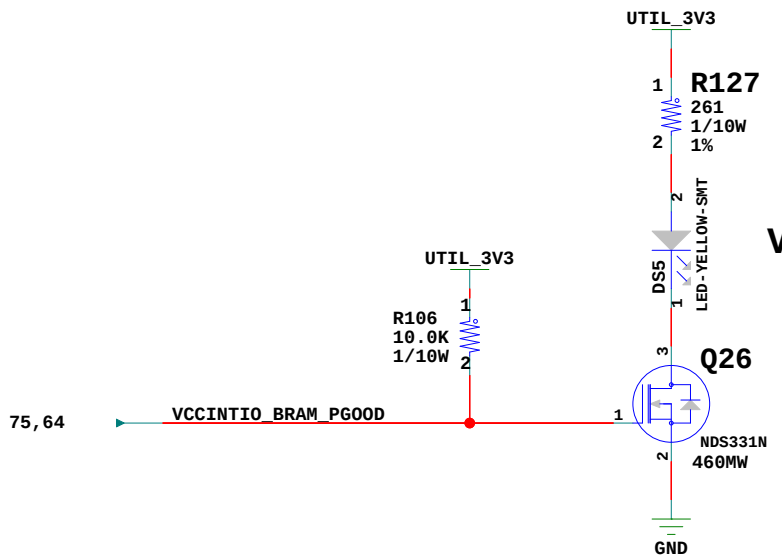
VADJ not included in POWER_GOOD logic



1V35 Power Good

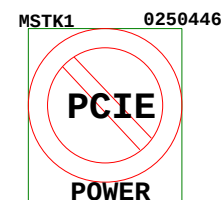
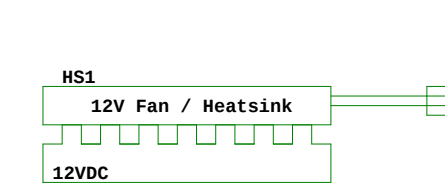


VCCINTIO_BRAM Power Good

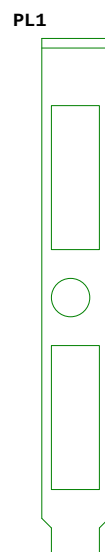


Power Status LEDs

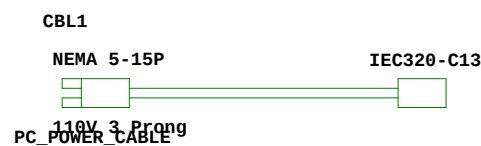
TITLE: Power Status LEDs SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_1	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 12/01/2016:13:37	VER: 1.1
SHEET SIZE: B	REV: 02
SHEET 75 OF 76	DRAWN BY: DN



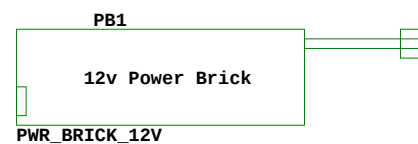
PCIE_POWER_STICKER



VCU118 PCIE PLATE



Power Cord



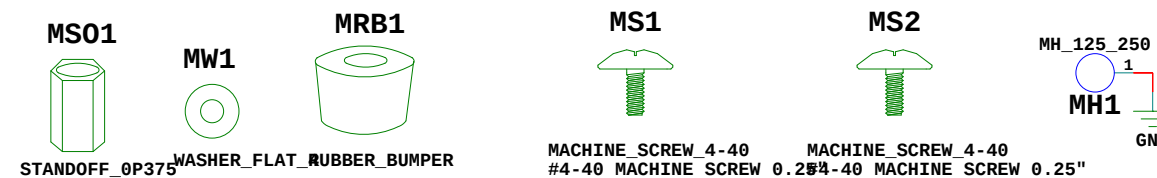
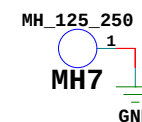
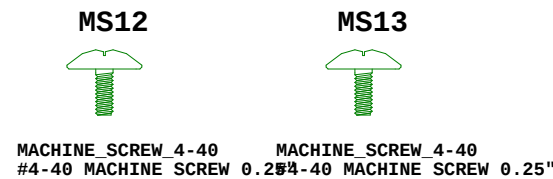
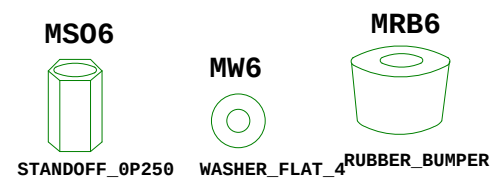
PWR_BRICK_12V



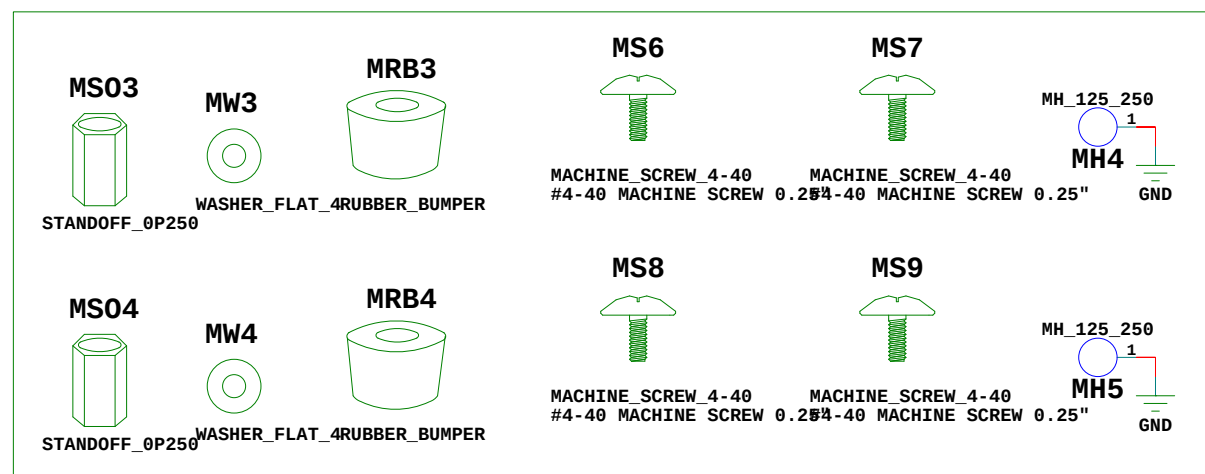
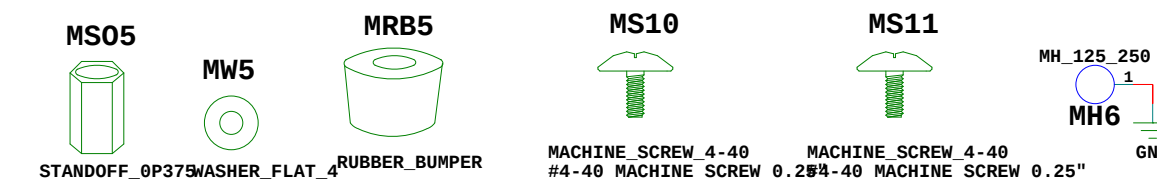
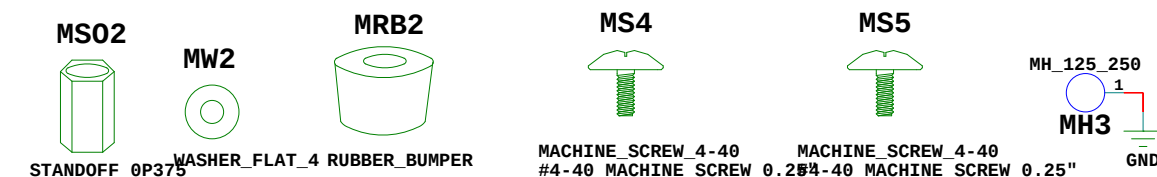
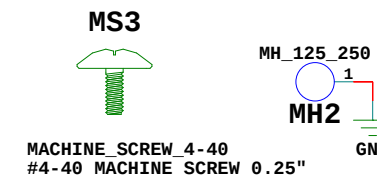
USB Micro-B Cable



USB Micro-B Cable



NOTE: No standoff foot on MH2



NOTE: The two standoffs above are shorter and must be connected to MH4 and MH5 to level the board properly

Mechanical Components



TITLE: Mechanical Components
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_1

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 12/01/2016:13:37

VER:	1.1
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SHEET SIZE: B

REV:	02
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SHEET **76** OF **76**

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